

## 21 - Lecture - Outline

- 1)   
 { conductor  
 insulator 绝缘体  
 semiconductor  
 transistor 晶体管.

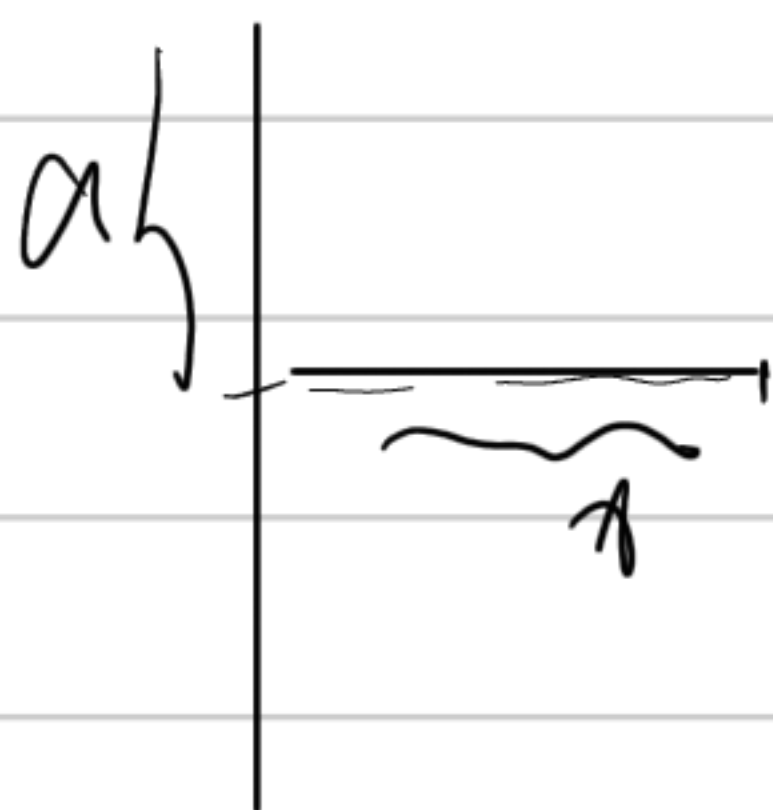
2) electrostatic painting

"positive charge is induced on surface of metal"

3)  $F_{1on2} = F_{2on1} = k \frac{|q_1 q_2|}{r^2}$

$$k = \frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

4) 充电线对点电荷造成的电场



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r\sqrt{r^2+a^2}} \hat{r}$$

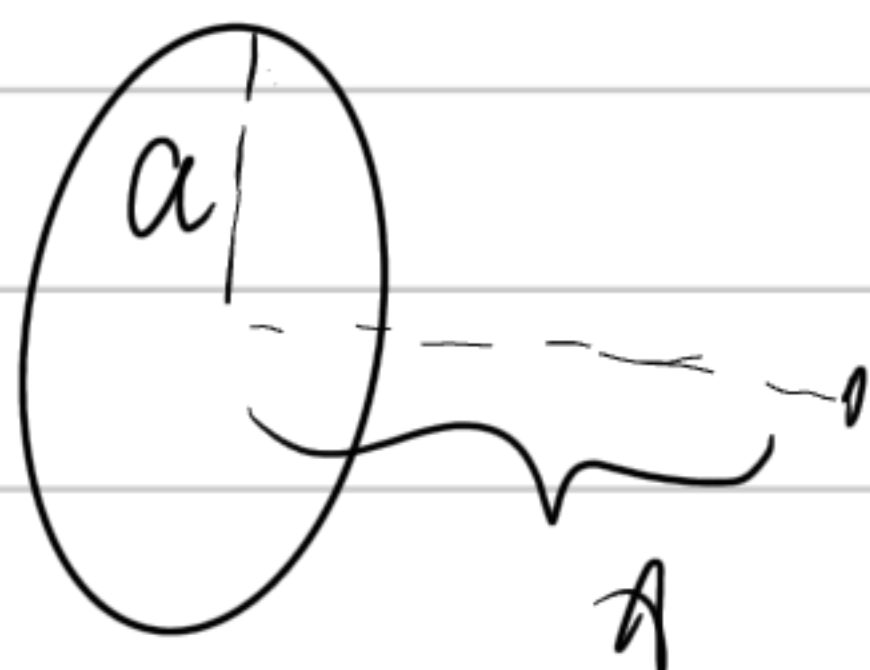
①  $r \gg a$

②  $a \gg r$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda a}{r\sqrt{r^2+a^2}} = \frac{\lambda}{2\pi\epsilon_0 r\sqrt{(\frac{a}{r})^2+1}} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

5) 充电圈



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Qr}{(r^2+a^2)^{3/2}} \hat{r}$$

b) Disk

$$dQ = \sigma dA$$

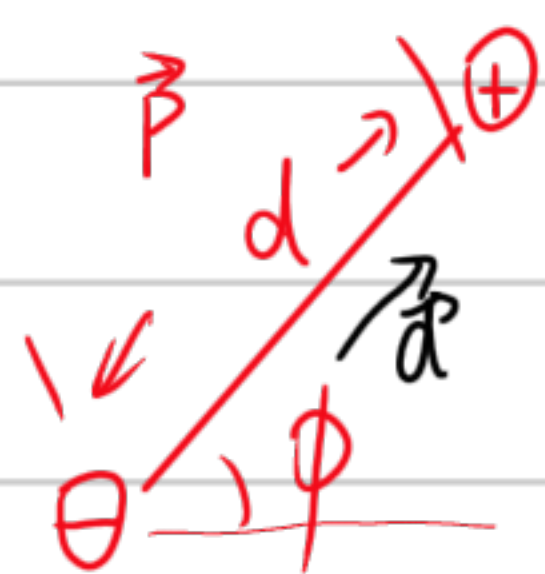
$$\sigma = \frac{Q}{\pi R^2}$$

$$dE = \frac{dQ}{4\pi\epsilon_0 (r^2+r'^2)} = \frac{\sigma}{4\pi\epsilon_0 (r^2+r'^2)} \overset{r dr d\theta}{\underset{\uparrow}{dA}} =$$



77 electric dipole

$\vec{E}$



所有量异电荷  
电偶极矩

electric dipole moment

$$\vec{p} = q\vec{d}$$

$\vec{d}$  从负电荷到正电荷的  
位移向量

$$\tau = pE \sin \phi$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$



21. lecture

P120 不懂

对比磁偶极矩  $\vec{\mu} = I\vec{A}$

22 - Lecture

Chap 22

Gauss's Law

Electric Flux through a cube:  $\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$

球体面积分  $dA = r^2 \sin \phi \, d\phi \, d\theta$

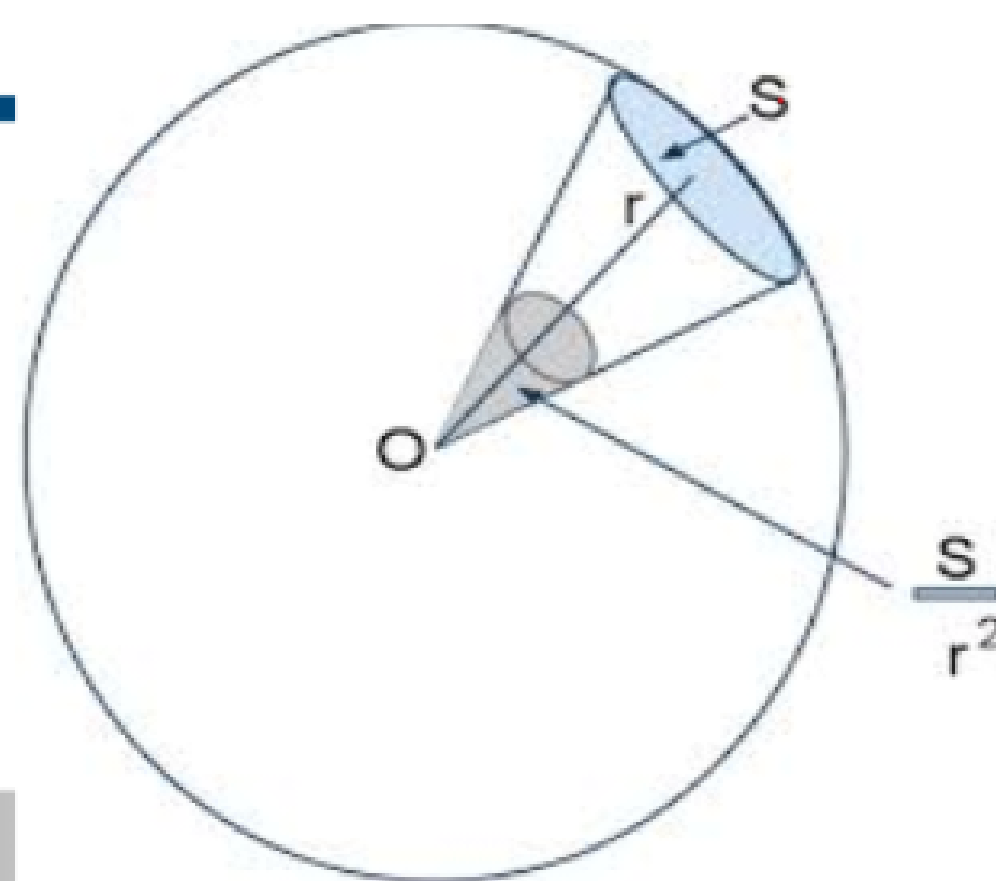
Gauss's Law

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{Gauss's law})$$

$$\underbrace{\nabla \cdot \vec{E}}_{\text{散度}} = \frac{1}{\epsilon_0} \rho$$



# Gauss's law



- Consider charge in a generic surface  $S$
- Surround charge with spherical surface  $S_1$  concentric to charge
- Consider cone of **solid angle**  $d\Omega$  from charge to surface  $S$  through the little sphere
- Electric flux through little sphere:

$$d\Phi_{S_1} = \vec{E} \cdot d\vec{A} = \left(\frac{q}{r^2} \hat{r}\right) (r^2 d\Omega \hat{r}) = q d\Omega$$

- Electric flux through surface  $S$ :

$$d\Phi_S = \vec{E} \cdot d\vec{A} = \left(\frac{q}{R^2} \hat{r}\right) \cdot \left(\frac{R^2 d\Omega}{\cos \theta} \hat{n}\right) = q d\Omega \frac{\hat{r} \cdot \hat{n}}{\cos \theta} = q d\Omega$$

- $d\Phi_S = d\Phi_{S_1} \rightarrow \Phi_S = \Phi_{S_1} = 4\pi Q$

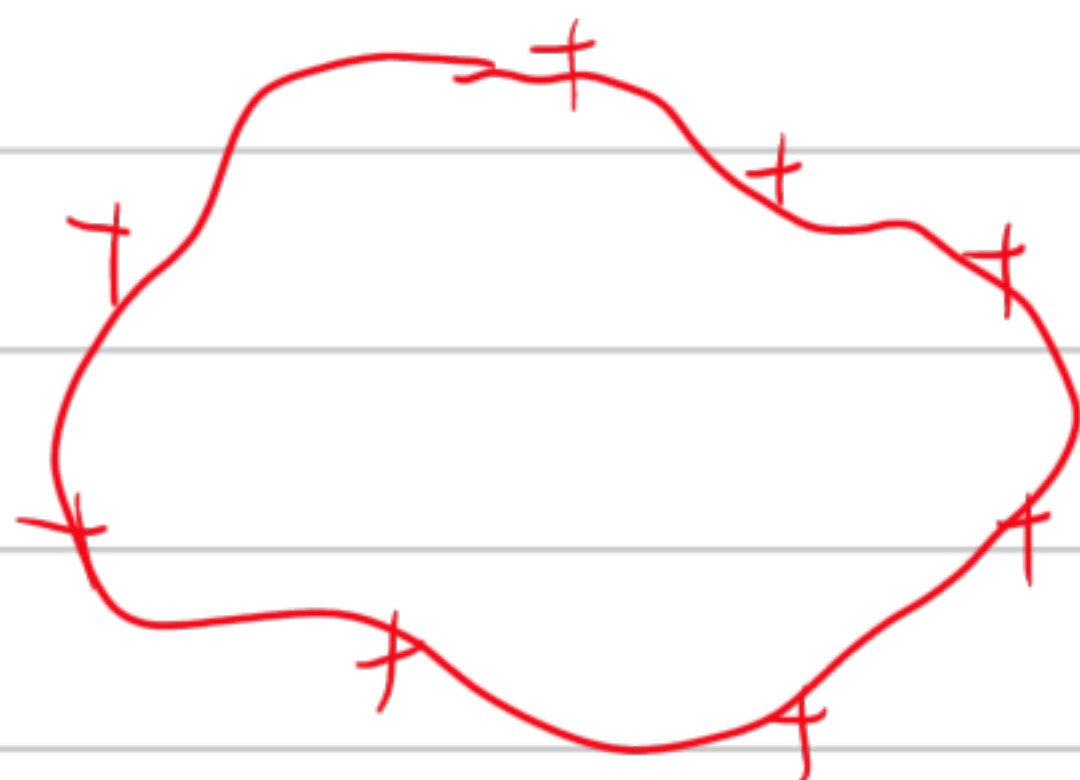
$$\Phi = \oint_S \vec{E} \cdot d\vec{A} = 4\pi Q_{encl} \text{ is valid for ANY shape } S.$$

以高斯单位制表示的

实心导体

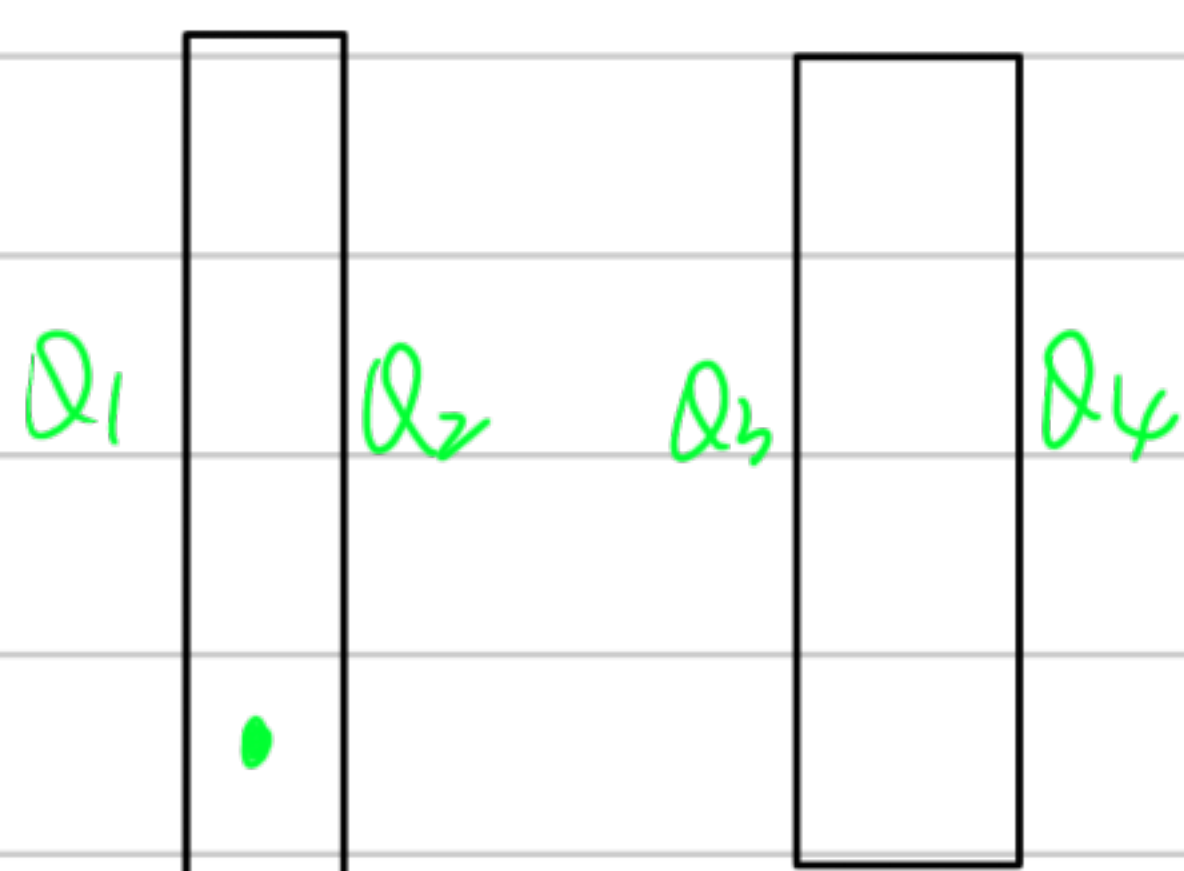
电荷均匀分布在表面  
内部任意点的电场为0

\*\*\*



$+3Q$

$+Q$



$$Q_1 + Q_2 = 3Q$$

$$Q_3 + Q_4 = Q$$

$$Q_2 + Q_3 = 0$$

$$E_1 = E_2 + E_3 + E_4$$

$$E = \frac{\rho}{2\epsilon_0} = \frac{Q}{2\epsilon_0 A}$$

$$Q_1 = Q_2 + Q_3 + Q_4$$



3. **立体角公式**: 从点  $(x_0, y_0, z_0)$  看向面积元素  $dA$  所张开的立体角  $d\Omega$  的表达式是:

$$d\Omega = \frac{dA \cos \theta}{|\vec{r}|^2}.$$

- $dA$ : 这是面积元素的大小。
- $\cos \theta$ : 这是面积元素的法向量和观察点方向之间的角度余弦, 表示方向上的投影。
- $|\vec{r}|^2$ : 这是面积元素与观察点之间距离的平方。

这个公式的意义是: 面积元素  $dA$  对点  $(x_0, y_0, z_0)$  所张开的立体角与面积元素的大小、方向和距离相关。直观地讲, 如果面积元素面向观察点 (即  $\theta = 0$ ,  $\cos \theta = 1$ ), 它对该点的立体角贡献更大; 如果它背向观察点 (即  $\theta = 90^\circ$ ,  $\cos \theta = 0$ ), 它的贡献则为零。

#### 如何应用于电场计算?

我们可以利用这个立体角来简化电场的计算。根据库仑定律, 微小电荷  $dq$  产生的电场  $d\vec{E}$  可以用立体角来表达, 从而避免复杂的三维积分。利用公式:

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\sigma dA \hat{r}}{|\vec{r}|^2} = \frac{1}{4\pi\epsilon_0} \sigma \hat{r} d\Omega,$$

这其中,  $\hat{r}$  是从  $dA$  指向观察点的单位向量, 而  $d\Omega$  是上面提到的立体角。这个公式表明, 电场的大小可以直接和立体角联系起来, 从而简化了计算过程。

希望这些解释能帮助你理解这个公式背后的物理意义! 如果还有其他问题, 随时告诉我!



# Electric Potential

1. 电势能  $U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$

$$W_{a \rightarrow b} = Fd = q_0 Ed = U_a - U_b$$

if a collection of charge  $q_i$

$$U = \frac{q_0}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

2. 电势  $V = \frac{U}{q_0} = \frac{q}{4\pi\epsilon_0 r}$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \quad V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

3. 电势差与场强积分  $V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E \cos\phi dl$

知



# Capacitors

<sup>相对</sup>  
 $K = \epsilon / \epsilon_0$  电常数

dielectric constant  $K = \frac{\epsilon}{\epsilon_0} > 1$   $K\epsilon_0 = \epsilon$ .

用介质会增加电容值 by a factor  $K$

但会减少两板之间电势差。  
↑  
一定程度

$$E = \frac{\sigma}{\epsilon_0} \quad \xrightarrow{+ \text{介质}}$$

$$E = \frac{\sigma}{\epsilon} \quad \begin{array}{l} \rightarrow \text{free surface charge.} \\ \checkmark \text{绝对电常数} \end{array}$$
$$\epsilon = K\epsilon_0$$

$\sigma$ : charge density on capacitor

$\sigma_i$ : induced charge density on dielectric

$\epsilon = K\epsilon_0$  新定义的绝对电常数

$$K = \epsilon_r = \frac{\epsilon}{\epsilon_0}$$

$$u = \frac{1}{2} K \epsilon_0 E^2 = \frac{1}{2} \epsilon E^2 \quad (\text{electric energy density in a dielectric}) \quad (24.20)$$

$$\oint K \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (\text{Gauss's law in a dielectric})$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon} \quad (\text{Gauss's law in a dielectric})$$

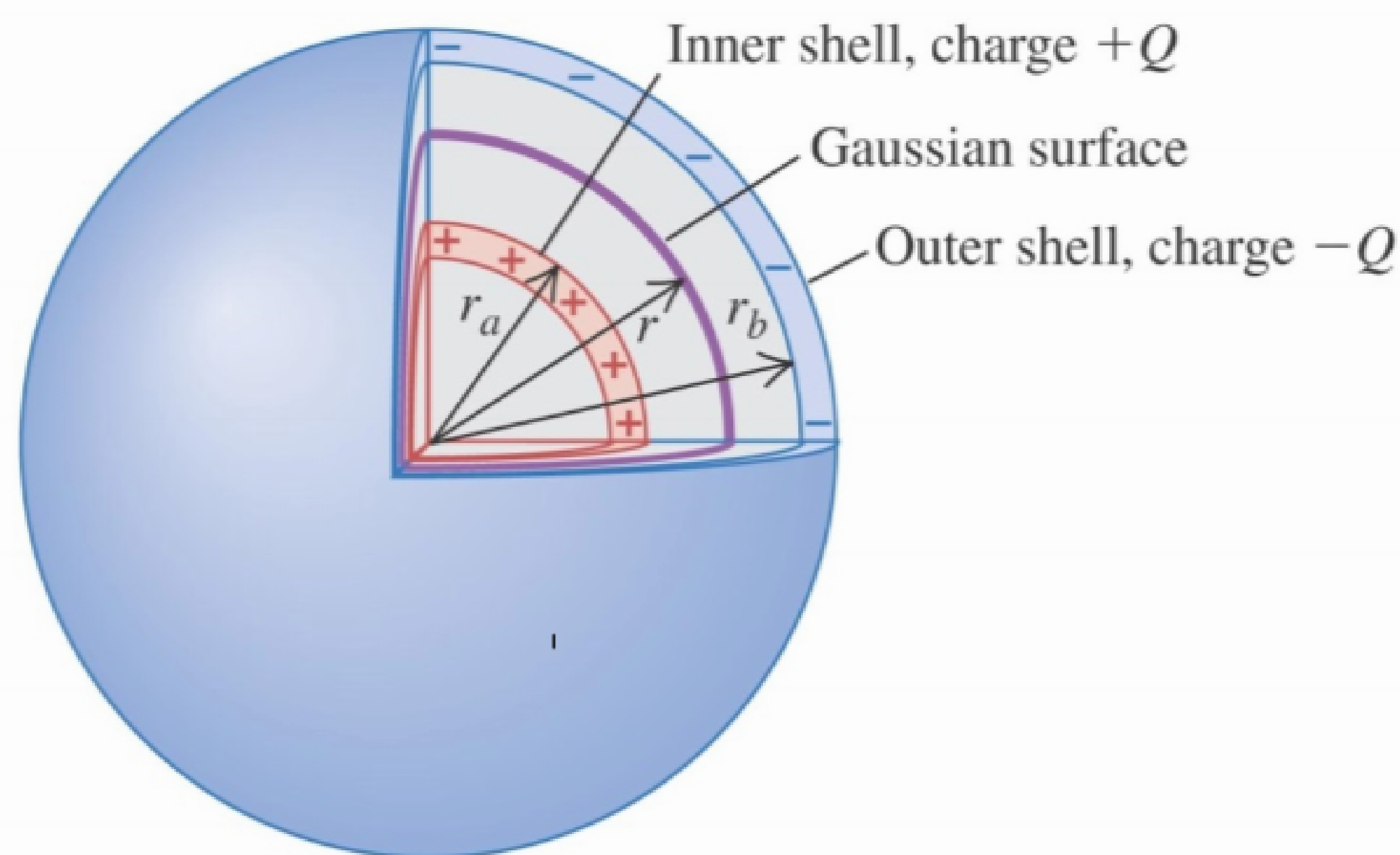


$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

$$C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (\text{capacitance of a parallel-plate capacitor in vacuum})$$

$$V_{ab} = Ed = \frac{1}{\epsilon_0} \frac{Qd}{A}$$

球壳类电容器



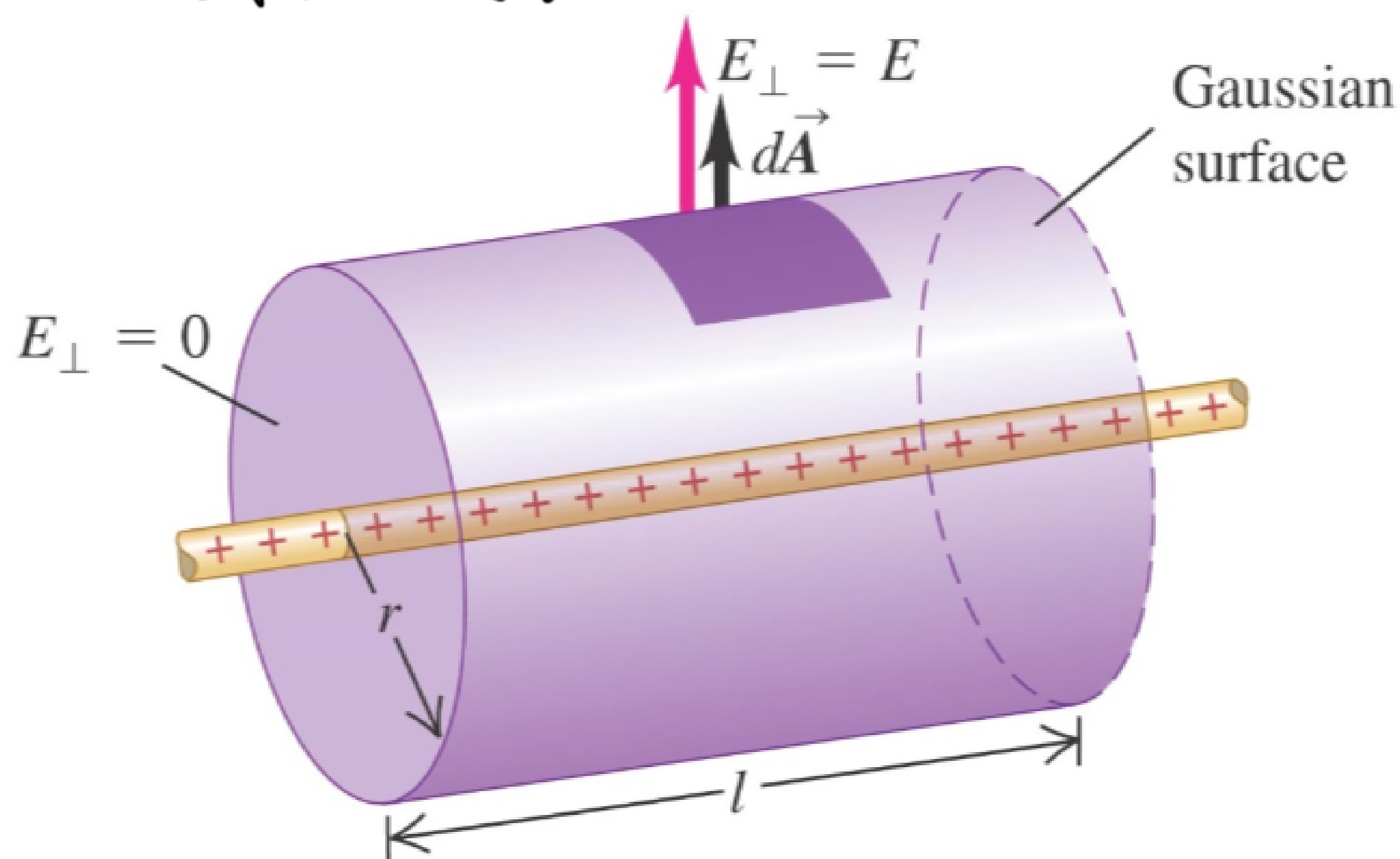
$$V_{ab} = V_a - V_b = \frac{Q}{4\pi\epsilon_0 r_a} - \frac{Q}{4\pi\epsilon_0 r_b}$$

$$= \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r_a} - \frac{1}{r_b} \right) = \frac{Q}{4\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b}$$

itance is then

$$C = \frac{Q}{V_{ab}} = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

圆柱类电容器、用线电荷的场强公式推导

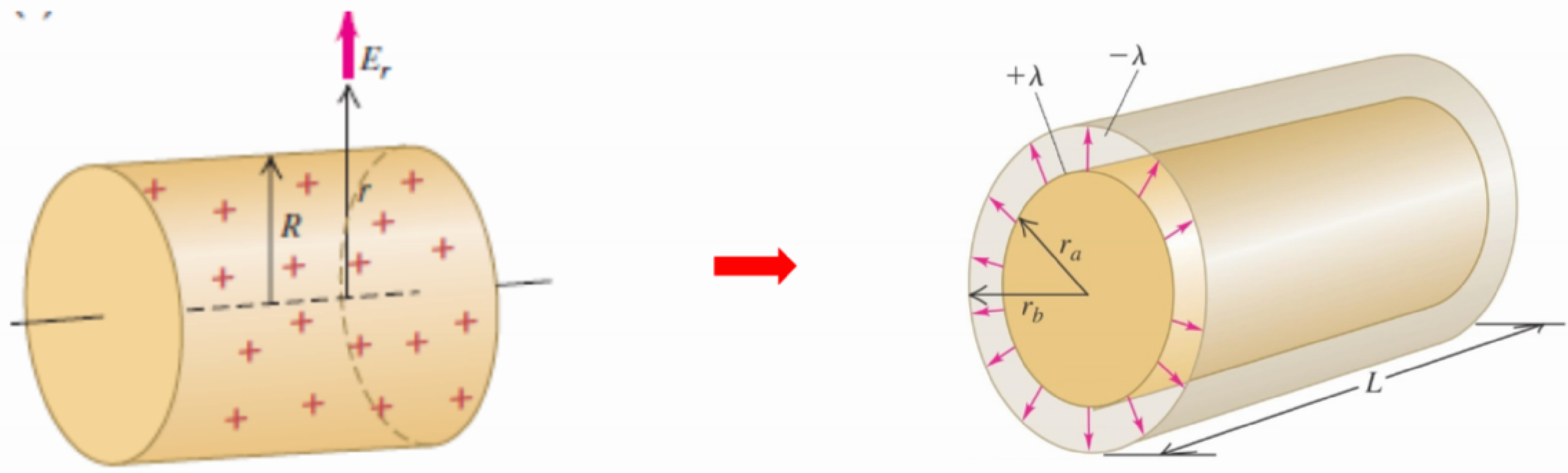


$Q_{\text{encl}} = \lambda l$ , and so from Gauss's law, Eq. (22.8),

$$\Phi_E = 2\pi r l E = \frac{\lambda l}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (\text{field of an infinite line of charge})$$





$$C = \frac{Q}{V_{ab}} = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}} = \frac{2\pi\epsilon_0 L}{\ln(r_b/r_a)}$$

The capacitance per unit length is

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln(r_b/r_a)}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (\text{field of an infinite line of charge})$$

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E_r dr = \frac{\lambda}{2\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

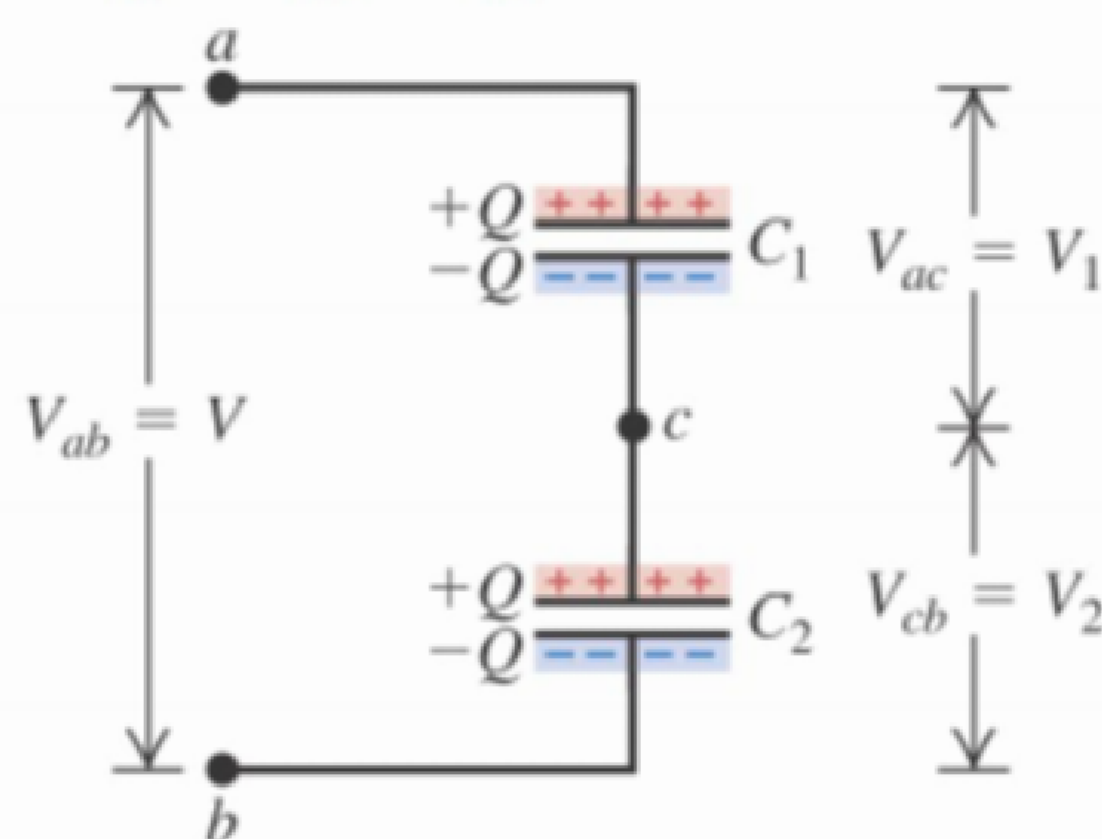
## 电容器的串并联

- The *equivalent capacitance* of a series combination is given by  $1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \dots$

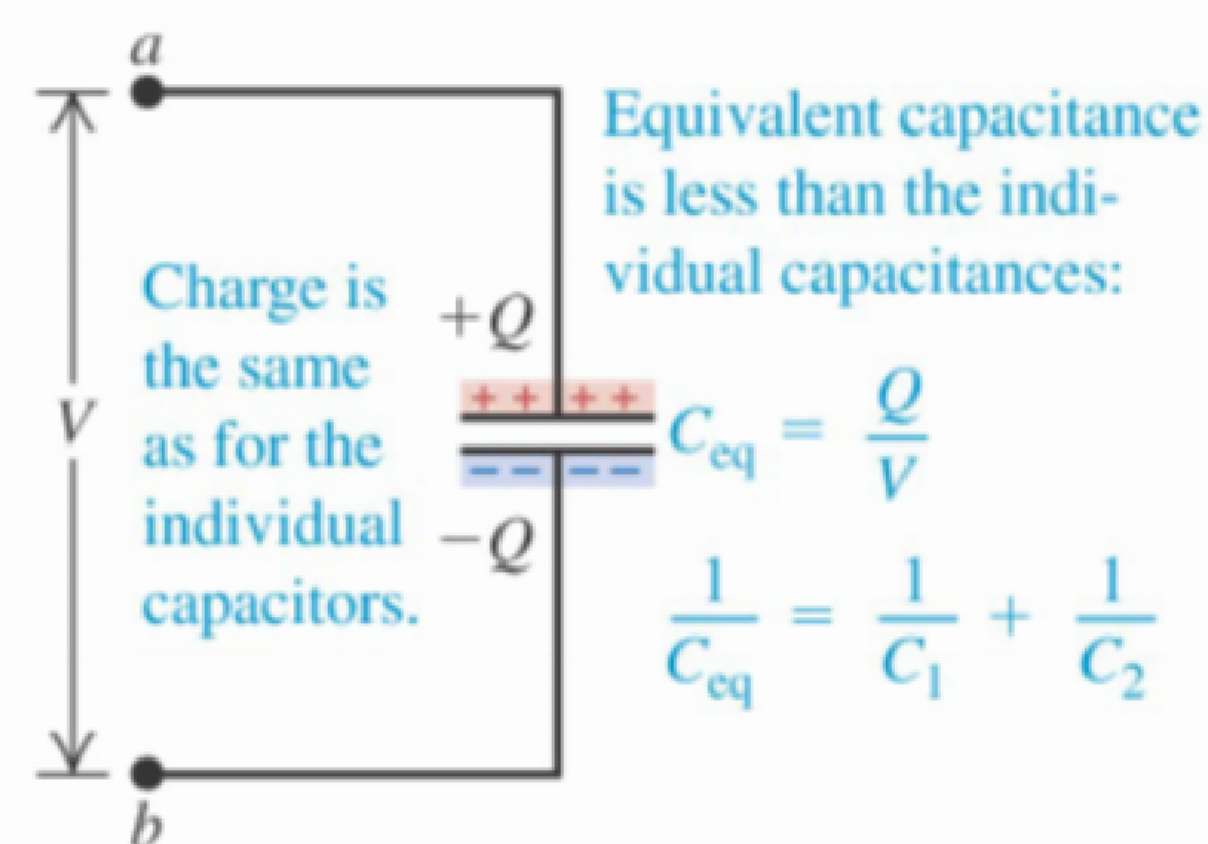
(a) Two capacitors in series

### Capacitors in series:

- The capacitors have the same charge  $Q$ .
- Their potential differences add:  
 $V_{ac} + V_{cb} = V_{ab}$ .



(b) The equivalent single capacitor



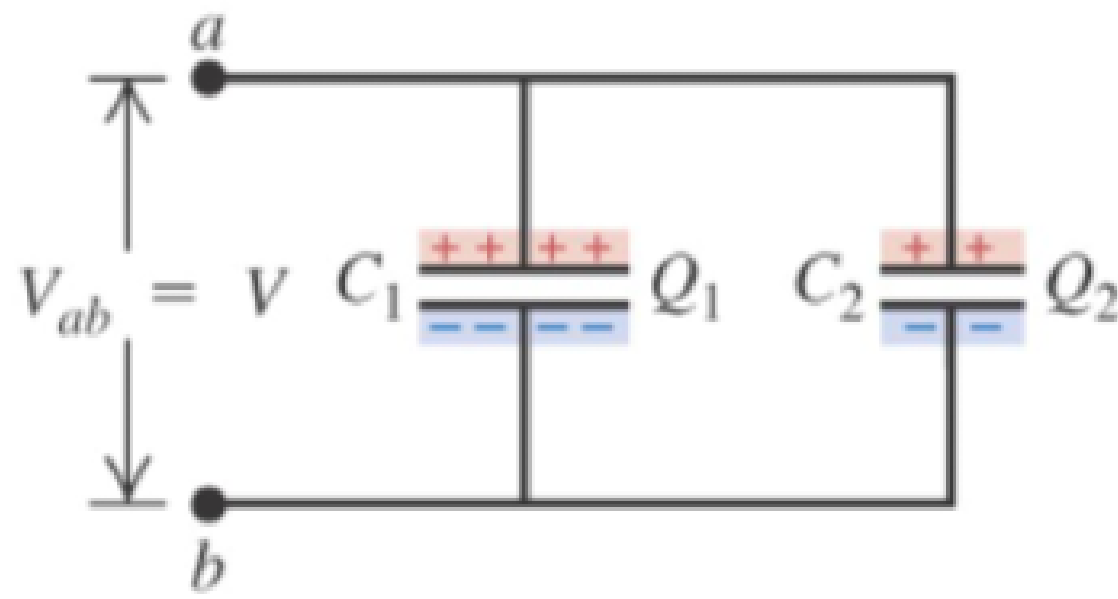


- The *equivalent capacitance* of a parallel combination is the *sum* of the individual capacitances:  $C_{eq} = C_1 + C_2 + C_3 + \dots$

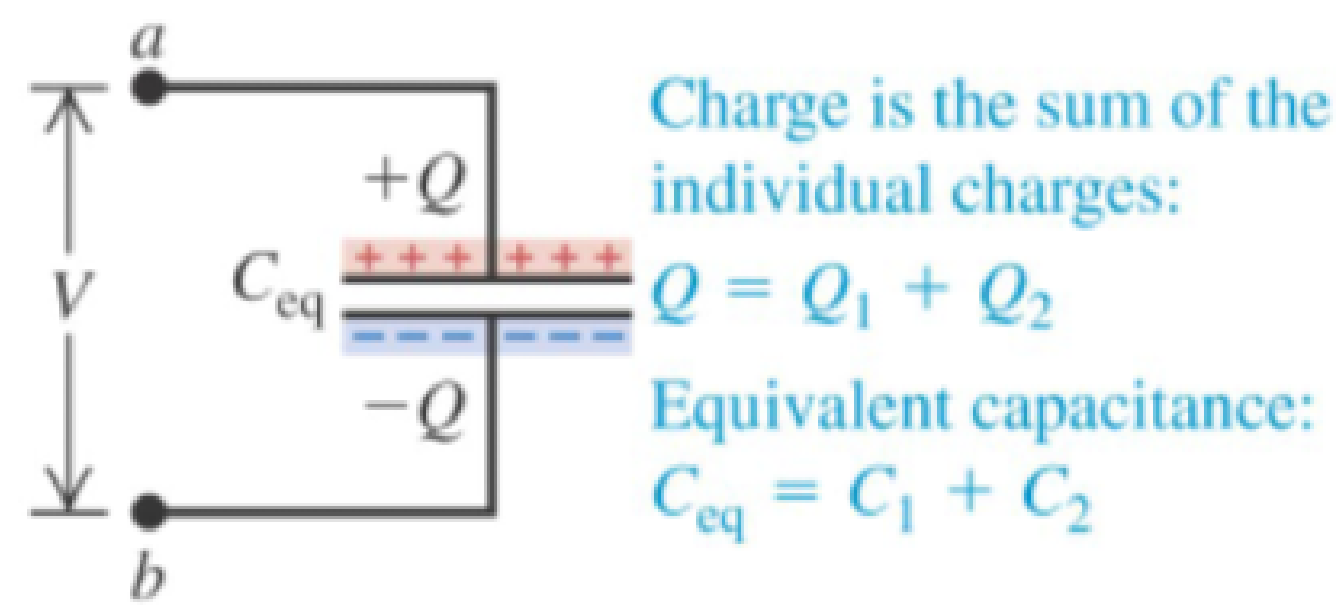
(a) Two capacitors in parallel

**Capacitors in parallel:**

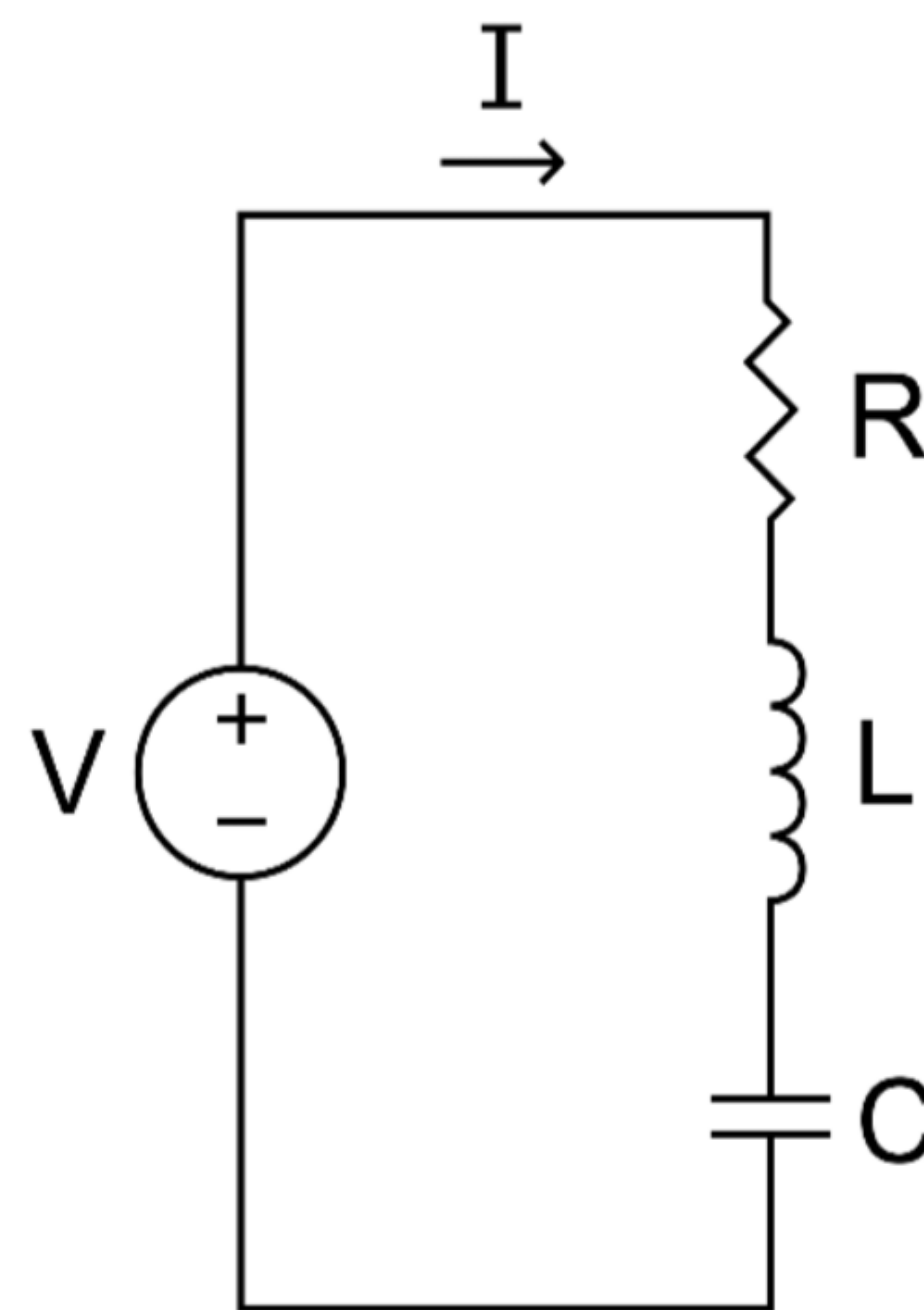
- The capacitors have the same potential  $V$ .
- The charge on each capacitor depends on its capacitance:  $Q_1 = C_1 V$ ,  $Q_2 = C_2 V$ .



(b) The equivalent single capacitor



$$\omega_0 = \frac{1}{\sqrt{LC}}$$



- The potential energy stored in a capacitor is

$$V = \frac{Q}{C}$$

↑  
电压

$$dW = v dq = \frac{q dq}{C}$$

$$W = \int_0^W dW = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C}$$

$$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$$

(potential energy stored in a capacitor)

- The *energy density* is  $u = \frac{1}{2} \epsilon_0 E^2$ .

$$u = \text{Energy density} = \frac{\frac{1}{2} CV^2}{Ad}$$

$$C = \epsilon_0 A/d.$$

$$V = Ed.$$

↓ 能量密度,  $\frac{1}{2} \epsilon_0 E^2$



# Current, drift velocity and current density

1.  $dQ = q(n A v_d dt) = nq v_d A dt$

$$I = \frac{dQ}{dt} = nq v_d A$$

$$\vec{J} = \frac{I}{A} = nq \vec{v}_d$$

concentration of moving charged particles  $\swarrow$   $\nearrow$  charge per particle

2. mean free time 推导漂移速度

$$\tau \quad \vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m}$$

电子在加速过程中，实际加速时间被控制在两次碰撞之间，即平均自由时间  $\tau$

$$\vec{v} = \vec{v}_0 + \vec{a}\tau$$

$$\text{若 } \vec{v}_0 = 0 \Rightarrow \vec{v}_d = \vec{v} = \vec{a}\tau = \frac{q\tau\vec{E}}{m}$$

$$\vec{J} = nq\vec{v}_d = \frac{nq^2\tau\vec{E}}{m}$$

电阻率

$$\rho = \frac{E}{J} = \frac{m}{nq^2\tau} \quad \text{对于金属电阻}$$

$$\rho = \frac{m}{ne^2\tau} \quad \text{单位 } \Omega \cdot m$$

$$R = \rho \frac{L}{S}$$

$$G = \frac{1}{R}$$

电导

$$\sigma = \frac{1}{\rho}$$

电导率



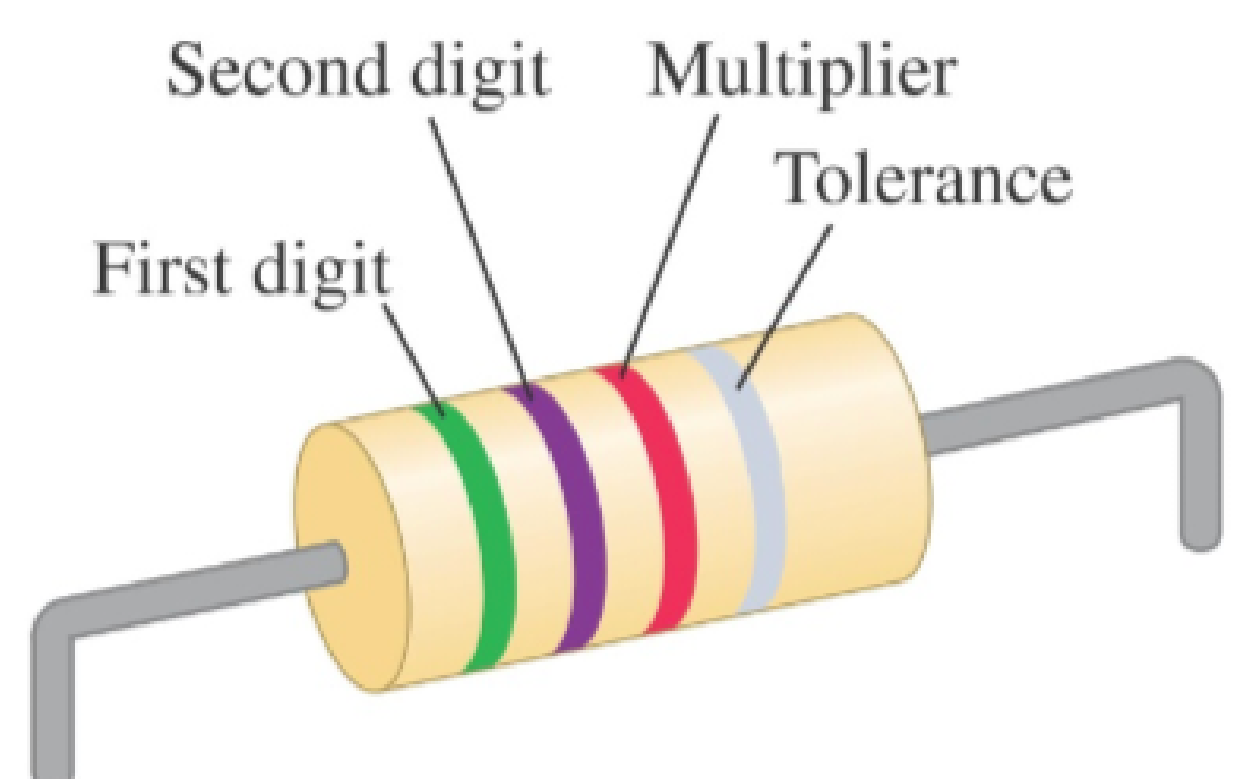
$$\rho(T) = \rho_0[1 + \alpha(T - T_0)] \quad (\text{temperature dependence of resistivity})$$

$$\rho = \frac{m}{ne^2\tau}$$

3. resistors are color-coded for easy identification
- This resistor has a resistance of  $5.7 \text{ k}\Omega$  with a tolerance of  $\pm 10\%$ .

**Table 25.3** Color Codes for Resistors

| Color  | Value as Digit | Value as Multiplier |
|--------|----------------|---------------------|
| Black  | 0              | 1                   |
| Brown  | 1              | 10                  |
| Red    | 2              | $10^2$              |
| Orange | 3              | $10^3$              |
| Yellow | 4              | $10^4$              |
| Green  | 5              | $10^5$              |
| Blue   | 6              | $10^6$              |
| Violet | 7              | $10^7$              |
| Gray   | 8              | $10^8$              |
| White  | 9              | $10^9$              |



4. 只有 closed loop 中才能形成 steady current

emf : electromotive force 电动势

5. 霍尔效应.  $qE = qv_d B$

$$E = \rho J = \rho nq v_d = v_d B \quad B = \rho nq d$$

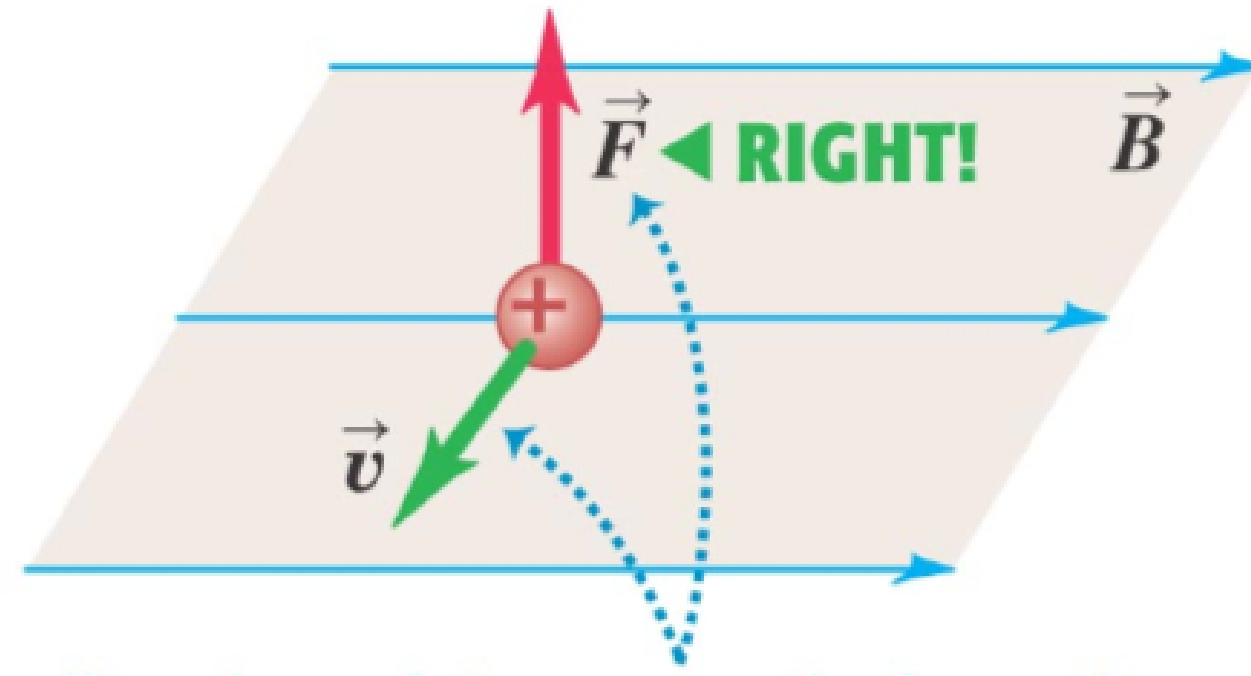
$$J = \frac{dI}{dA} = nqv$$

$$I = \frac{dq}{dt} = \frac{d(vtSnq)}{dt} = vAnq$$



# Magnetic Field and Magnetic Forces

- It is important to remember that magnetic field lines are *not* lines of magnetic force. (See Figure 27.12 below.)

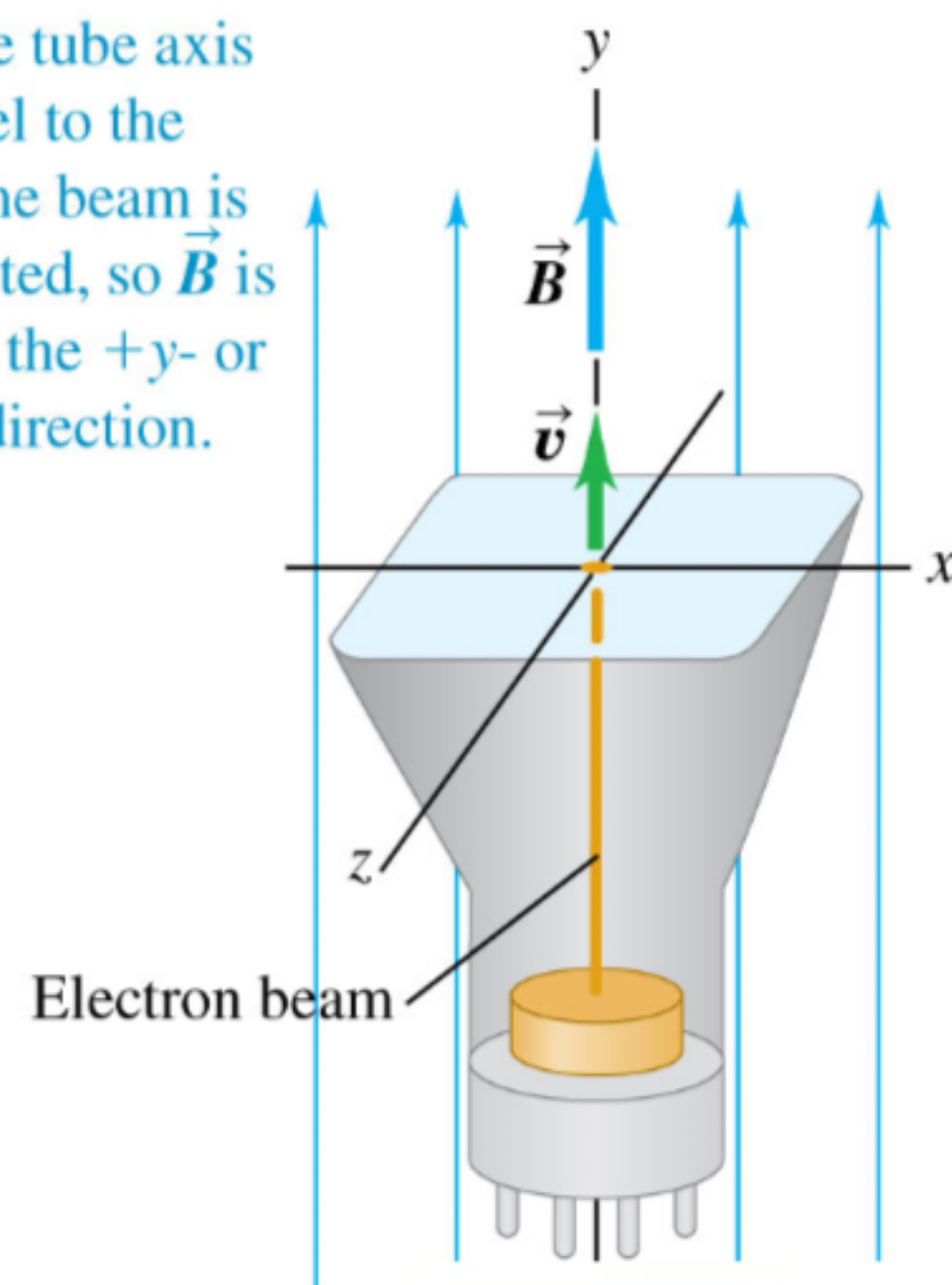


The direction of the magnetic force depends on the velocity  $\vec{v}$ , as expressed by the magnetic force law  $\vec{F} = q\vec{v} \times \vec{B}$ .

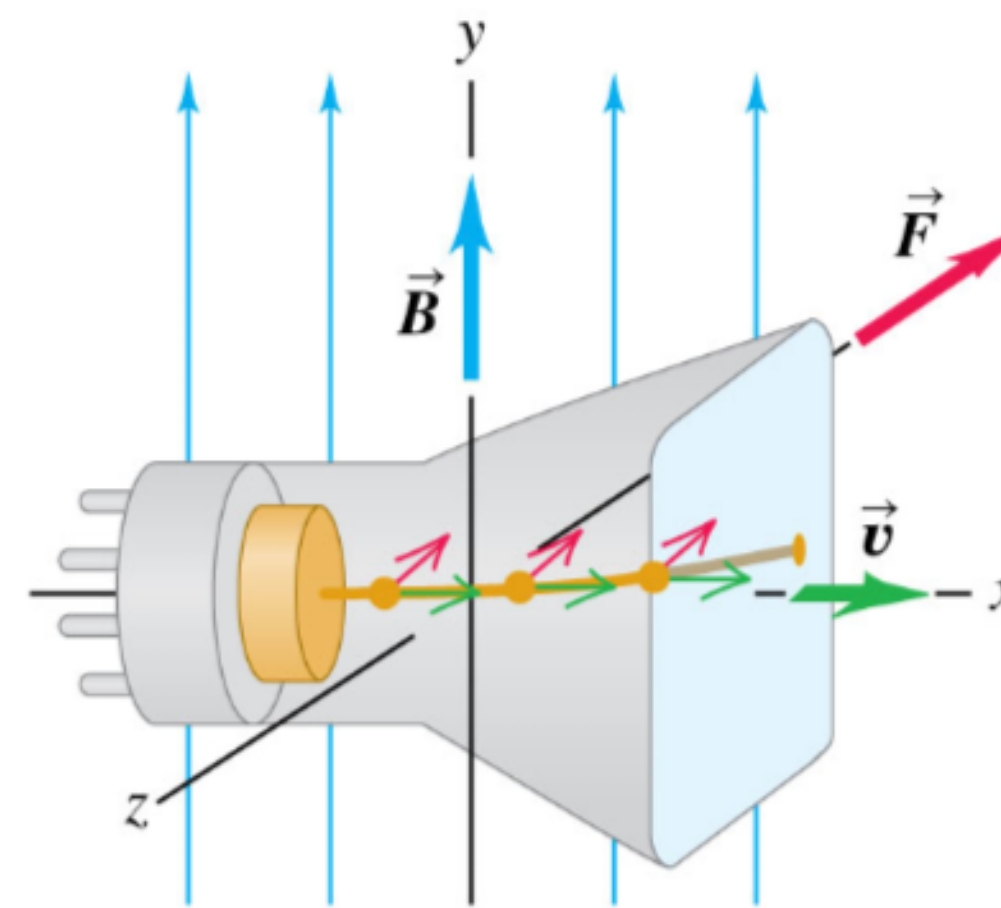
闭合曲面的磁通量为 0

$$\oint \vec{B} \cdot d\vec{A} = 0$$

(a) If the tube axis is parallel to the  $y$ -axis, the beam is undeflected, so  $\vec{B}$  is in either the  $+y$ - or the  $-y$ -direction.



(b) If the tube axis is parallel to the  $x$ -axis, the beam is deflected in the  $-z$ -direction, so  $\vec{B}$  is in the  $+y$ -direction.



$$F = qv \quad B = m \frac{v^2}{R}$$

$$\omega = \frac{v}{R} = \frac{qB}{m} \quad \text{角频率}$$

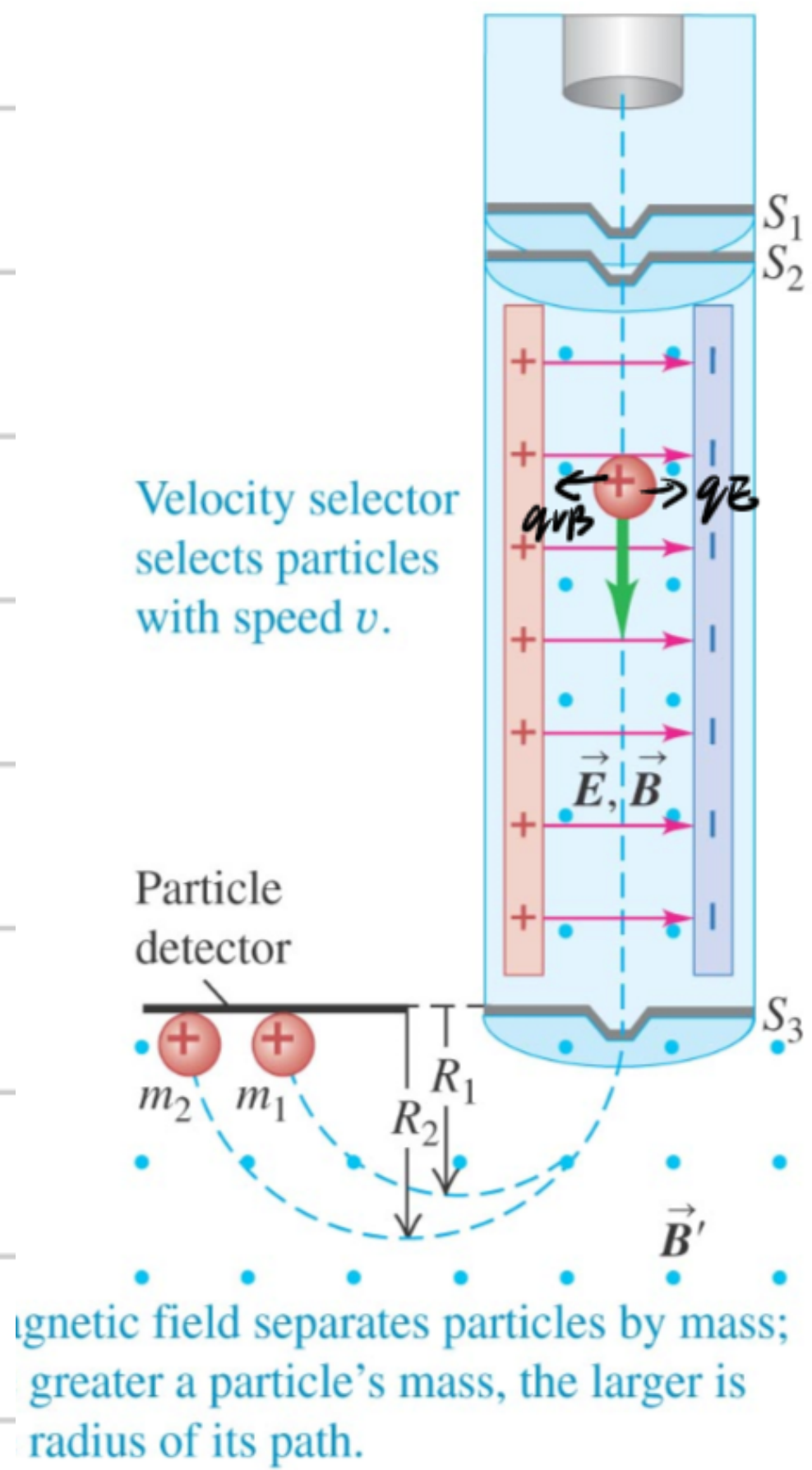
$$f = \frac{\omega}{2\pi}$$

revolution per unit time  
频率

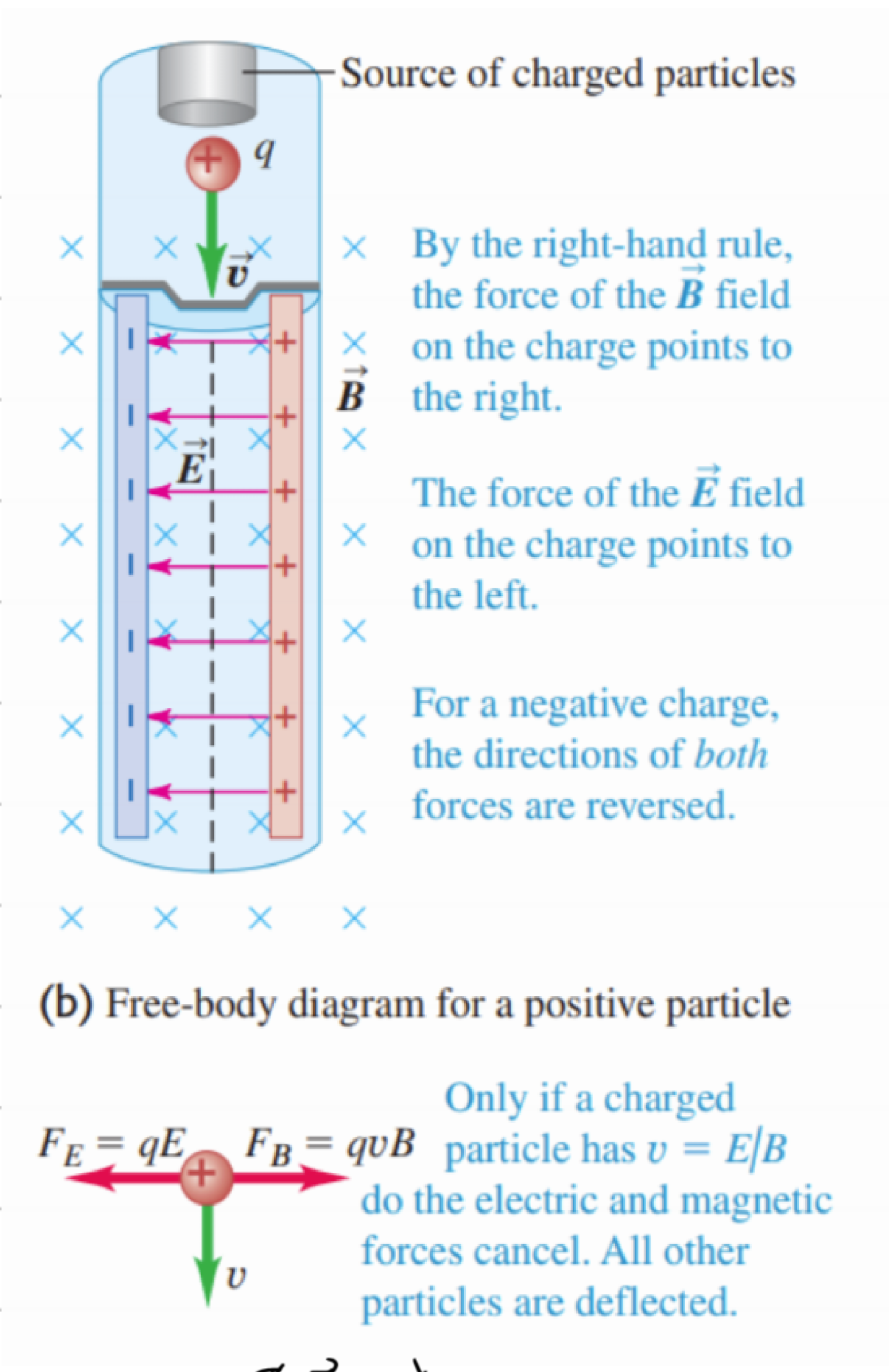
? cyclotron radiation



# 质量选择仪

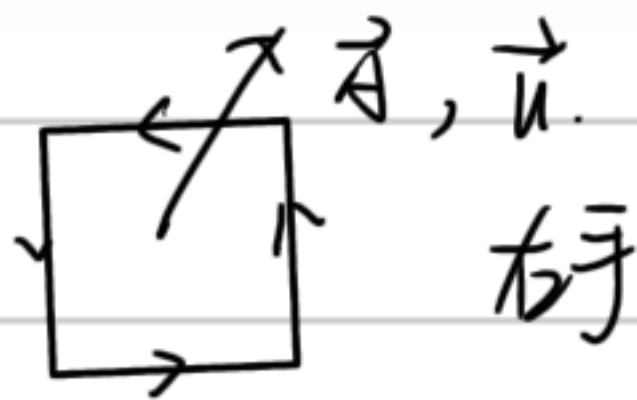


# 速度选择仪



磁偶极矩  $\vec{\mu} = I \cdot \vec{A}$   
magnetic dipole moment

对于  $N$  个 coil 而言  $\mu = NIA$



磁力矩  $\vec{\tau} = \vec{\mu} \times \vec{B}$



# Chap 28 磁场强度

通电导线  $B = \frac{\mu_0 I}{2\pi r}$

匀速点电荷  $\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \vec{r}}{r^2}$

小段导线  $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^2}$

长直导线  $B = \frac{\mu_0 I}{2\pi r}$

(参考点距离导线长度很大时可用)

两根长直导线的相互作用力  $F = I'LB = \frac{\mu_0 I I' L}{2\pi r}$

$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$

同向电流相互吸引, 异向电流相互排斥

环形导线对轴线上一点  $B_x = \frac{\mu_0 I a^2}{2(\pi^2 + a^2)^{\frac{3}{2}}}$  — radius of loop

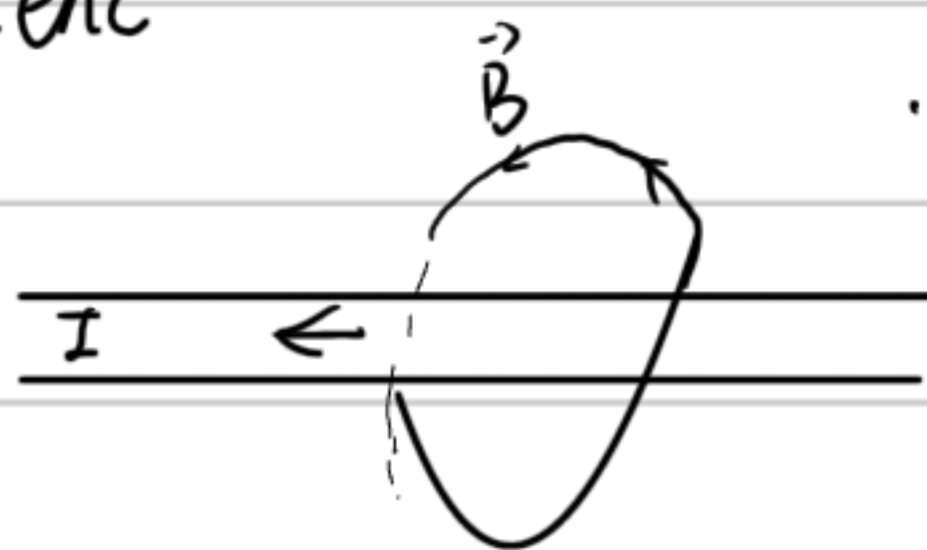
↓

多层环形导线

$B_x = \frac{\mu_0 I a^2}{2(\pi^2 + a^2)^{\frac{3}{2}}}$

$\pi = 0 \quad B_{x\max} = \frac{\mu_0 N I}{2a}$

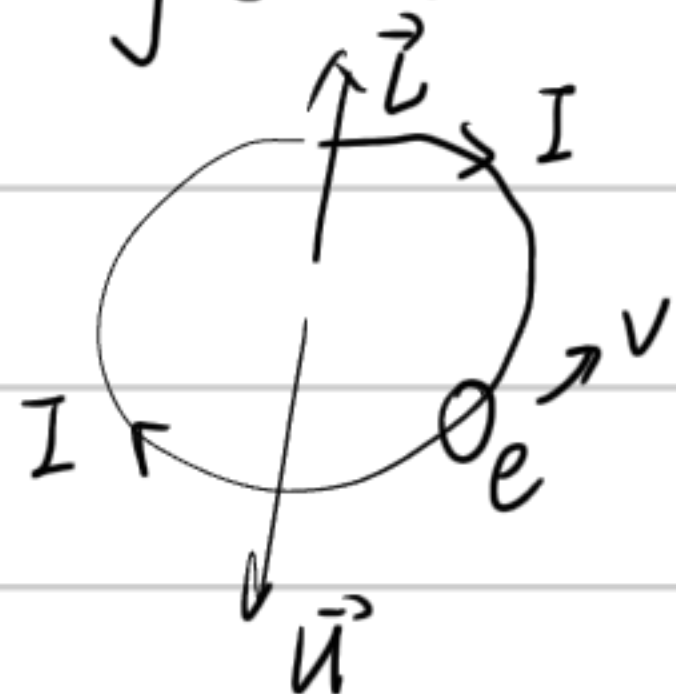
安培定律  $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$



理想螺线管

内部磁场均匀  
外部磁场为0

Bohr Magneton  $\mu = IA \quad A = \pi r^2 \quad T = \frac{2\pi r}{v} \quad I = \frac{e}{T} = \frac{ev}{2\pi r}$



$\mu = \frac{ev}{2\pi r} \pi r^2 = \frac{evr}{2}$

since  $L = mvr \quad \mu = \frac{e}{2m} L$

$\mu_B = \frac{eh}{2me}$

由于电子的角动量是量子化的, 特定方向上是  $\frac{h}{2\pi}$  整数倍

顺磁性 逆磁性



# Chap 29 Electromagnetic Induction

1. Induced Current

2. magnetic flux

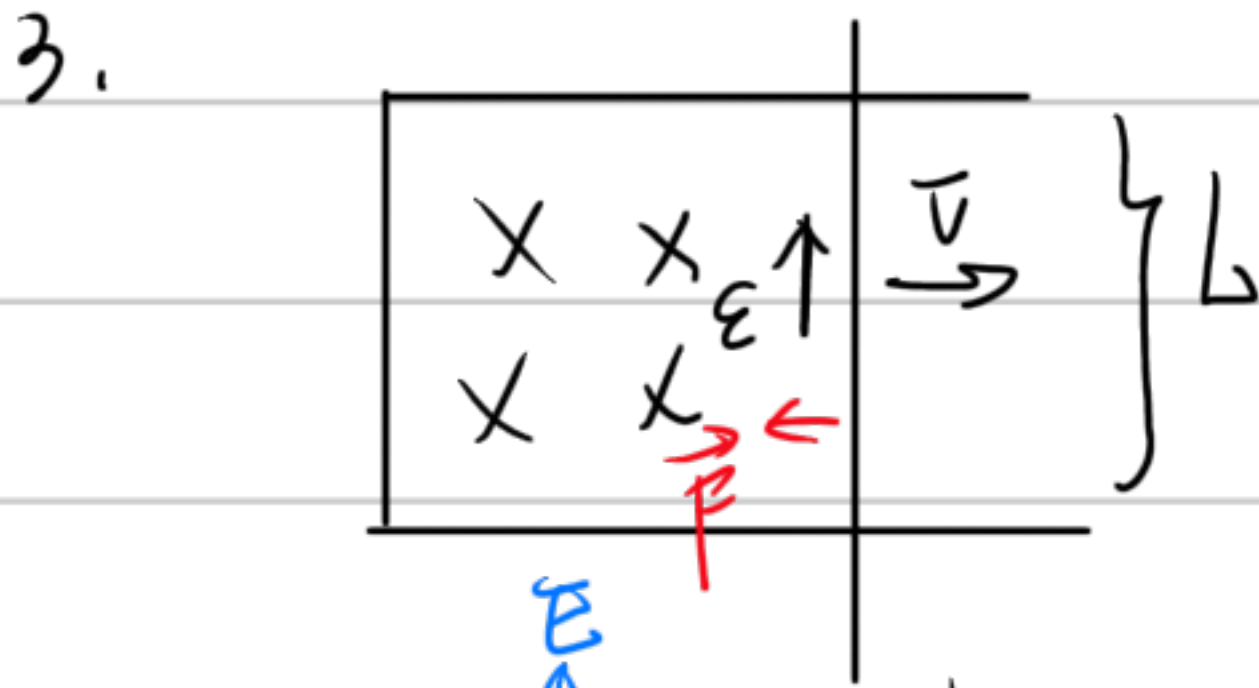
$$d\Phi_B = \vec{B} \cdot d\vec{A} = B_L dA = B dA \cos\theta$$

Induced emf  $\int \vec{E} \cdot d\vec{l} = \mathcal{E} = -\frac{d\Phi_B}{dt}$  (Faraday's law of induction)

磁通量变化引发的电动势

感应电流方向：右手螺旋定则，为了抑制原来的变化

感应电流正方向：右手四指方向，但是负值



$$\mathcal{E} = -\frac{d\Phi}{dt} = -B \frac{Lv}{dt} = -BLv$$

$$d\vec{\Phi} = I d\vec{l} \times \vec{B}$$

$$I = \frac{|\mathcal{E}|}{R} = \frac{BLv}{R}$$

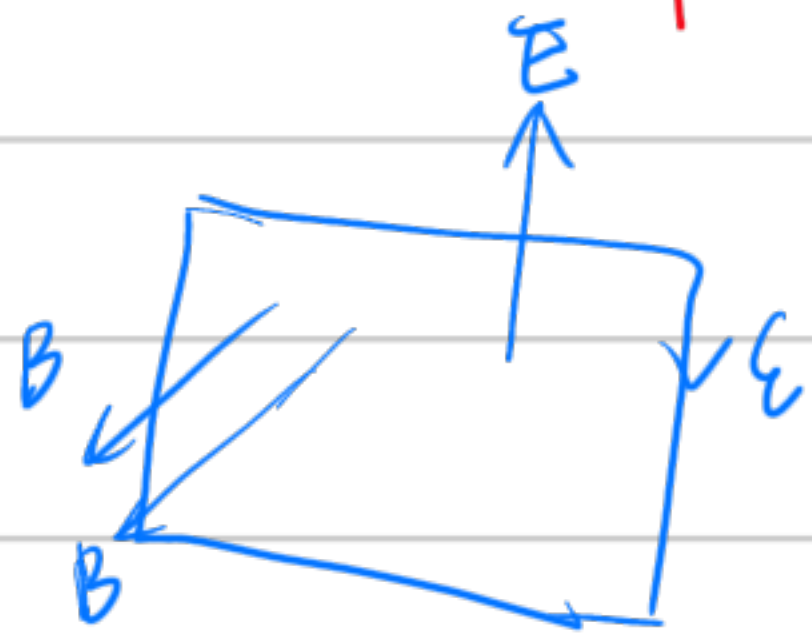
耗散功率：

$$P_{\text{dissipated}} = I^2 R = \left(\frac{BLv}{R}\right)^2 R = \frac{B^2 L^2 v^2}{R}$$

做功功率

$$F = ILB = \frac{BLv}{R} LB = \frac{B^2 L^2 v}{R}$$

$$P_{\text{applied}} = Fv = \frac{B^2 L^2 v^2}{R}$$



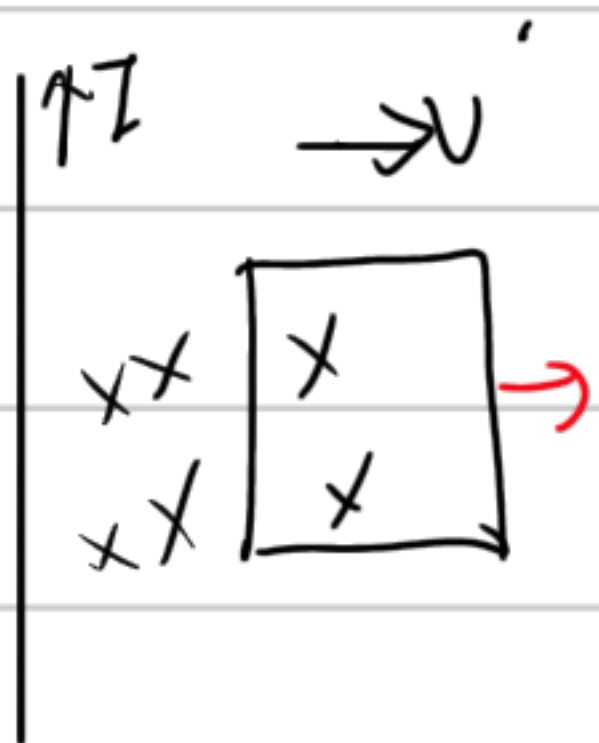
$\oint \vec{E} \cdot d\vec{l}$  积分方向是B的右手方向

加负号后的逆向积分结果才是B导致的

4. motional emf

$$d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

$$\mathcal{E} = BLv$$

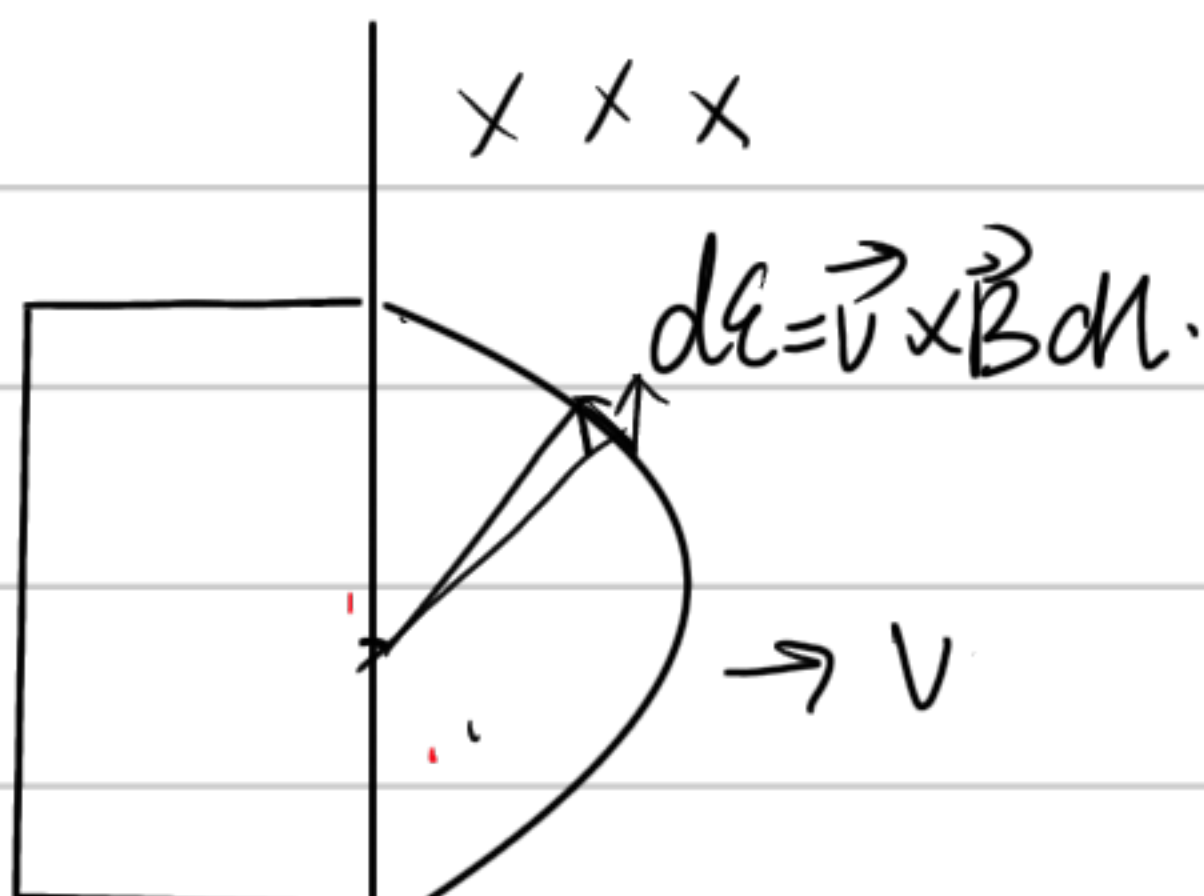


$$B = \frac{\mu_0 I}{2\pi r}$$

$$\mathcal{E}_{\text{tot}} = \mathcal{E}_1 - \mathcal{E}_2$$

$$= B_1 L v - B_2 L v$$

$$= \left(\frac{\mu_0 I}{2\pi r_1} - \frac{\mu_0 I}{2\pi r_2}\right) L v$$



$$B v dl \cos\theta$$

5. 常识题：安培：eddy current

generate eddy current  $\rightarrow$  eddy current cause  $B'$   
 $\rightarrow$  detect  $B'$

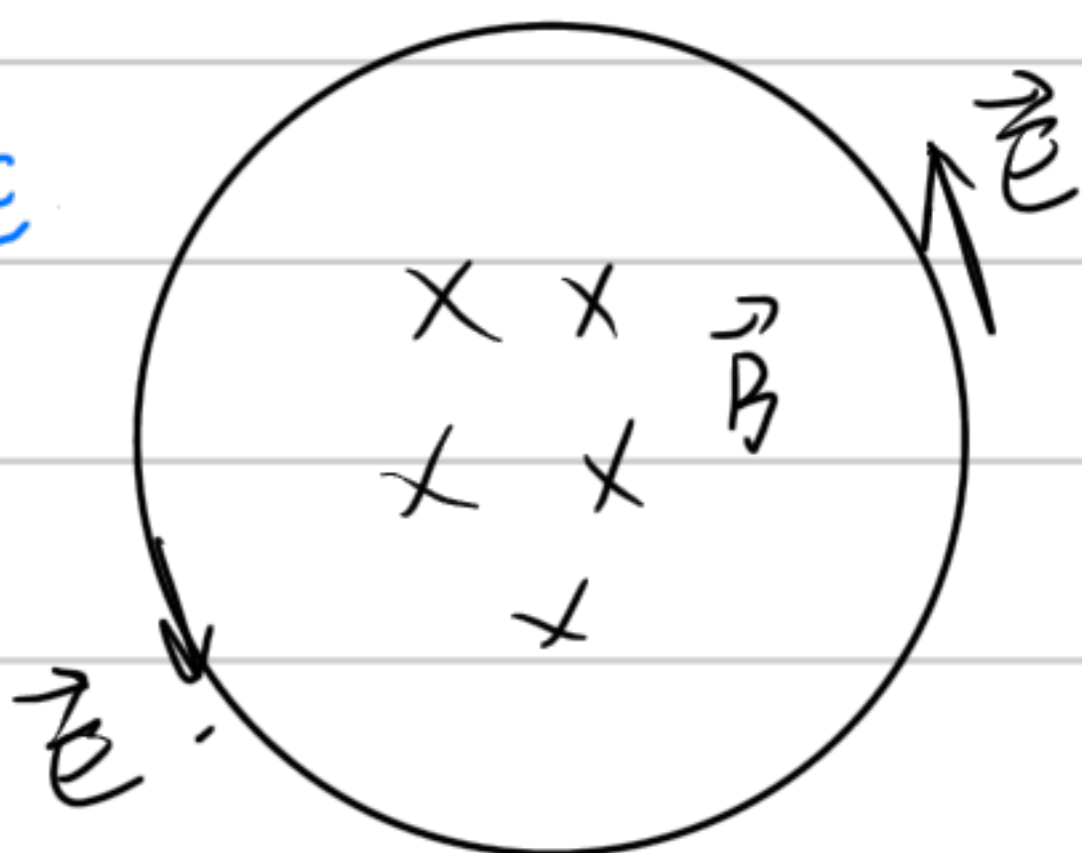


perfect diamagnetism: eddy current run forever  
产生的磁场可以完全抵消原磁场  
 $\vec{B} = \vec{B}_0 + \mu_0 \vec{M}$

Meissner effect:

b. induced electric fields:

$$\frac{d\vec{B}}{dt} = \odot E$$



$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt} = \mathcal{E}$$

其实就是法拉第电磁感应定律.

eg.  $\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$   
↓ 积分方向也要考虑楞次定律  
 $2\pi r E \Rightarrow E = \frac{1}{2\pi r} \left| \frac{d\Phi_B}{dt} \right|$

7. Maxwell's equations.

✓ Gauss's law for the electric field  $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$  (Gauss's law for  $\vec{E}$ )

✓ Gauss's law for the magnetic field  $\oint \vec{B} \cdot d\vec{A} = 0$  (Gauss's law for  $\vec{B}$ )

✓ Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}}$$
 (Ampere's law)

假设1-中只空气

✓ Faraday's law.

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$
 (Faraday's law)

安培定律修正版:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}}$$

$$\vec{i}_p = \epsilon \frac{d\Phi_E}{dt}$$

displacement current

generalized Ampere's law:  $\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_c + i_p)_{\text{encl}}$

(i)  $\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} \rho$  (Gauss's law),

(ii)  $\nabla \cdot \vec{B} = 0$  (no name),

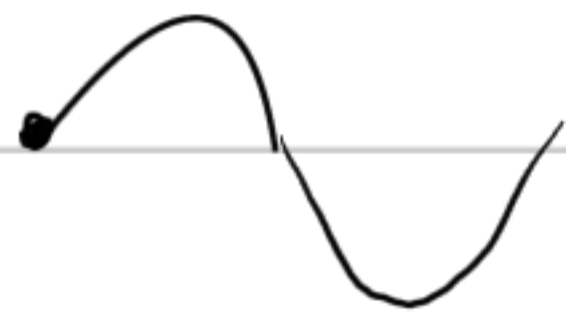
(iii)  $\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$  (Faraday's law),

(iv)  $\nabla \times \vec{B} = \mu_0 \vec{J}$  (Ampère's law).



RC

Q0

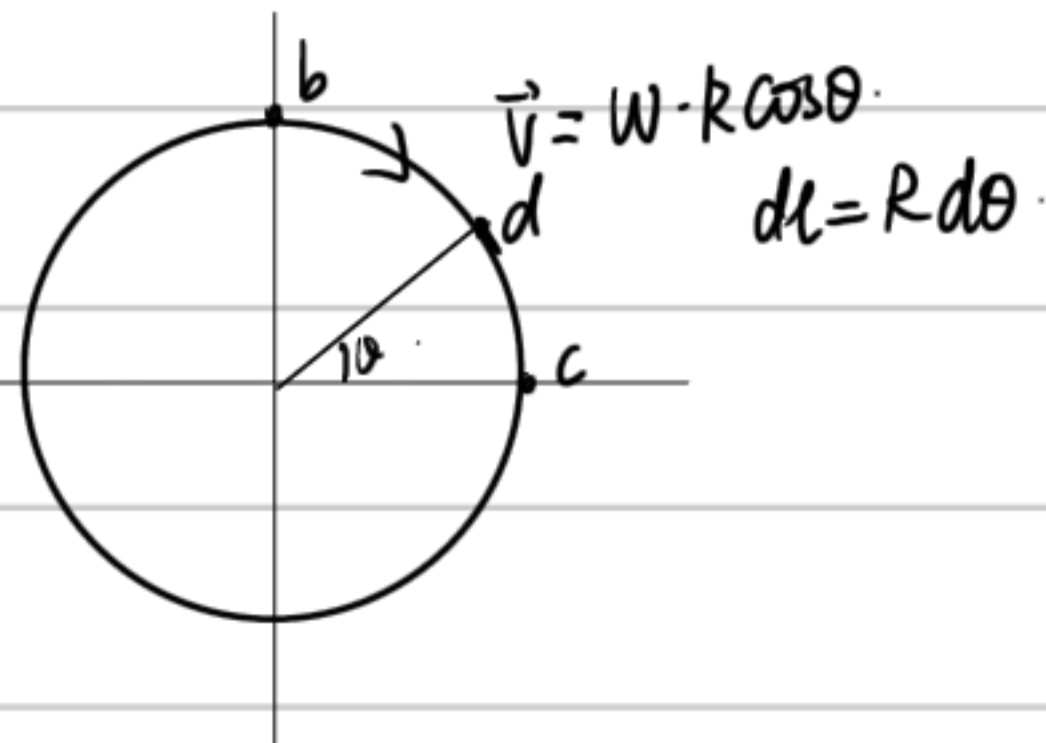


$$B = B_0 \sin \omega t$$

$$\Phi = B \cdot S \cdot \cos \theta$$

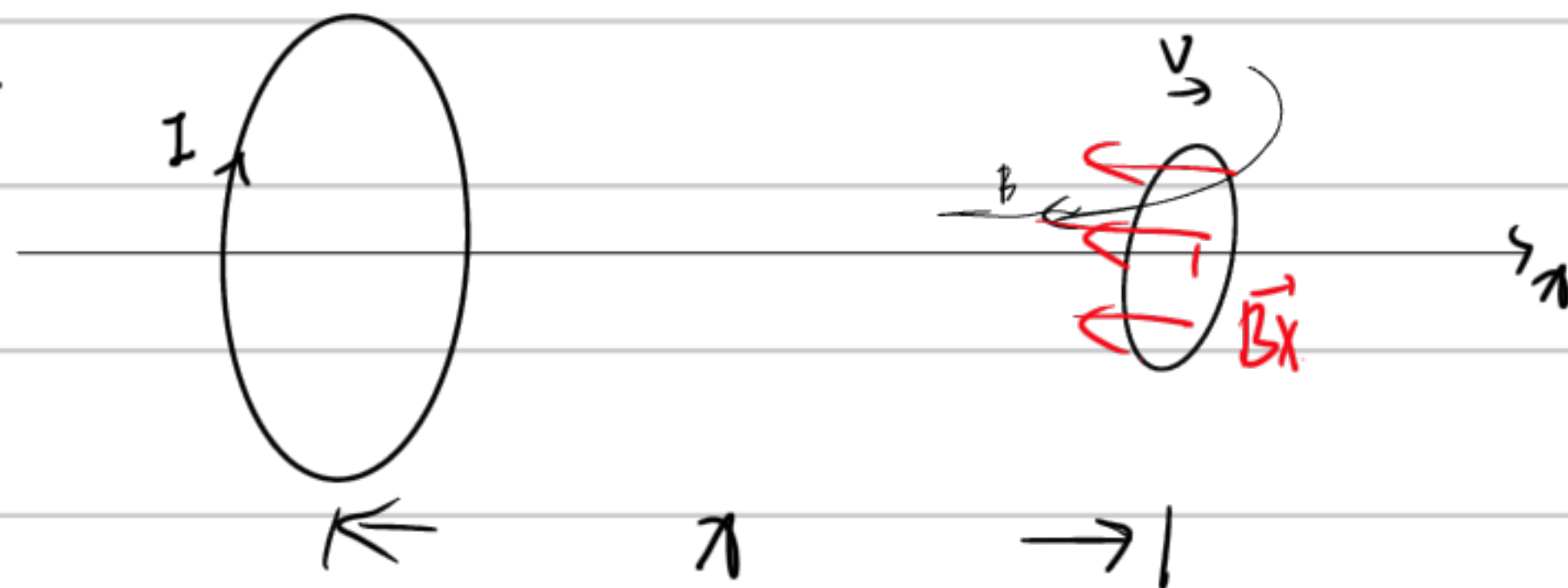
$$\frac{d\Phi}{dt} = B S \cos \theta$$

$$d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$$



$$\begin{aligned} d\mathcal{E} &= \omega R \cos \theta \cdot B \cdot R d\theta \\ \mathcal{E} &= \omega R^2 B \int \cos \theta d\theta \\ &= \omega R^2 B \sin \theta \Big|_{\theta}^0 \\ &= -\omega R^2 B \sin \theta \end{aligned}$$

Q.1.



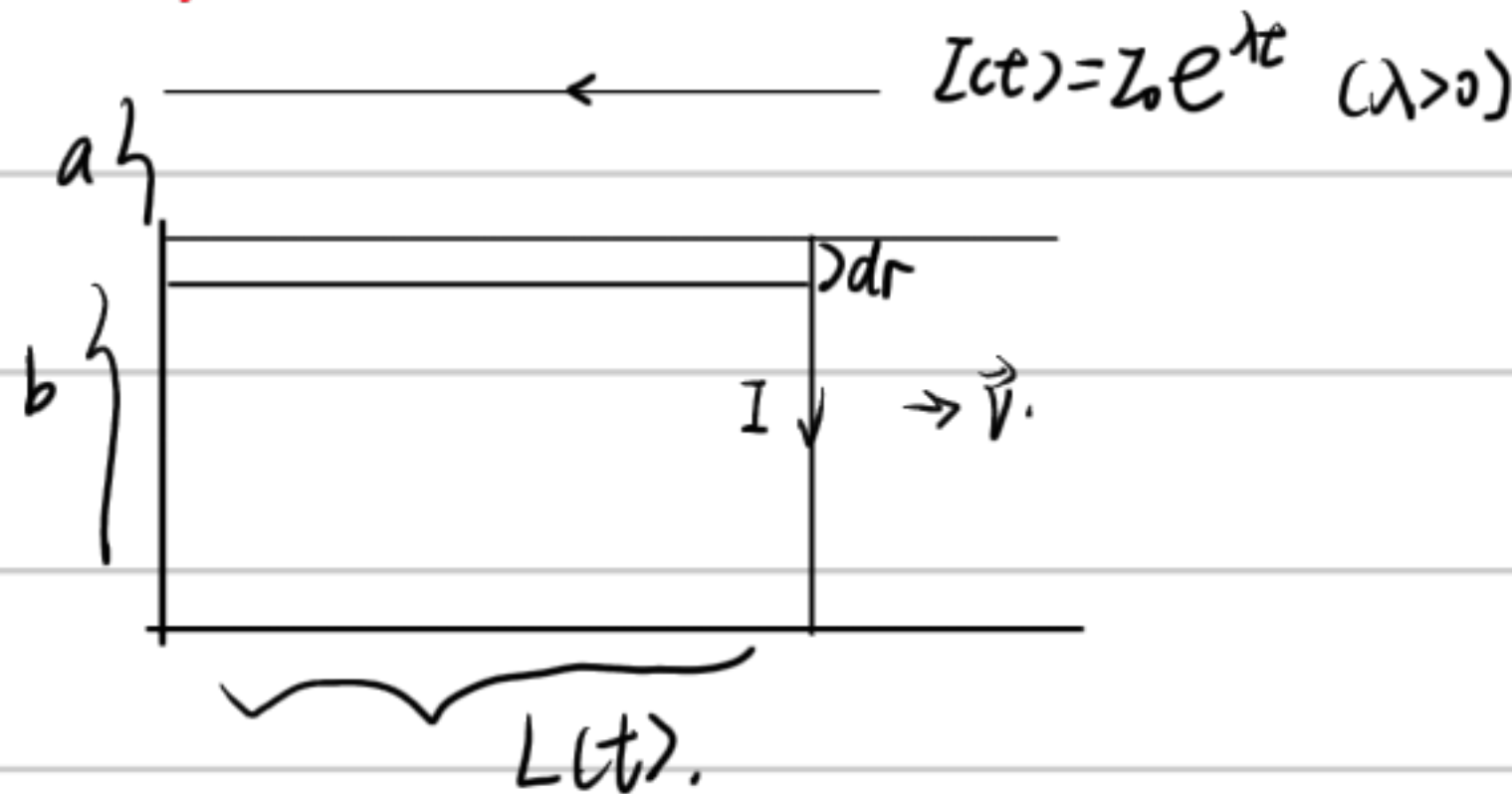
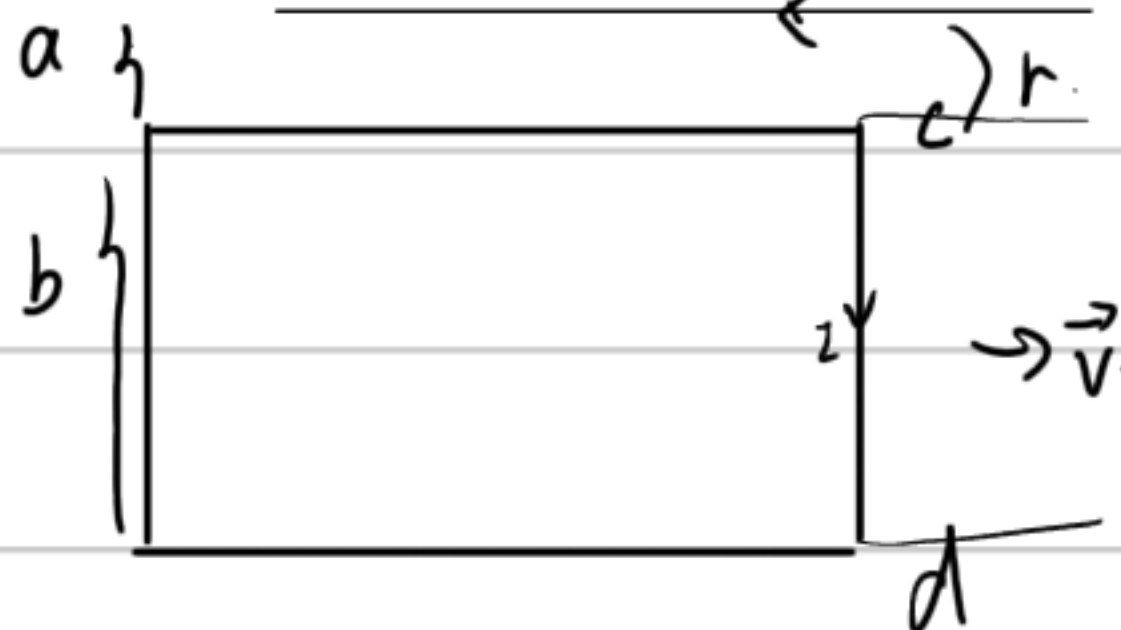
Since  $x \gg R$  consider B as uniform

$$B_x = \frac{\mu_0 I b^2}{2(x^2 + b^2)^{3/2}}$$

$$\Phi_x = B_x \cdot S = B_x \cdot \pi a^2$$

$$\frac{\Delta \Phi}{\Delta t} = S \frac{\partial B}{\partial t} = S \frac{\partial B}{\partial x} \frac{\partial x}{\partial t} = -S \frac{\mu_0 I b^2}{2} \cdot \frac{3}{2} (x^2 + b^2)^{-5/2} v$$

Q2



~~$$\begin{aligned} B &= \frac{\mu_0 I}{2\pi r} \\ d\mathcal{E} &= B dr \cdot v = v \frac{\mu_0 I}{2\pi r} dr \\ \mathcal{E} &= \frac{v \mu_0 I}{2\pi} \ln r \Big|_a^{a+b} \\ &= \frac{v \mu_0 I}{2\pi} \ln \frac{a+b}{a} \end{aligned}$$~~

变化的磁场中不能用公式  $\mathcal{E} = BLv$

$$d\Phi = \frac{\mu_0 I}{2\pi r} L dr$$

$$\Phi = \frac{\mu_0 I L}{2\pi} \ln r \Big|_a^{a+b}$$

$$= \frac{\mu_0 I L}{2\pi} \ln \frac{a+b}{a}$$

$$= \frac{\mu_0}{2\pi} \ln \frac{a+b}{a} I(t) L(t) v$$

$$\begin{aligned} \frac{d\Phi}{dt} &= \frac{\mu_0}{2\pi} \ln \frac{a+b}{a} (I' v + I v') \\ &= \frac{\mu_0}{2\pi} \ln \frac{a+b}{a} (-\lambda I e^{\lambda t} v + I_0 e^{\lambda t} v) \end{aligned}$$



# Chap 32 Electromagnetic Waves

$\epsilon$  电容率 permittivity

$$\epsilon = \kappa \epsilon_0 \rightarrow \text{真空介电常数}$$

↓  
相对介电常数  
dielectric constant

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$\mu$  磁导率 permeability

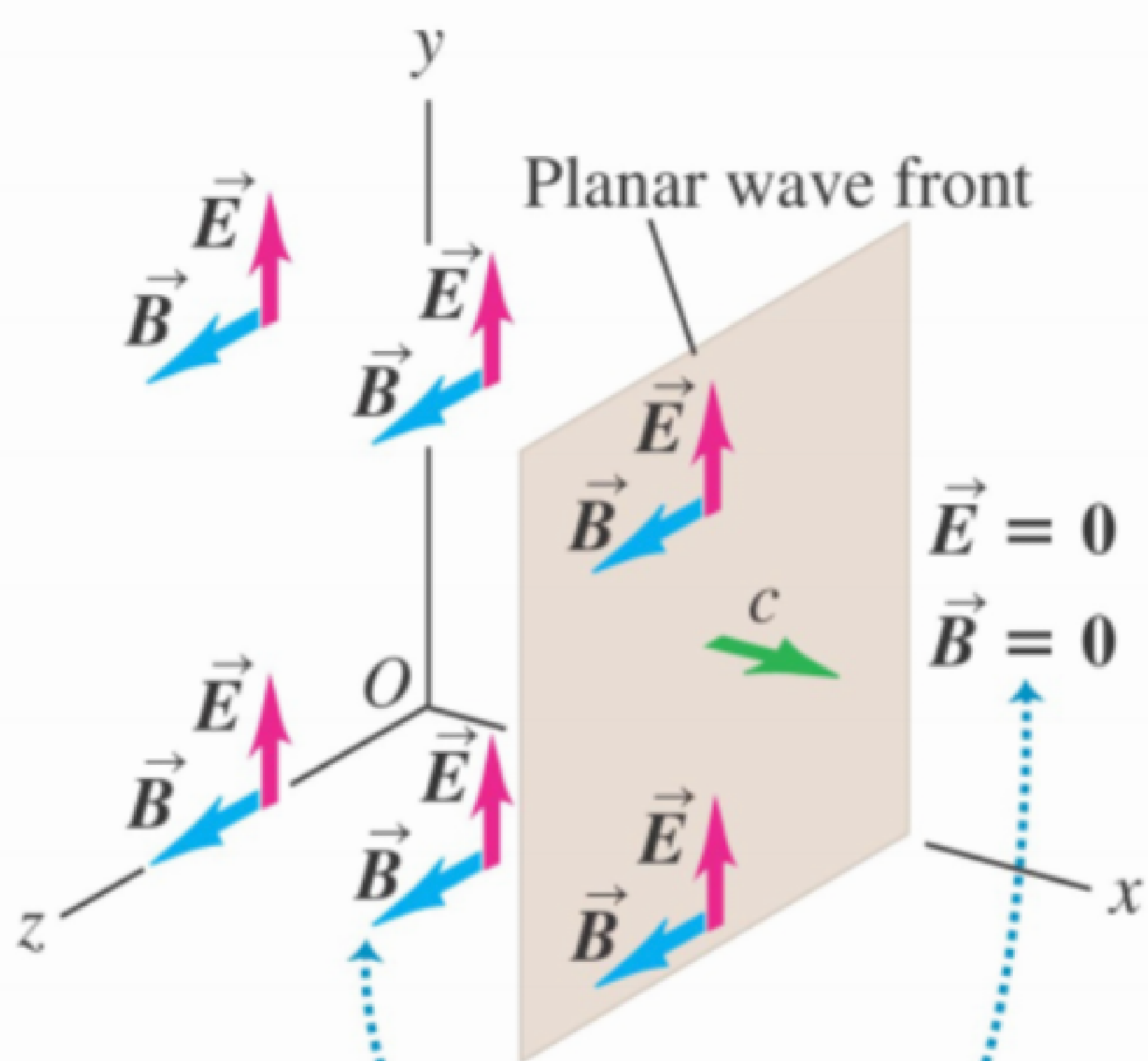
$$\mu = \mu_r \mu_0 \text{ 或 } \kappa_m \mu_0 \rightarrow \text{真空磁导率}$$

↓  
相对磁导率  
relative permeability

direction of propagation  
 $\vec{E} \times \vec{B}$

$$v = \frac{1}{\sqrt{\epsilon \mu}} = \frac{1}{\kappa \kappa_m} \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{c}{\kappa \kappa_m}$$

$$\frac{c}{v} = n = \sqrt{\kappa \kappa_m} \approx \sqrt{\kappa}$$



The electric and magnetic fields are uniform behind the advancing wave front and zero in front of it.

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{Gauss's law for } \vec{E})$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Gauss's law for } \vec{B})$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (\text{Ampere's law})$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (\text{Faraday's law})$$

★ free-space only.

$$E = cB \quad (\text{electromagnetic wave in vacuum})$$

$$B = \epsilon_0 \mu_0 c E \quad (\text{electromagnetic wave in vacuum})$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (\text{speed of electromagnetic waves in vacuum})$$

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2}$$

(electromagnetic wave equation in vacuum)

$$\frac{\partial E_y(x, t)}{\partial x} = - \frac{\partial B_z(x, t)}{\partial t}$$



$$\left. \begin{array}{ll} \text{(i)} & \nabla \cdot \mathbf{E} = 0, \quad \text{(iii)} & \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} & \nabla \cdot \mathbf{B} = 0, \quad \text{(iv)} & \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{array} \right\}$$

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = \nabla \times \left( -\frac{\partial \mathbf{B}}{\partial t} \right)$$

$$= -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2},$$

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = \nabla \times \left( \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$= \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.$$

$$\Rightarrow \nabla \cdot \mathbf{E} = 0 \text{ and } \nabla \cdot \mathbf{B} = 0,$$

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.}$$

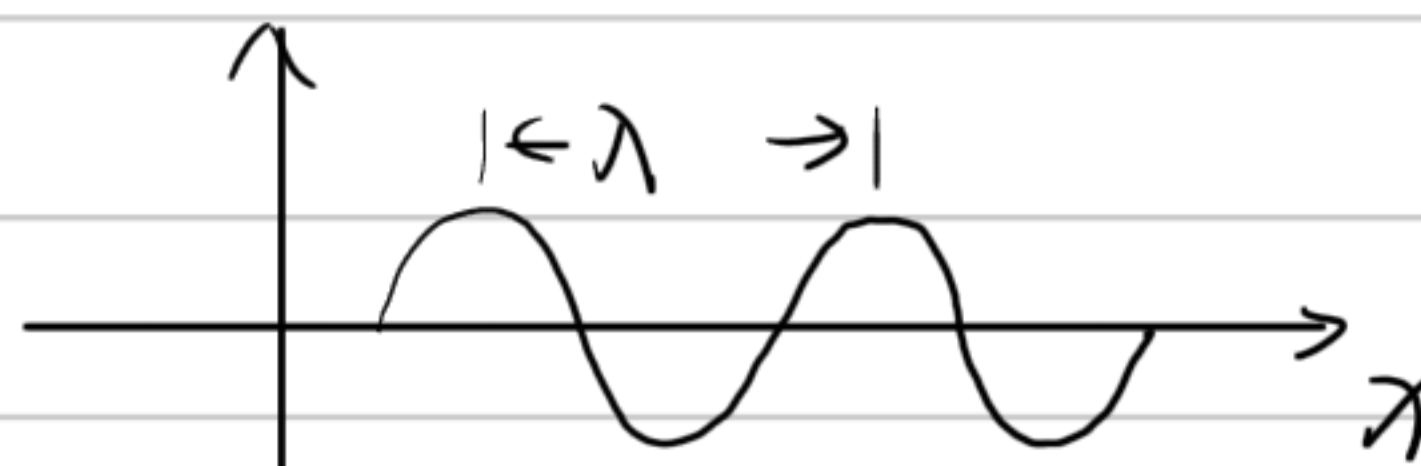
$$\vec{E}(x, t) = \hat{j} E_{\max} \cos(kx - \omega t)$$

$$\vec{B}(x, t) = \hat{k} B_{\max} \cos(kx - \omega t)$$



$$E_{\max} = c B_{\max}$$

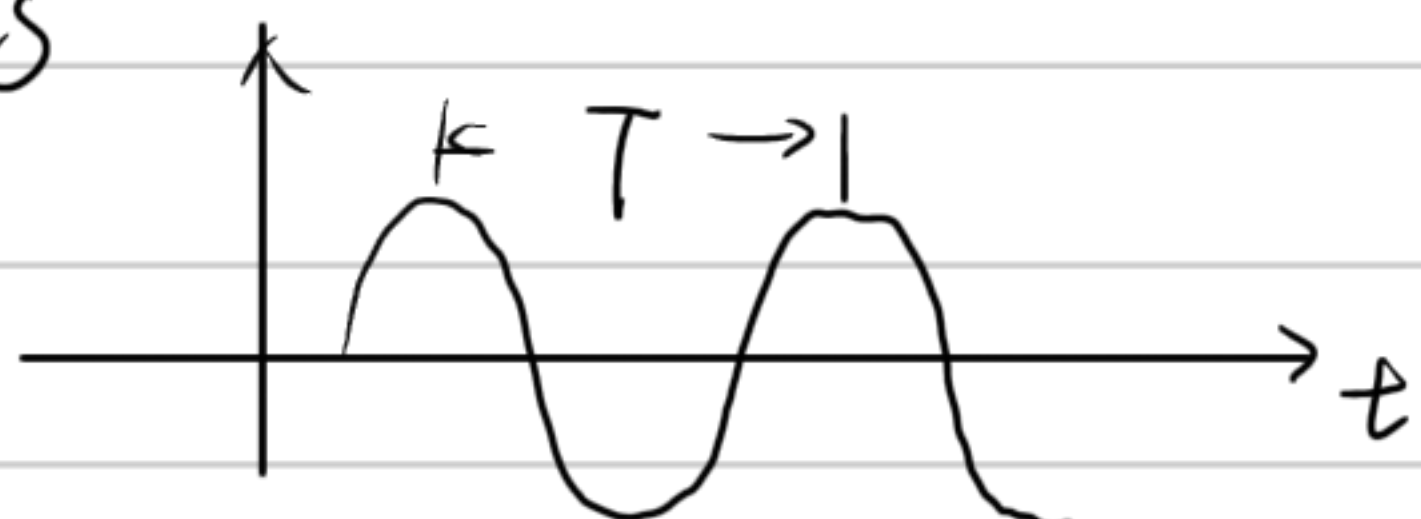
spatial quantities



$$k = \frac{2\pi}{\lambda}$$

$$\cos(kx)$$

temporal quantities



$$\cos(\omega t)$$

$$E(x, t) = A \cos(\varphi)$$

$$\varphi = kx - \omega t - \theta$$

$$\omega = -\frac{\partial \varphi}{\partial t}$$

$$k = \frac{\partial \varphi}{\partial x}$$

$$v = \frac{-\partial \varphi / \partial t}{\partial \varphi / \partial x} = \frac{\omega}{k}$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = \frac{2\pi}{T}$$

$$f = \frac{1}{T}$$



Ampere's law the displacement current

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\epsilon \frac{d\Phi_E}{dt} = K\epsilon_0 \frac{d\Phi_E}{dt},$$

$$\mu = K_m \mu_0$$

$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_1} \quad (\text{Gauss's law in a dielectric})$$

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2} \quad (\text{electromagnetic wave equation in vacuum})$$

$$\frac{1}{v^2} = \epsilon_0 \mu_0 \quad \text{or} \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$v = \frac{1}{\sqrt{\epsilon \mu}} = \frac{1}{\sqrt{KK_m}} \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{c}{\sqrt{KK_m}} \quad (\text{speed of electromagnetic waves in a dielectric})$$

$$K_m \cong 1, \quad \frac{c}{v} = n = \sqrt{KK_m} \cong \sqrt{K}$$

$$\vec{v} = \frac{1}{\lambda} \vec{A} \quad h\nu$$

negative index

Energy in electromagnetic waves

for conductor  $+-$

$$u = \frac{1}{2} \epsilon_0 E^2 \quad \text{能量密度}$$

for inductor  $a \text{---} b$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$u = \frac{B^2}{2\mu_0}$$

for electromagnetic waves:  $u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \quad B = \frac{E}{c} = \sqrt{\epsilon_0 \mu_0} E$   
 $\downarrow$   
 $= \epsilon_0 E^2$

自感系数  $L = \frac{N\Phi_B}{i} = \frac{NBA}{i} = \frac{N \mu_0 N i A}{i \cdot l} = \frac{N^2 \mu_0 A}{l}$   
 $\downarrow$   
 定义式



# Poynting Vector:

先定义 - 1 S : energy per unit time per unit area

$$S = \frac{1}{A} \frac{dU}{dt} = \epsilon_0 c E^2 = \frac{EB}{\mu_0}$$

$$dU = u dV = (\epsilon_0 E^2) (A c dt)$$

## 定义 Poynting Vector:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$P = \oint \vec{S} \cdot d\vec{A}$$

pointing to the direction of propagation

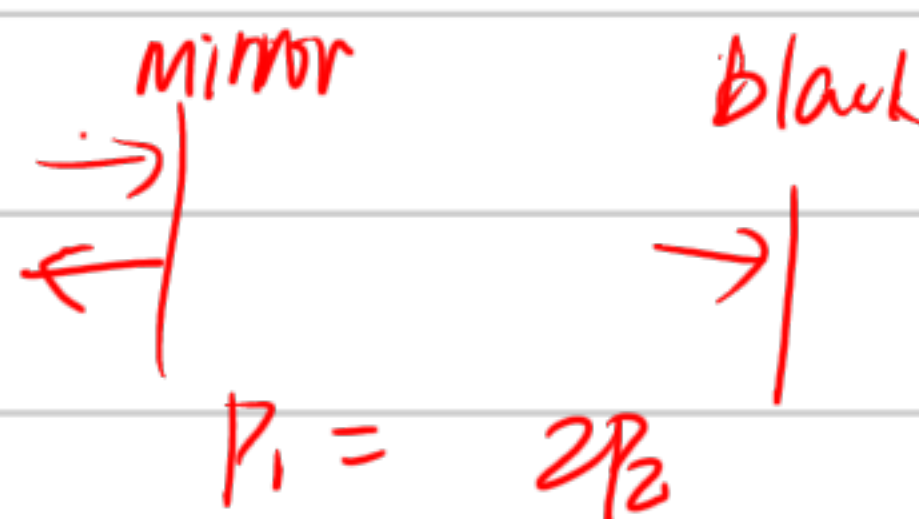
$$I = S_{av} = \frac{E_{max} B_{max}}{2\mu_0} = \frac{E_{max}^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_{max}^2 = \frac{1}{2} \epsilon_0 c E_{max}^2$$

electromagnetic momentum

$$\frac{1}{A} \frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c}$$

pressure

change 对于不同材料上是不一样的



## List of Parameter Terminologies

- Space:
  - $k$ : Wave number
  - $\lambda$ : Wavelength
- Time:
  - $\omega$ : Angular frequency
  - $T$ : Period
  - $f$ : Frequency
- Other:
  - $3 \times 10^8$  m/s
  - $c$  or  $v$ : Speed of EM wave propagation
  - $E_{max}$  and  $B_{max}$ : Amplitude (i.e. maximum value) of fields

$$\vec{E}(x, t) = \hat{j} E_{max} \cos(kx - \omega t)$$

$$\vec{B}(x, t) = \hat{k} B_{max} \cos(kx - \omega t)$$

## Consider velocity of wave propagation

- $c = \frac{E}{B}$
- $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
- $B_{max} = \epsilon_0 \mu_0 c E_{max}$
- $v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{\sqrt{K K_m}}$  (When not in vacuum)

$$f(\mu_0, \epsilon_0) = c$$

$$\mu = \mu_r \mu_0 \longleftrightarrow \mu = K_m \mu_0$$

$$\epsilon = \epsilon_r \epsilon_0 \longleftrightarrow \epsilon = K \epsilon_0$$

## Parameter Relationships

- $k = \frac{2\pi}{\lambda}$
- $\omega = \frac{2\pi}{T}, f = \frac{1}{T}$

antinodal plane, nodal plane

for absorb materials:  $P_{rad} = \frac{S_{av}}{c} = \frac{I}{c}$

for reflect materials:  $P_{rad} = \frac{2S_{av}}{c} = \frac{2I}{c}$

驻波条件: 已在反射导体面处为0; 距离半波长整数倍.

11.14 后面的课没听



$$\begin{aligned}
 & E = E_m \cos(kx + \omega t + \varphi) \\
 & \left. \begin{aligned}
 & \text{某点 } E = E_m \cos(\omega t + \varphi) \\
 & \text{某时刻 } E = E_m \cos(kx + \varphi)
 \end{aligned} \right\}
 \end{aligned}$$

$\downarrow$   
 $\begin{array}{ccc} \bullet x_1 & \xrightarrow[L \quad \Delta t = \frac{L}{c}]{} & \bullet x_2 \end{array}$ 
 $t \rightarrow t + \Delta t$ 
 $\textcircled{?}$

$E_{x_1} = E_m \cos(\omega t + \varphi)$ 
 $\textcircled{?}$  What is the magnitude<sup>as</sup> function of  $t$ .

$$I = \bar{S} = \frac{P_w}{A}$$

$$P_{\text{right}} = \frac{P_{\text{power}}}{A} = A \frac{S}{c}$$

$$P_{\text{right, pre}} = \frac{S}{2c}$$



# Chap 34 Geometric Optics

1.  $\frac{\sin \theta_a}{\sin \theta_b} = \frac{n_b}{n_a}$  折射定律 Snell's law

$$n = \frac{c}{v} \quad \lambda = \frac{\lambda_0}{n}$$

2. ~~Example~~ plane of incidence : 由入射光和界面法线<sup>线</sup>确定的平面  
 interface 发生折射、反射的平面

"S" 偏振:  $E$  垂直于 plane of incidence  
 (perpendicular)

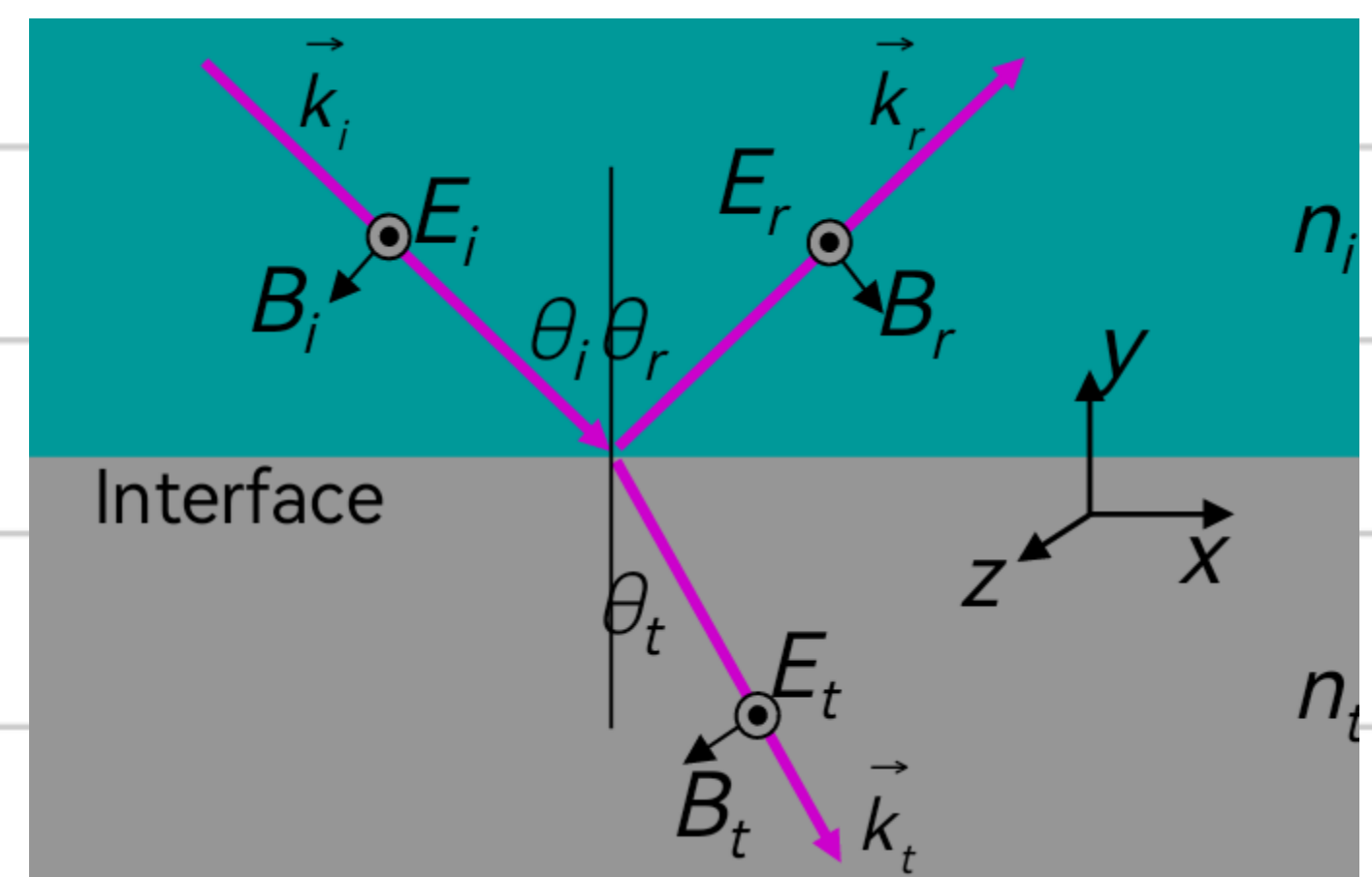
"P" 偏振: "平行于" ...  
 (parallel)

3. Brewster's Angle: P 偏振光完全不反射

4. Fresnel Equation:

反射系数  $r_{\perp} = \frac{E_{or}}{E_{oi}}$   $r_{\parallel} = \frac{E_{or}}{E_{oi}}$

透射系数  $t_{\perp} = \frac{E_{ot}}{E_{oi}}$   $t_{\parallel} = \frac{E_{ot}}{E_{oi}}$



$$r^2 + t^2 = 1 \Rightarrow \frac{E_r^2}{E_i^2} + \frac{E_t^2}{E_i^2} = 1$$

$$\Rightarrow E_r^2 + E_t^2 = E_i^2$$

$$\propto I_r + I_t = I_i$$

( $E$  都是 field complex amplitudes)

垂直方向上  $y=0$  处  $E_i + E_r = E_t \rightarrow$  有点反直觉的

$B$  在  $x, y$  平面,  $x$  方向上连续  $-B_{oi} \cos \theta_i + B_{or} \cos \theta_r = -B_{ot} \cos \theta_t$

$$E = cB = \frac{\omega}{n} B \quad \theta_r = \theta_i$$

$$\downarrow B = \frac{nE}{c_0}$$

最终整理得到

$$r_{\perp} = E_{or} / E_{oi} = [n_i \cos(\theta_i) - n_t \cos(\theta_t)] / [n_i \cos(\theta_i) + n_t \cos(\theta_t)]$$

$$t_{\perp} = E_{ot} / E_{oi} = 2n_i \cos(\theta_i) / [n_i \cos(\theta_i) + n_t \cos(\theta_t)]$$

$$r_{\parallel} = E_{or} / E_{oi} = [n_i \cos(\theta_t) - n_t \cos(\theta_i)] / [n_i \cos(\theta_t) + n_t \cos(\theta_i)]$$

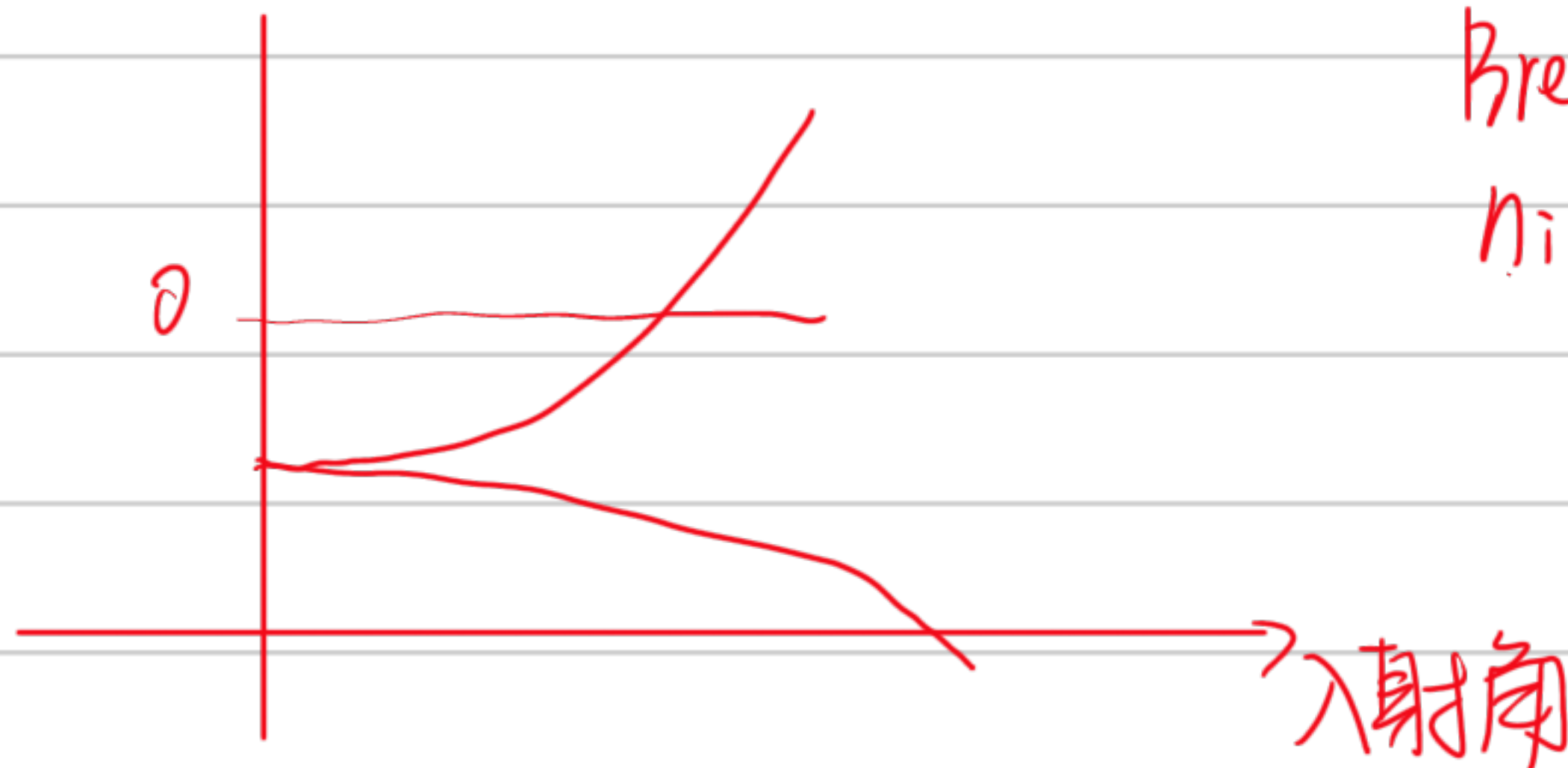
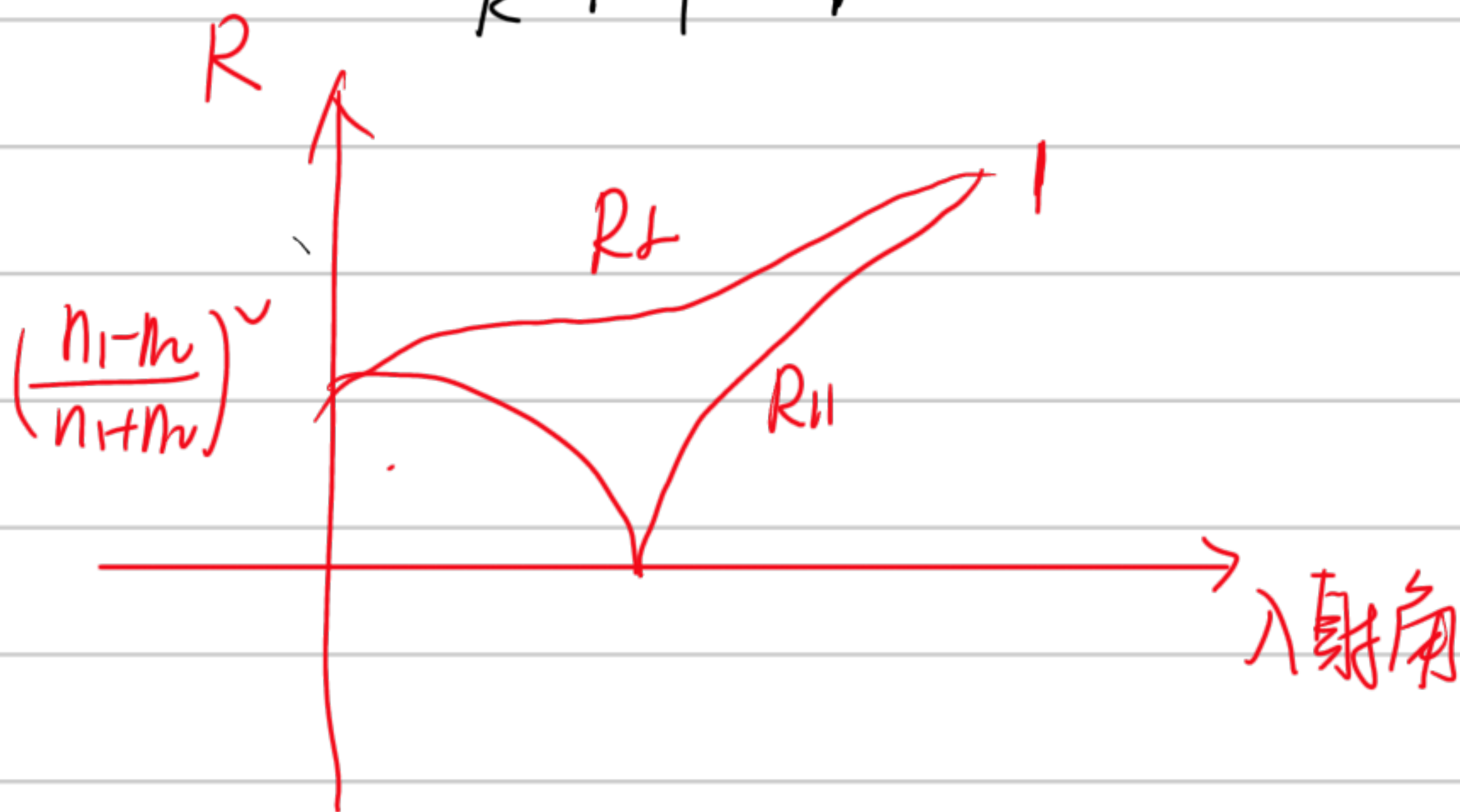
$$t_{\parallel} = E_{ot} / E_{oi} = 2n_i \cos(\theta_i) / [n_i \cos(\theta_t) + n_t \cos(\theta_i)]$$



折射率  $R$ : 光强之比, 正比于振幅平方  $R = |r|^2$

反射率  $T$   $T = |t|^2$

$$R + T = 1$$



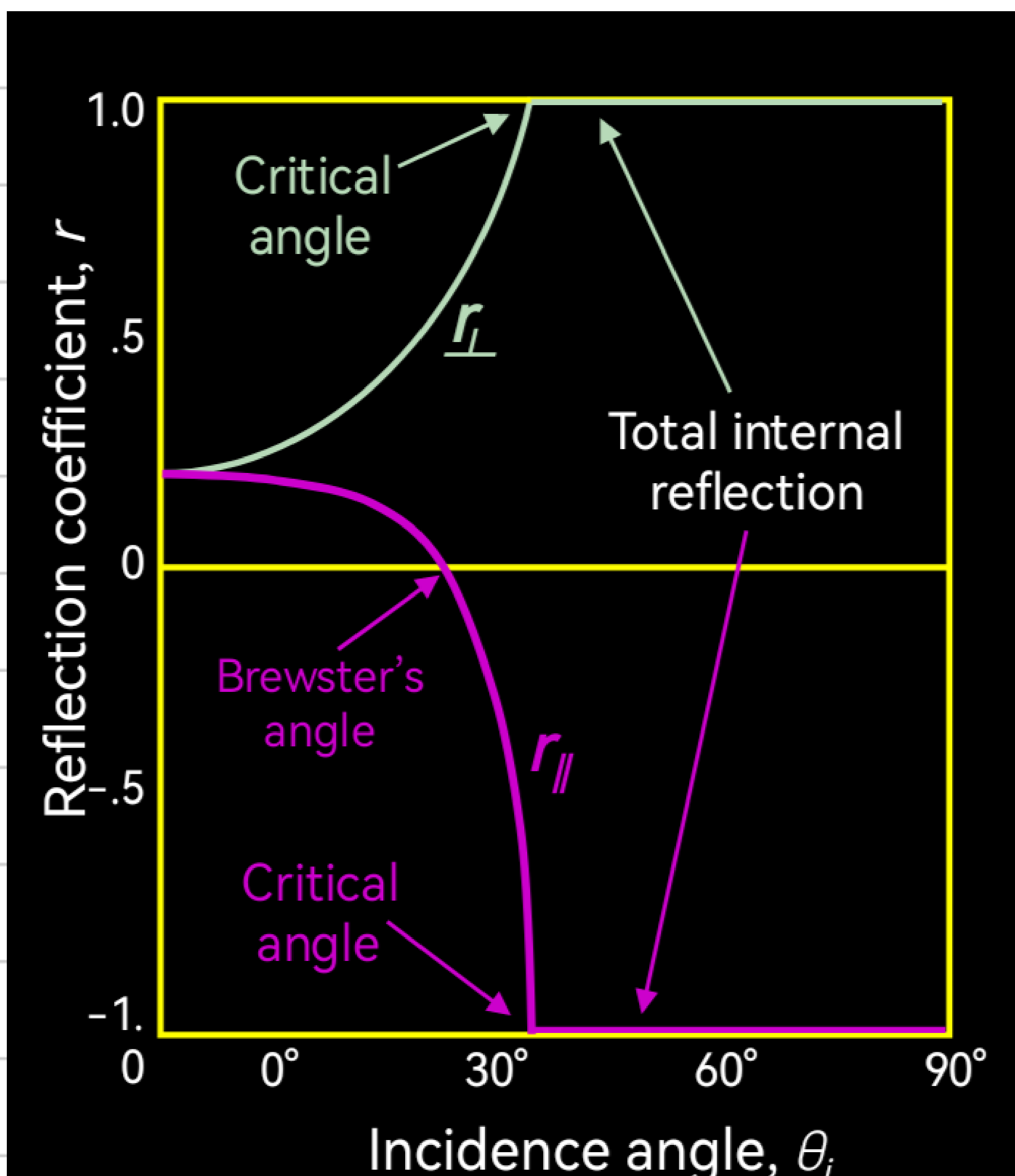
Brewster Angle 布儒斯特角  
 $n_i \cos \theta_t = n_t \cos \theta_i$

或者

$$\theta_i + \theta_t = 90^\circ$$

↓

$$\tan \theta_i = \frac{n_t}{n_i}$$



$$\theta_{crit} \equiv \arcsin(n_t / n_i)$$



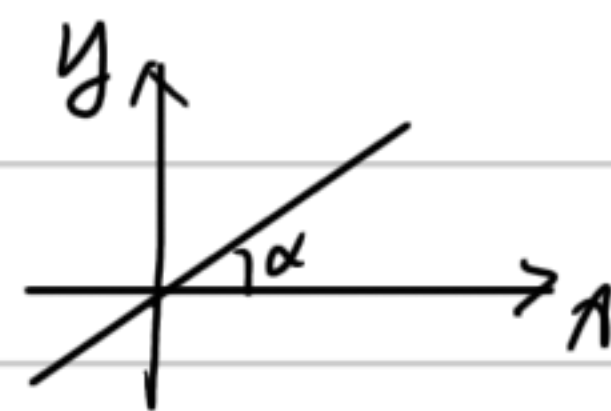
Malus's Law :  $I = I_{\max} \cos^2 \phi$

Polarization } ① Linear polarized :  $E$  has only one component

②  $45^\circ$  polarized : no phase difference between  $E_x, E_y$

$$E_x(z, t) = \text{Re} \{ \tilde{E}_0 \exp[i(kz - \omega t)] \}$$

$$E_y(z, t) = \text{Re} \{ \tilde{E}_0 \exp[i(kz - \omega t)] \}$$



③ circular (or helical) polarization  
90° 相位差.

counter clockwise  
↓

$$E_x(z, t) = E_0 \cos(kz - \omega t)$$

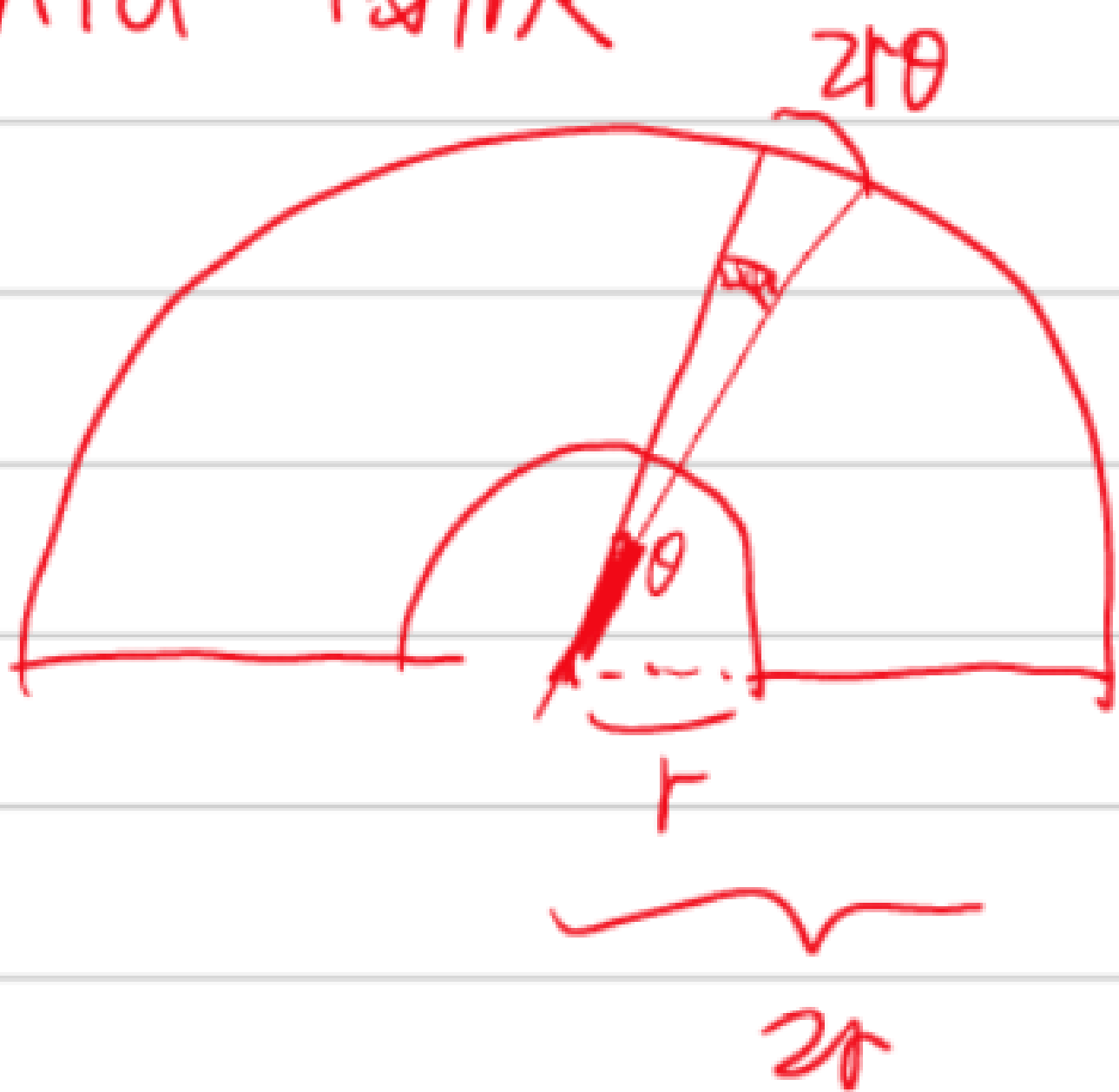
$$E_y(z, t) = E_0 \sin(kz - \omega t)$$

④

PM 2.5 mm

天为什么 blue

Mid 回顾



$$ds = r d\theta dr$$

$$\frac{2r\theta \cdot 2r}{2} = \frac{\theta r \cdot r}{2}$$

$$(r, \theta) \rightarrow \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \rightarrow \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

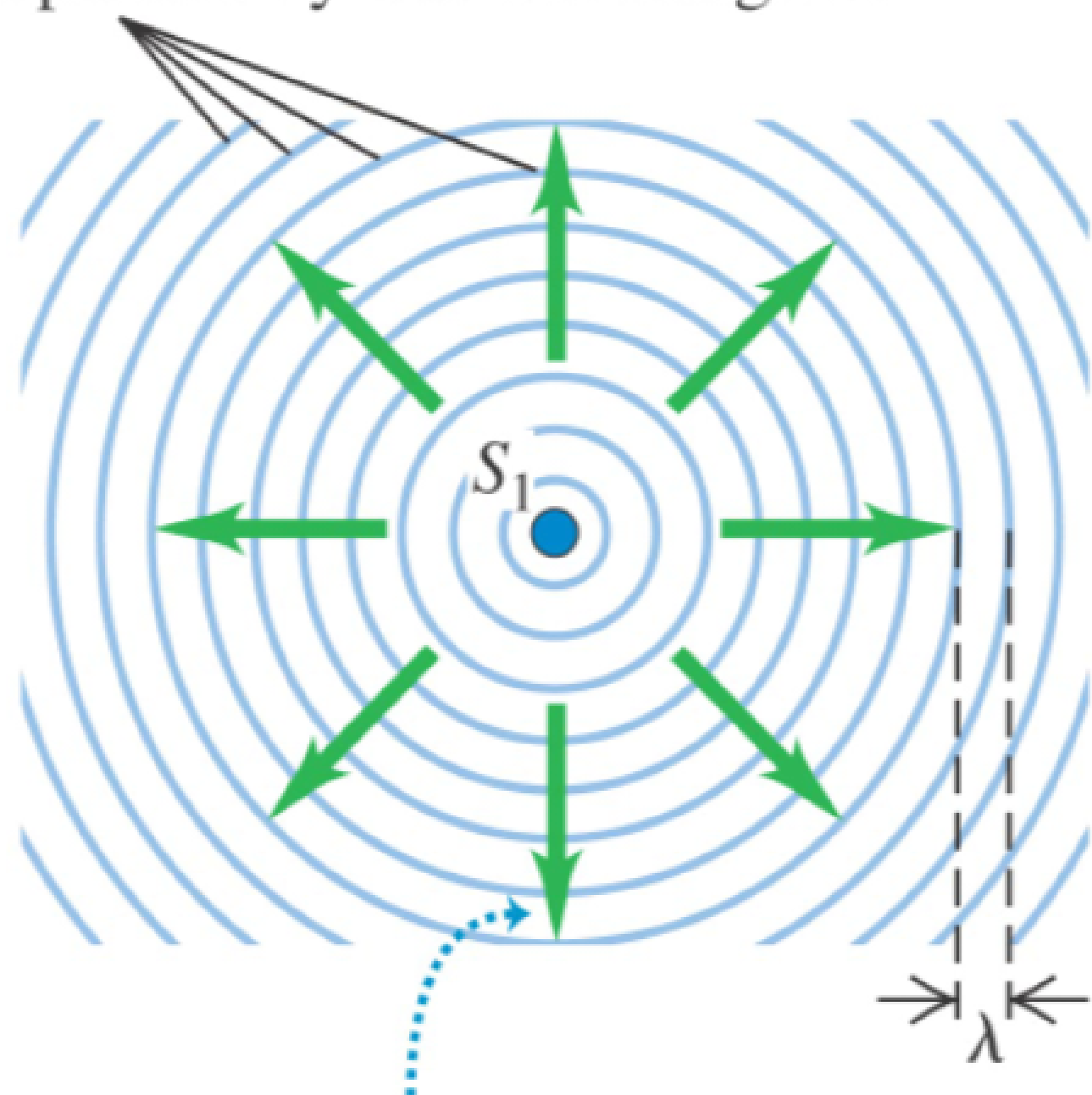
$$\det = 1$$



# Chap 35 Interference

## 1. Spherical wave

Wave fronts: crests of the wave (frequency  $f$ ) separated by one wavelength  $\lambda$



The wave fronts move outward from source  $S_1$  at the wave speed  $v = f\lambda$ .

$$E(x, t) = A \cos \varphi \rightarrow \text{某方向上的}$$

$$\varphi = kx - \omega t - \theta$$

$$k = \frac{2\pi}{\lambda}$$

↓

但在三维空间中

$$E(\vec{r}, t) = E_0 \cos(\vec{k} \cdot \vec{r} - \omega t - \theta)$$

$\hat{r}$  unit vector of direction

$$E(\vec{r}, t) = E_0 \cos(\vec{k} \cdot \vec{r} - \omega t - \theta) = E_0 \cos(k_x x + k_y y + k_z z - \omega t - \theta)$$

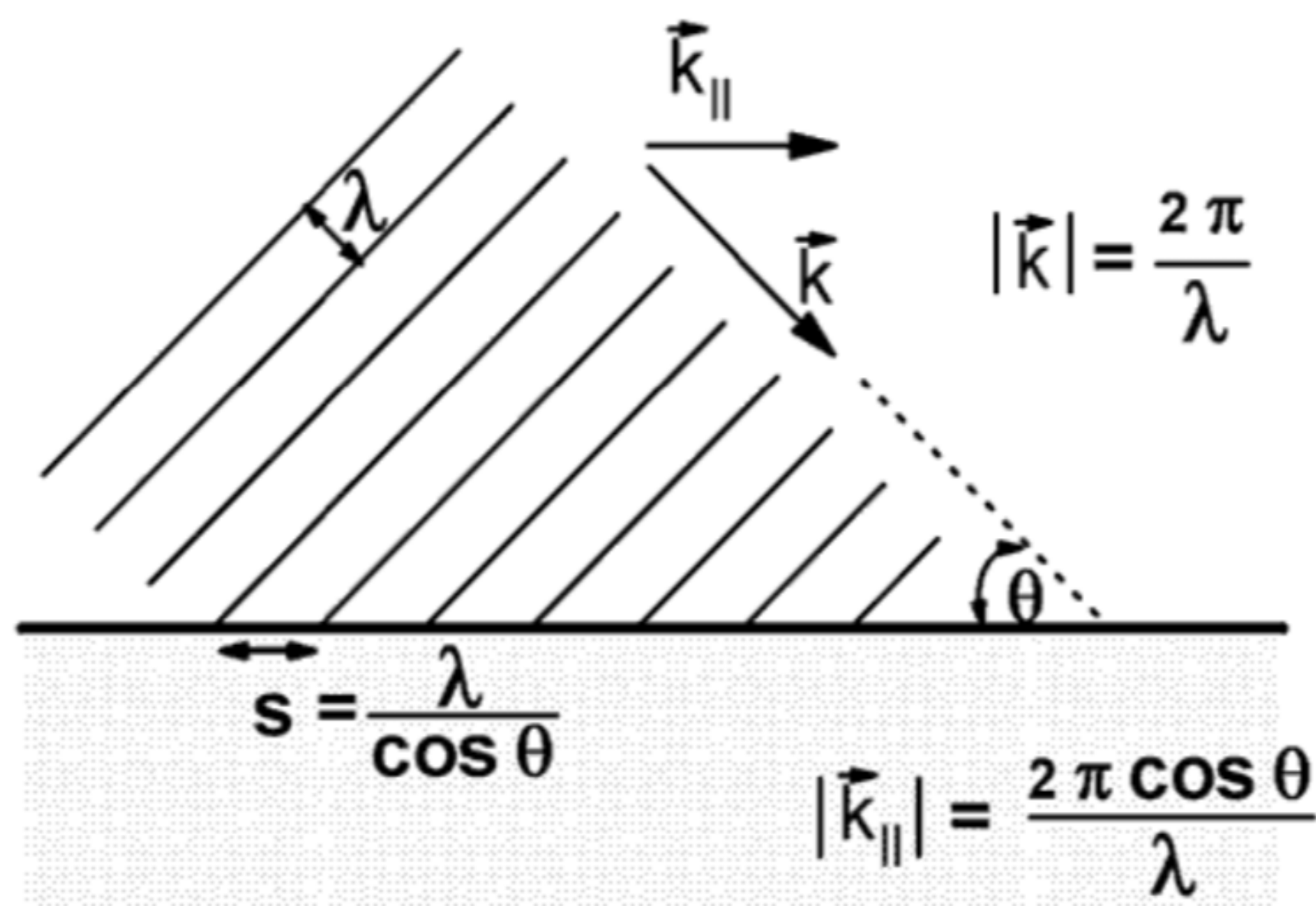
$$k_x^2 + k_y^2 + k_z^2 = |\mathbf{k}|^2 = \left(\frac{2\pi}{\lambda}\right)^2$$

延伸出有效波长

$$k_{\parallel} = \frac{2\pi \cos \theta}{\lambda}$$

$$= \frac{2\pi}{\left[\frac{\lambda}{\cos \theta}\right]}$$

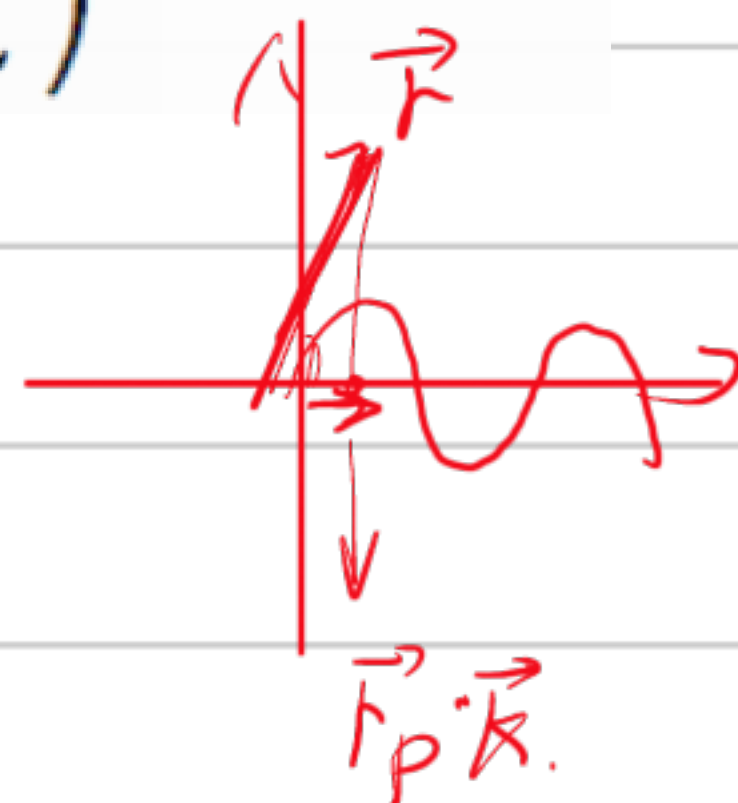
有效波长  $s$



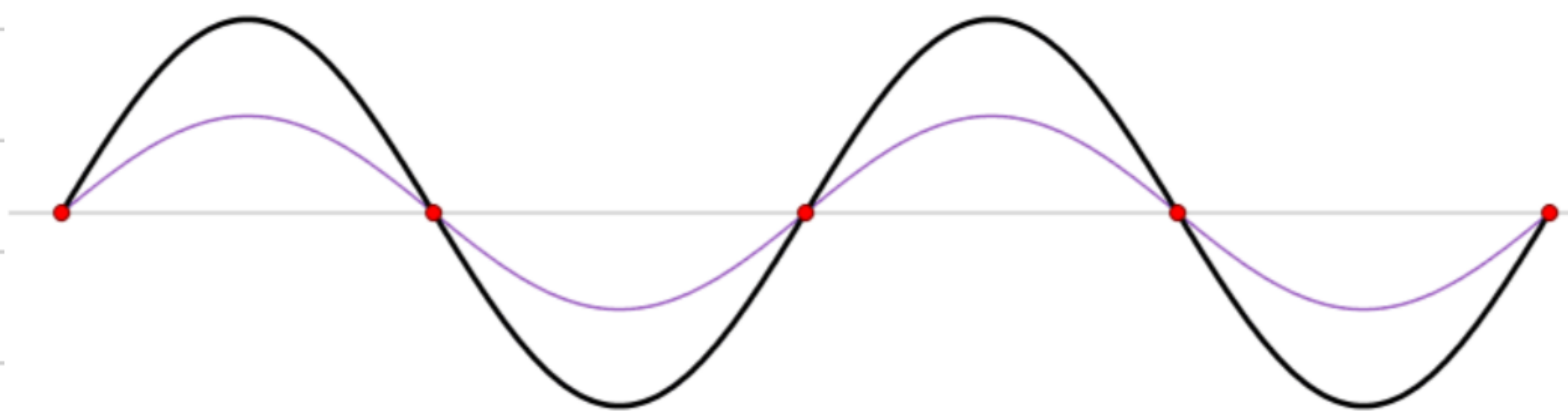
$$|\vec{k}_{\parallel}| = |\vec{k}| \cos \theta$$

$$|\vec{k}| = \frac{2\pi}{\lambda}$$

$$|\vec{k}_{\parallel}| = \frac{2\pi \cos \theta}{\lambda}$$







$$E_1 \cos(kx - \omega t - \theta_1) \quad \longrightarrow$$

$$\longleftarrow E_2 \cos(-kx - \omega t - \theta_2)$$

$$E_y(x, t) = E_{\max} [\cos(kx + \omega t) - \cos(kx - \omega t)]$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

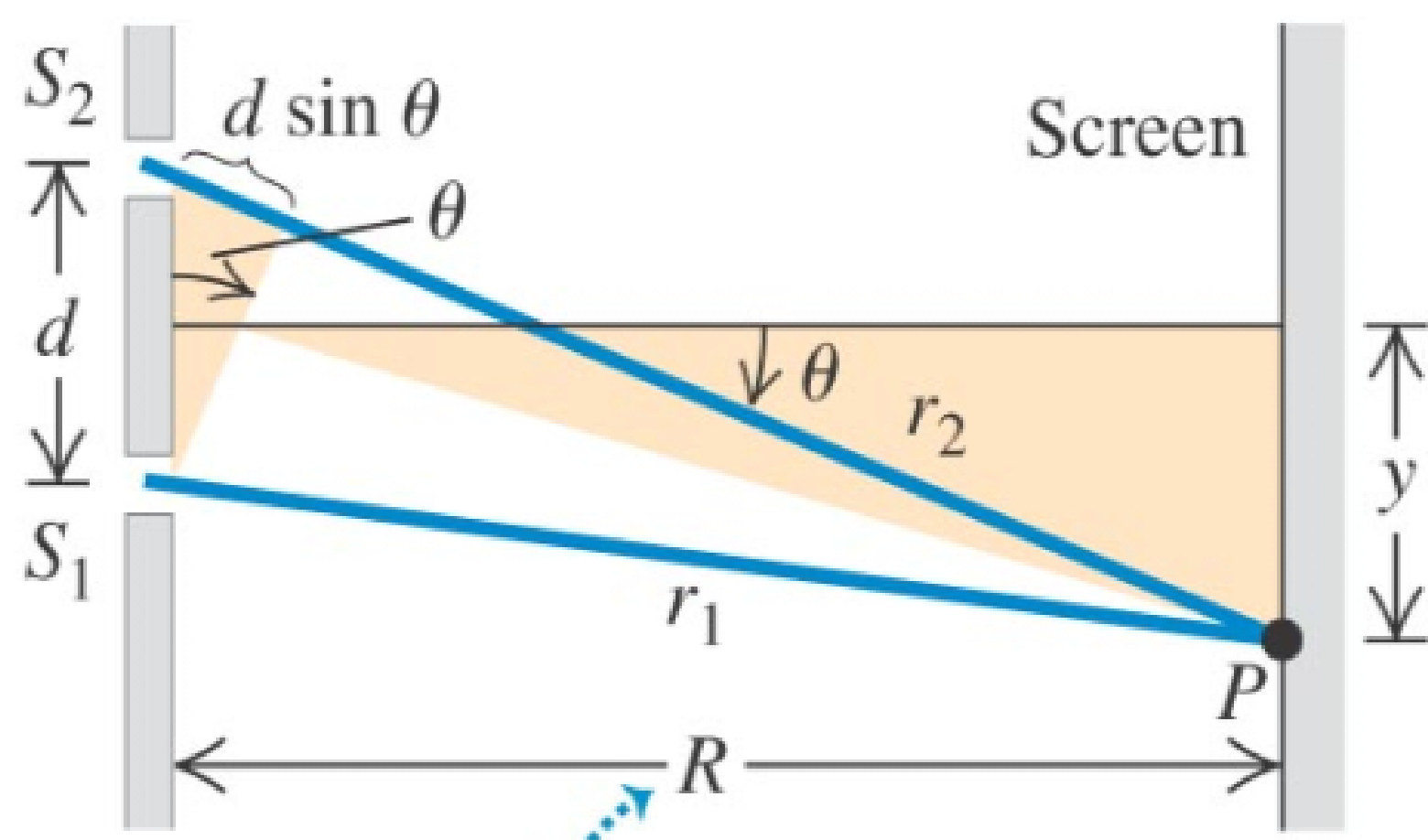
$$E_y(x, t) = -2E_{\max} \sin kx \sin \omega t$$

$S_1$

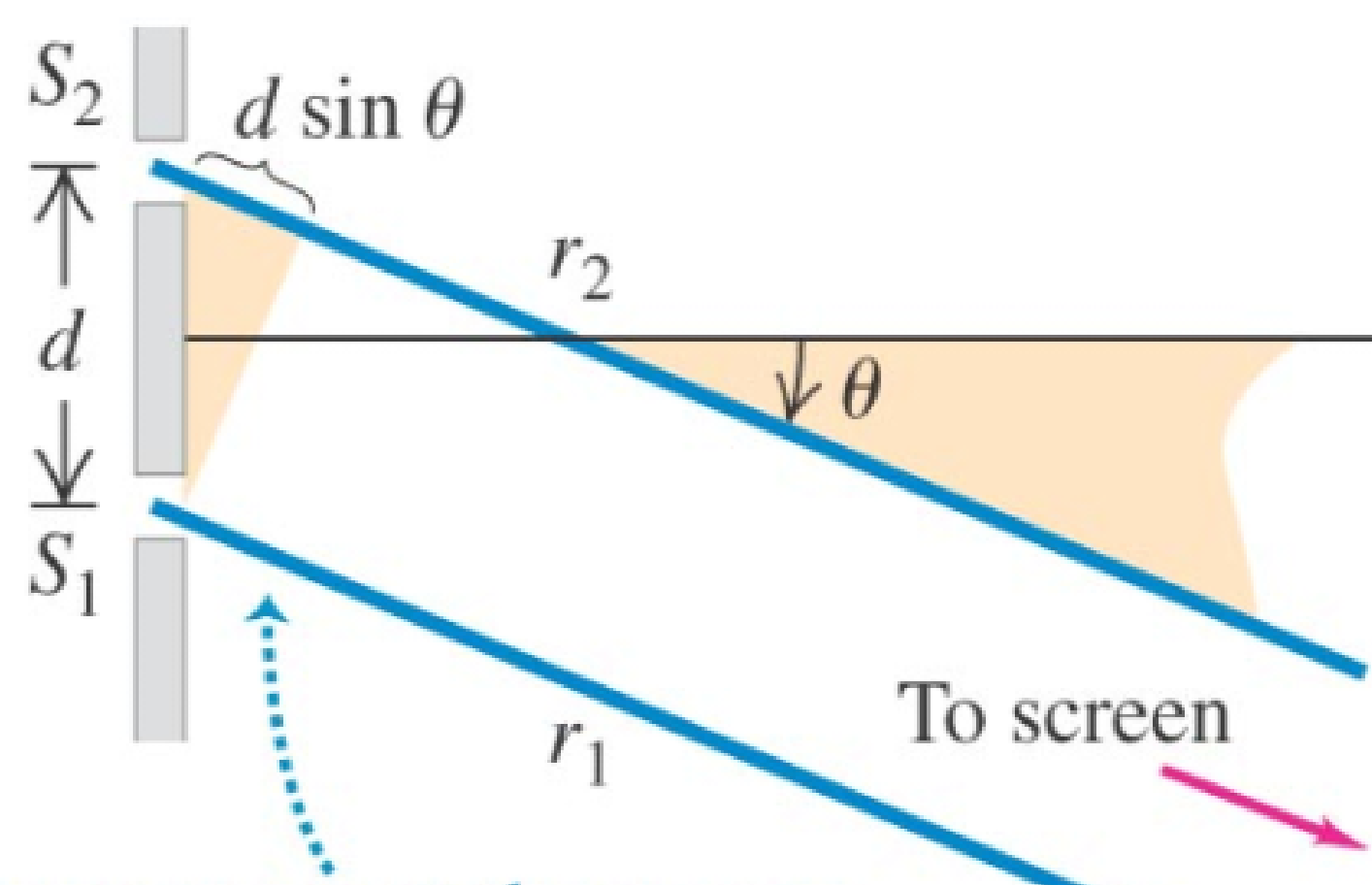
constructive interference.

$S_2$

(b) Actual geometry (seen from the side)



(c) Approximate geometry



$$r_2 - r_1 = d \sin \theta$$

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad \text{(constructive interference, two slits)}$$

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad \text{(destructive interference, two slits)}$$

日很小的时候才可近似

$$y_m = R \tan \theta_m$$

$$y_m = R \sin \theta_m$$

$$y_m = R \frac{m\lambda}{d} \quad \text{(constructive interference in Young's experiment)}$$



$$E_1(t) = E \cos(\omega t + \phi)$$

$$E_2(t) = E \cos \omega t$$

$$\begin{aligned} E_P^2 &= E^2 + E^2 - 2E^2 \cos(\pi - \phi) \\ &= E^2 + E^2 + 2E^2 \cos \phi \end{aligned}$$

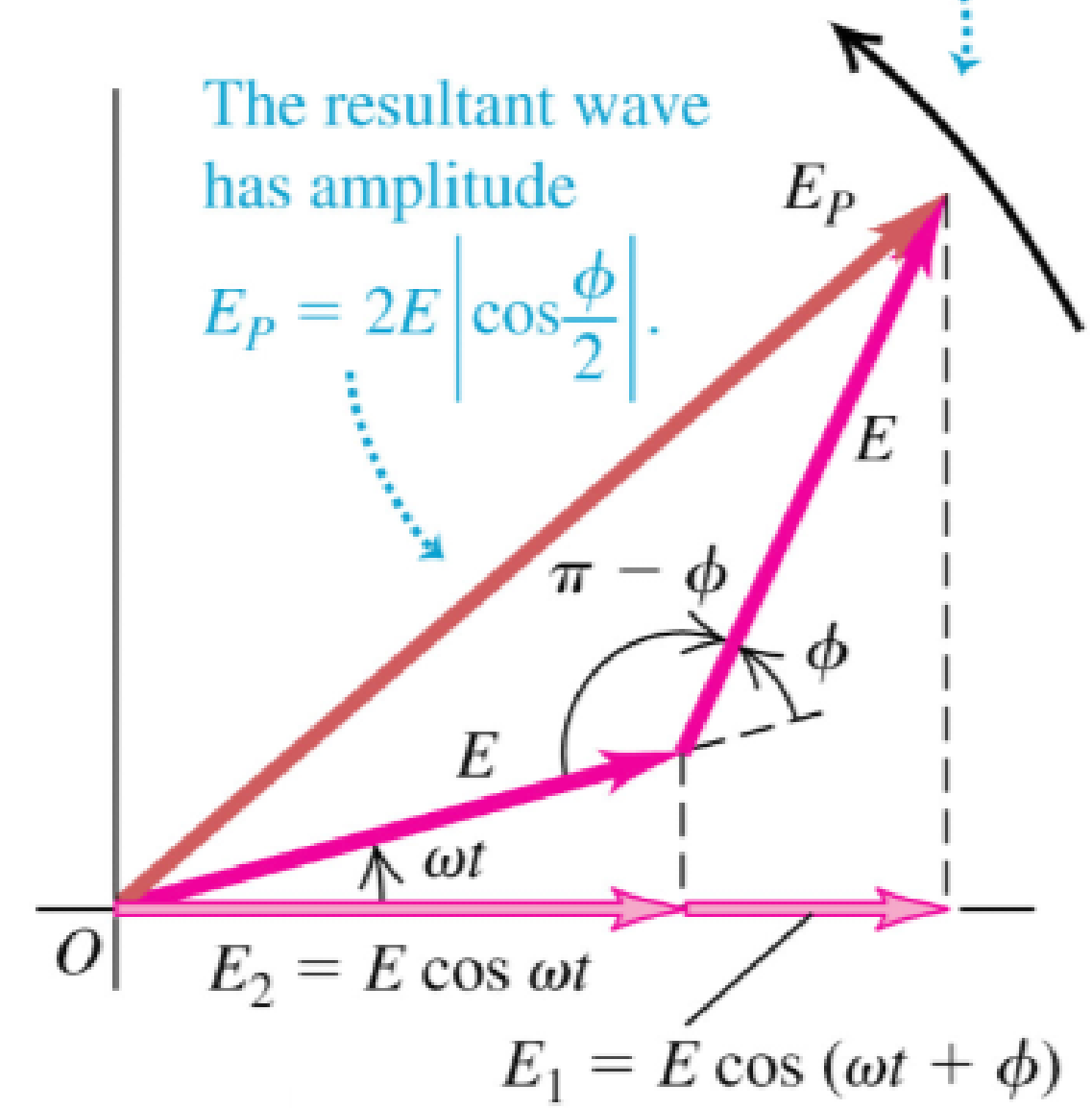
entity  $1 + \cos \phi = 2 \cos^2(\phi/2)$ , we obtain

$$E_P^2 = 2E^2(1 + \cos \phi) = 4E^2 \cos^2\left(\frac{\phi}{2}\right)$$

$$E_P = 2E \left| \cos \frac{\phi}{2} \right| \quad (\text{amplitude in two-source interference})$$

$\frac{1}{2} \epsilon_0 c E^2$  from each individual source.

All phasors rotate counterclockwise with angular speed  $\omega$ .



The resultant wave has amplitude

$$E_P = 2E \left| \cos \frac{\phi}{2} \right|$$

## Intensity in Two-Source Interference

$$I = S_{\text{av}} = \frac{E_P^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_P^2 = \frac{1}{2} \epsilon_0 c E_P^2$$

$$I = \frac{1}{2} \epsilon_0 c E_P^2 = 2\epsilon_0 c E^2 \cos^2 \frac{\phi}{2}$$

the maximum intensity  $I_0 = 2\epsilon_0 c E^2$

$$I = I_0 \cos^2 \frac{\phi}{2} \quad (\text{intensity in two-source interference})$$

$$I = \epsilon_0 \langle \vec{E}^2 \rangle$$

$$\propto \langle \vec{E}^2 \rangle$$

$$\vec{E}_1(\vec{r}, t) = \vec{E}_{0,1} \cos(\vec{k}_1 \cdot \vec{r} - \omega t + \epsilon_1)$$

$$\vec{E}_2(\vec{r}, t) = \vec{E}_{0,2} \cos(\vec{k}_2 \cdot \vec{r} - \omega t + \epsilon_2)$$

$$I = \langle \vec{E}^2 \rangle = \int_0^t \vec{E}^2 dt$$

$$= \langle (\vec{E}_1 + \vec{E}_2)^2 \rangle$$

$$= \langle (\vec{E}_1 + \vec{E}_2) \cdot (\vec{E}_1 + \vec{E}_2) \rangle$$

$$= \langle \vec{E}_1^2 \rangle + \langle \vec{E}_2^2 \rangle + 2\langle \vec{E}_1 \cdot \vec{E}_2 \rangle$$

$$= I_1 + I_2 + I_{12}$$

$$\vec{E}_1 \cdot \vec{E}_2 = \vec{E}_{0,1} \cdot \vec{E}_{0,2} \cos(\vec{k}_1 \cdot \vec{r} - \omega t + \epsilon_1) \cos(\vec{k}_2 \cdot \vec{r} - \omega t + \epsilon_2)$$

$$= \vec{E}_{0,1} \cdot \vec{E}_{0,2} [\cos(\vec{k}_1 \cdot \vec{r} + \epsilon_1) \cos \omega t + \sin(\vec{k}_1 \cdot \vec{r} + \epsilon_1) \sin \omega t]$$

$$\times [\cos(\vec{k}_2 \cdot \vec{r} + \epsilon_2) \cos \omega t + \sin(\vec{k}_2 \cdot \vec{r} + \epsilon_2) \sin \omega t]$$

$$\langle f(t) \rangle = \frac{1}{T} \int_t^{t+T} f(t') dt'$$

$$\langle \cos^2 \omega t \rangle = \frac{1}{2}$$

$$\langle \sin^2 \omega t \rangle = \frac{1}{2}$$

$$\langle \cos \omega t \sin \omega t \rangle = 0$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\langle \vec{E}_1 \cdot \vec{E}_2 \rangle = \frac{1}{2} \vec{E}_{0,1} \cdot \vec{E}_{0,2} \cos(\vec{k}_1 \cdot \vec{r} + \epsilon_1 - \vec{k}_2 \cdot \vec{r} - \epsilon_2)$$

$$I_{12} = \vec{E}_{0,1} \cdot \vec{E}_{0,2} \cos \delta$$



$$\delta = \vec{k}_1 \cdot \vec{r} + \epsilon_1 - \vec{k}_2 \cdot \vec{r} - \epsilon_2$$

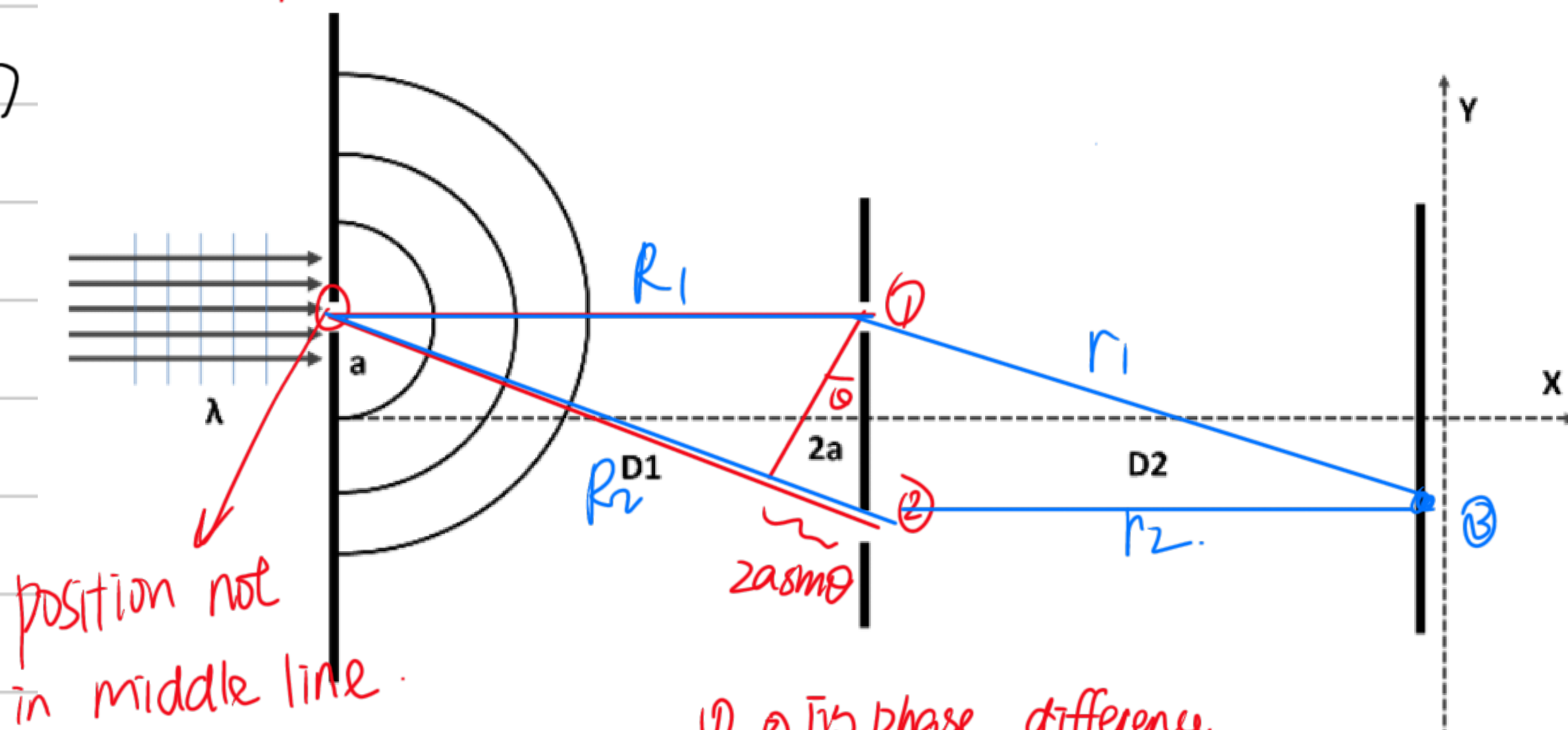
$\vec{k}_1 \cdot \vec{r} - \vec{k}_2 \cdot \vec{r}$  paths

$\epsilon_1 - \epsilon_2$  initial phase difference



可能是final题

17

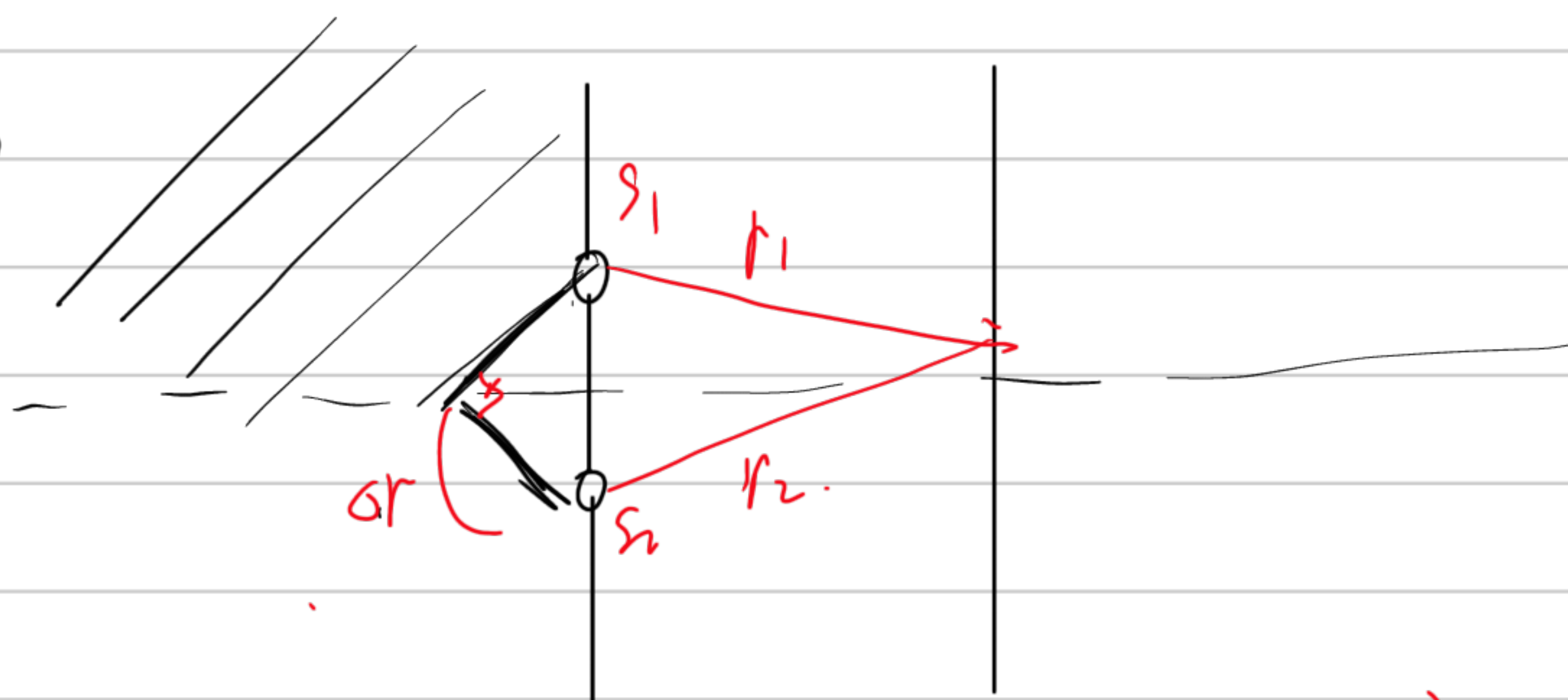


①. ② path difference

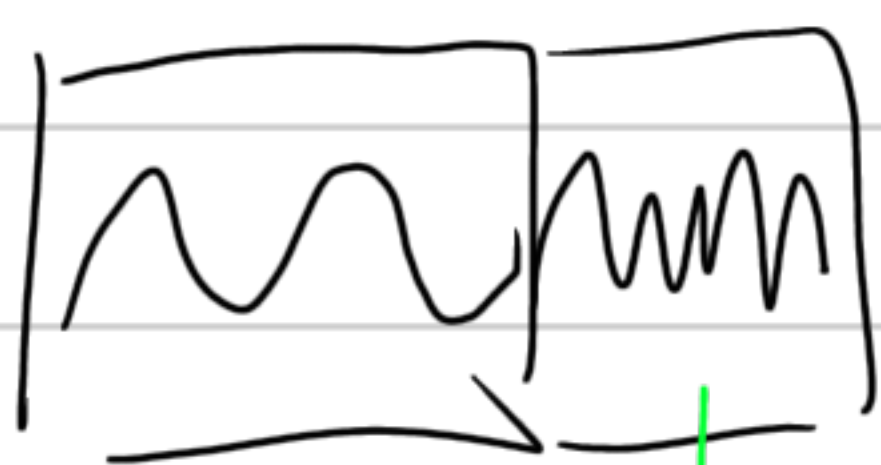
$$\Delta \theta = 2a \sin \theta$$

法二: 算  $k(R_1 + r_2) - k(R_2 + r_2)$

27



path difference:  $k(r_1 - (\Delta r + r_2))$



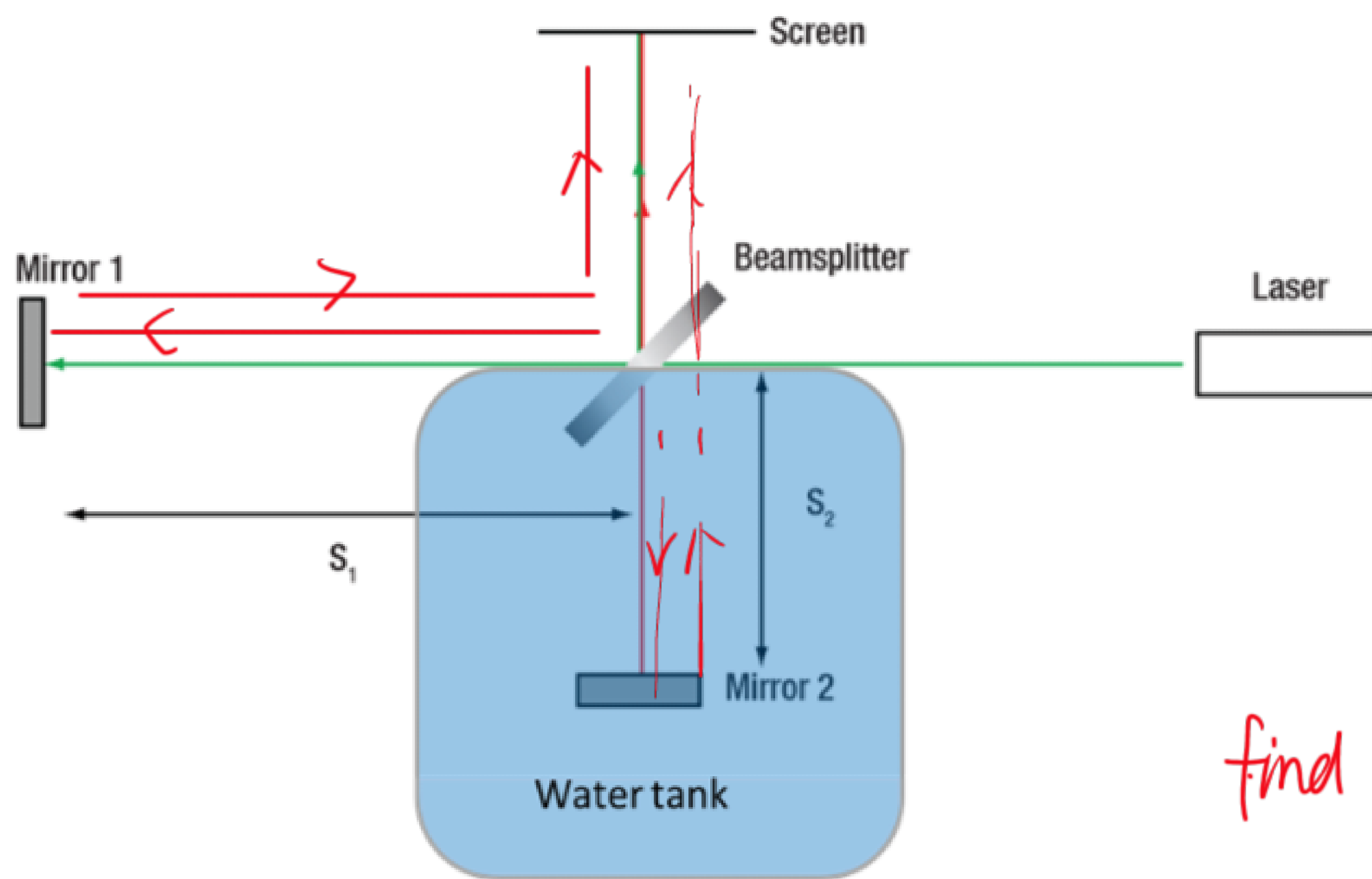
$$\lambda = \frac{\lambda_0}{n}$$

$v \downarrow$   $n \uparrow$   $\lambda \downarrow$   $\uparrow k = \frac{2\pi}{\lambda}$

$$n = \frac{c}{v}$$



★ [15%] In an interferometer, one arm is embedded in the water (index of refraction  $n_{\text{water}}=1.3$ ), while the other one is in the air ( $n_{\text{air}}=1$ ). Suppose Mirror 2 is movable, find the separation  $S_2$ 's value such that the center of the interference pattern on the Screen is dark (the complete destructive interference point). Assuming the interferometer is perfectly aligned, no phase jump on the mirrors or beamsplitter. The beamsplitter is 50% transparent and 50% reflective.



★  $k$  正比于  $n$ ,  $k\lambda - \omega t + \theta$   
 $\downarrow$   
 会不一样

$$\cos(k_1 r - \omega t + \theta_1)$$

$$\cos(k_2 r - \omega t + \theta_2)$$

$\Downarrow$

$$\text{find } 2k_1 S_1 - 2k_2 S_2$$

$\parallel$

$$\frac{2\pi}{\lambda_1} S_1 - \frac{2\pi}{\lambda_2} S_2$$

$\parallel$

$$\frac{2\pi}{\lambda_0} (n_1 S_1 - n_2 S_2)$$

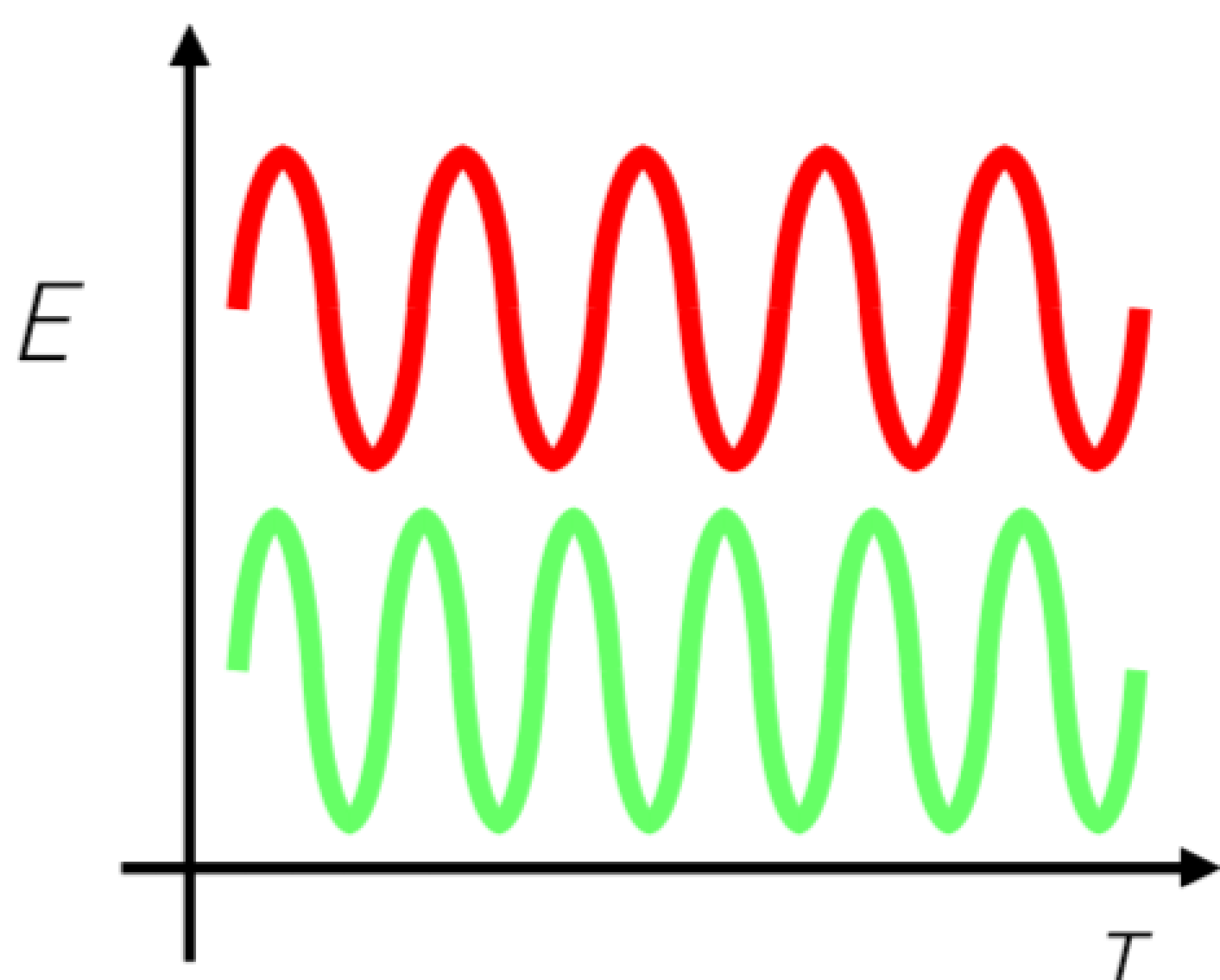
易错

$k, r, \theta$  could be different.

## How quickly will a light wave deviate from a perfect sine wave in time?

Suppose the light wave has two frequencies:

$$E_{\text{tot}}(x, t) = E_0 \exp i(k_1 x - \omega_1 t) + E_0 \exp i(k_2 x - \omega_2 t)$$



The two frequencies will become significantly out of phase with each other in a time,  $t_c$ :

$$\omega_1 T_c - \omega_2 T_c = 2\pi$$

$$\Rightarrow T_c = 2\pi / (\omega_1 - \omega_2)$$

So the phase will drift on a time scale of:  $\sim 2\pi / \Delta\omega = 1 / \Delta\nu$

where:  $\Delta\omega = \omega_1 - \omega_2 = 2\pi \Delta\nu$



相干时间

# The coherence time is the reciprocal of the bandwidth.

The largest frequency difference in the light wave will yield the shortest phase-drift time, which we call the **coherence time**:

带宽越长, 光波覆盖频率范围越广, 相干时间越短.

$$\tau_c = 1 / \Delta \nu$$

where  $\Delta \nu$  is the light bandwidth (the width of the spectrum).

Sunlight and light bulbs are temporally very **incoherent**—and have very small coherence times (a few fs)—because their bandwidths are very large (the entire visible spectrum).

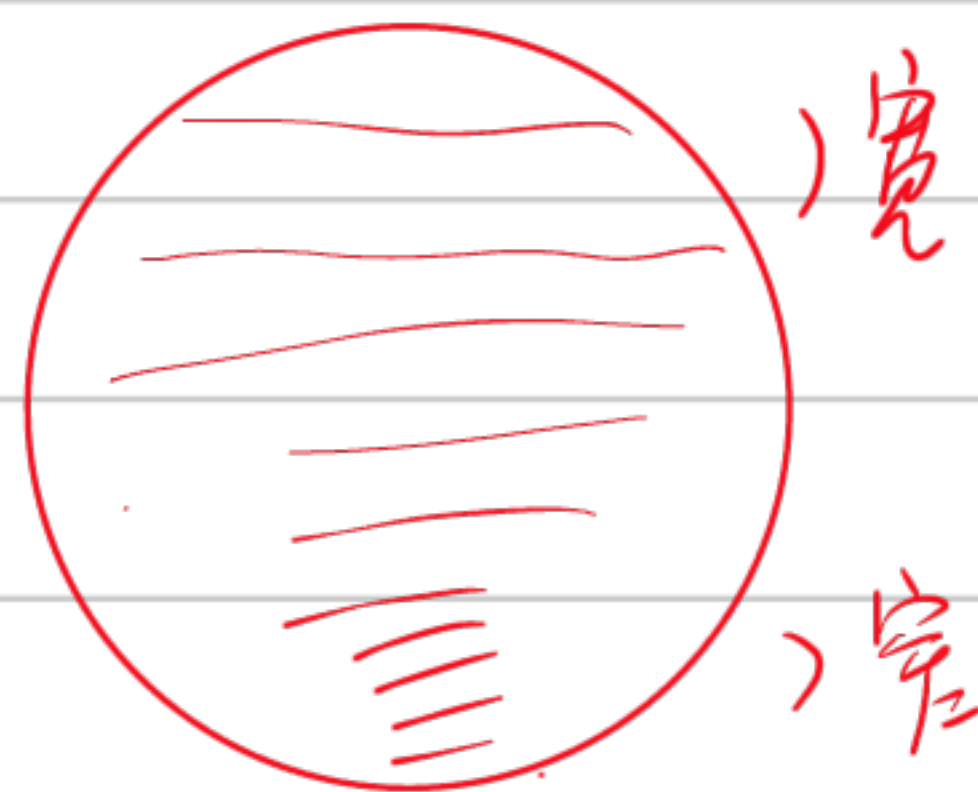
Lasers can have much longer coherence times—as long as about a second, which is amazing; that's  $>10^{14}$  cycles!

air bubble thickness  $< L_c$

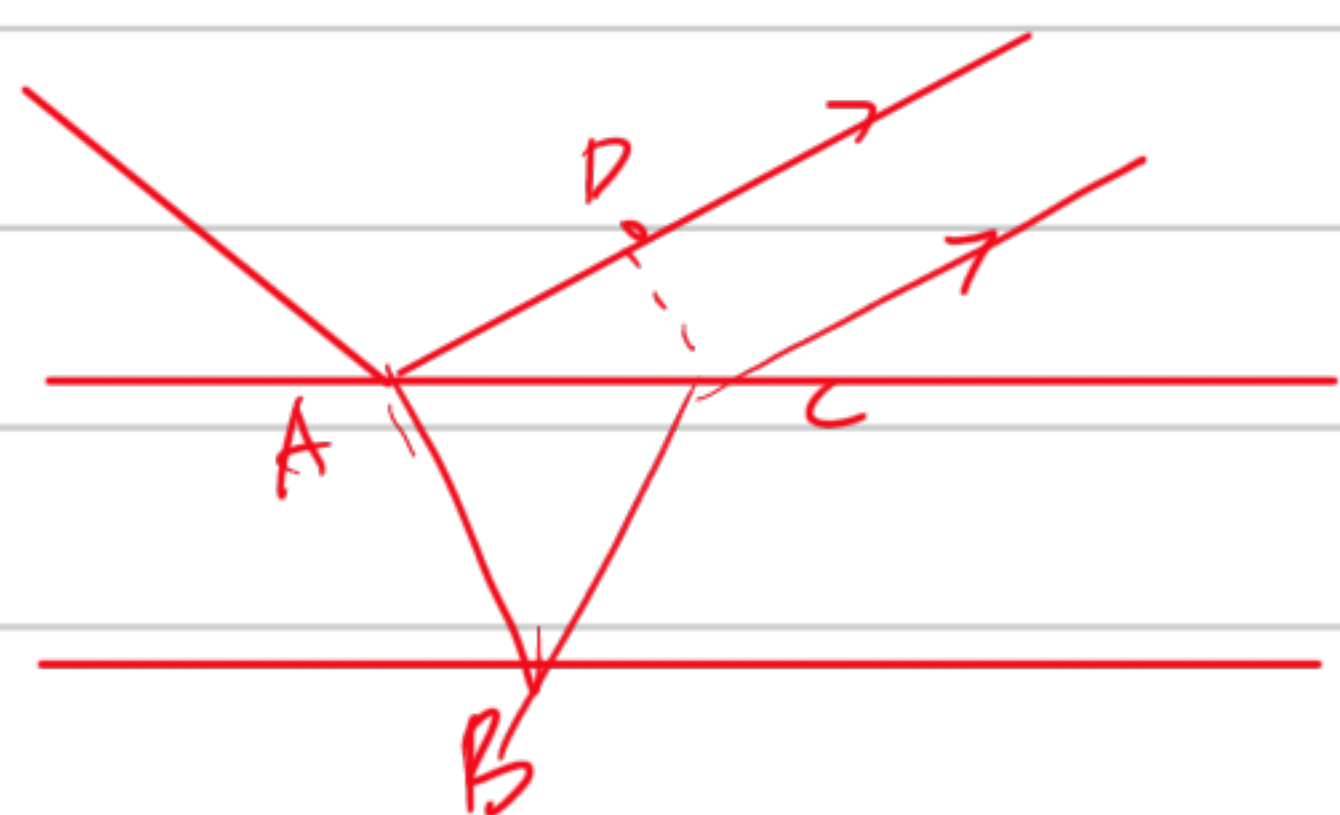
↓

interference

一个泡泡膜







光从一种介质进入另一种：  
频率不变，波长、传播速度变

易错： $\nabla k(AD - ABC)$

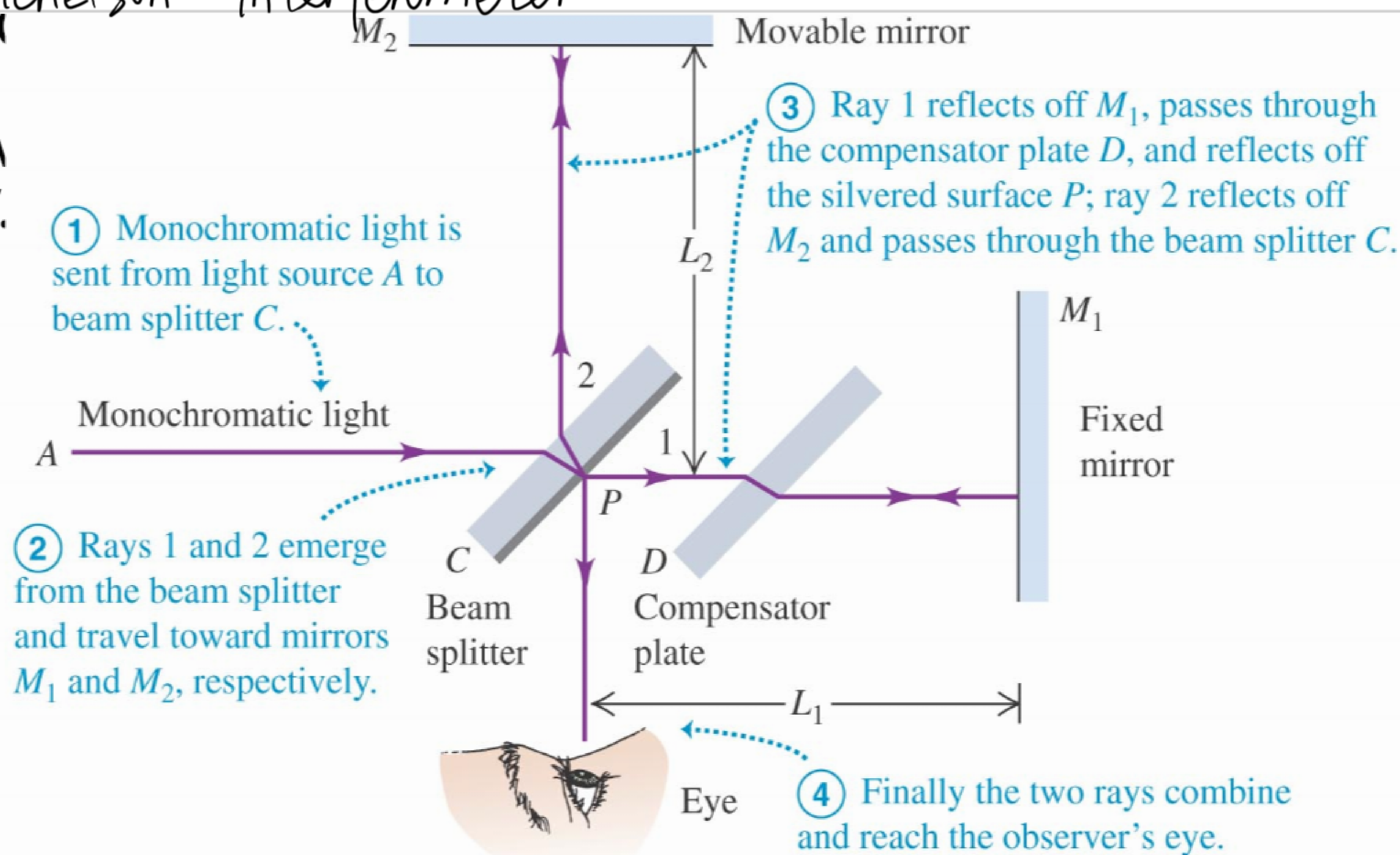
instead:  $\Phi_{AD} - \Phi_{ABC}$   
 $= k_0 AD - k_1 ABC$   
 $= \frac{2\pi}{\lambda_0} AD - \frac{2\pi}{\lambda_1} ABC$

$n = \frac{c}{v_{\text{medium}}} = \frac{f\lambda_0}{f\lambda_1}$        $\lambda_1 = \frac{\lambda_0}{n}$

constructive reflect  
destructive

photonic crystal 光子晶体 通过干涉来引导光路

Michelson interferometer





# lecture 36

# Diffraction

## 1. 惠更斯原理

Huygens' Principle says that every point along a wave-front emits a spherical wave that interferes with all others.

## 2. { Fresnel (near-field) Frauenhofer (far-field)

$$\textcircled{1} \quad E(x', y') = \frac{\exp(ikz)}{i\lambda z} \exp\left[ik \frac{x'^2 + y'^2}{2z}\right] \iint \exp\left\{ik \left[\frac{(-2xx' - 2yy')}{2z} + \frac{(x^2 + y^2)}{2z}\right]\right\} t(x, y) E(x, y) dx dy$$

This is the **Fresnel integral**.

It yields the light wave field at the distance  $z$  from the screen.

## ② far-field

Let  $D$  be the size of the aperture:  $D^2 \geq x^2 + y^2$ .

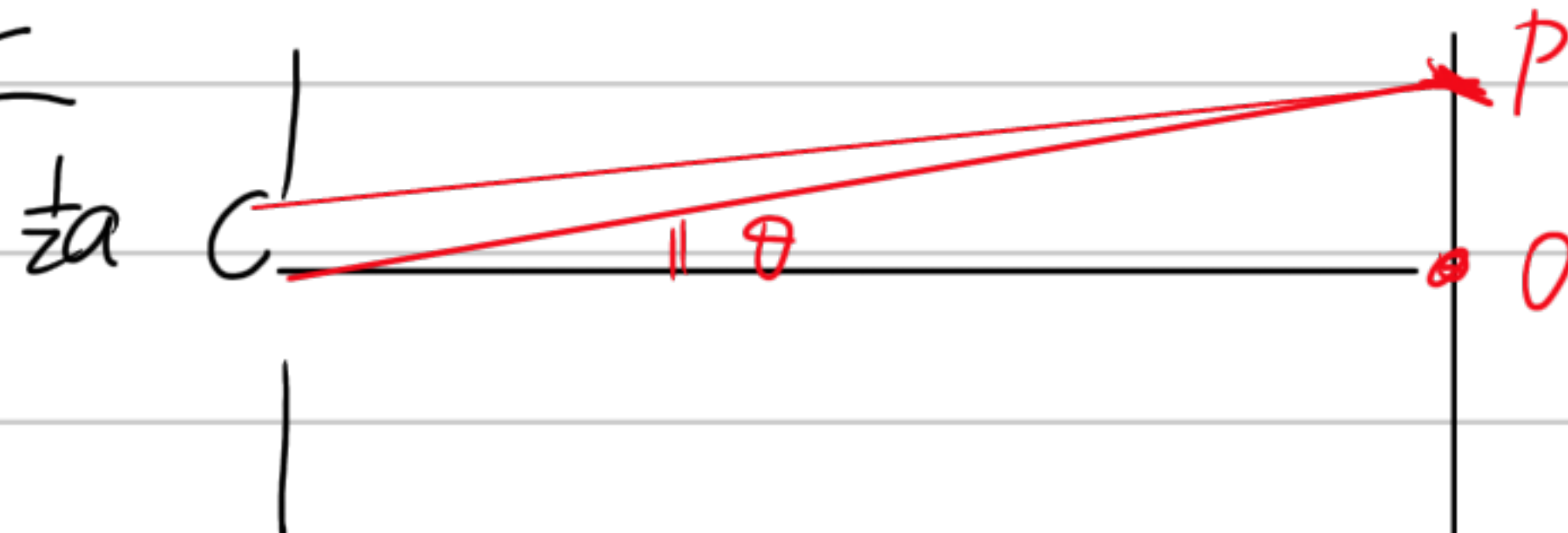
When  $kD^2/2z \ll 1$ , the quadratic terms  $\ll 1$ , so we can neglect them.

This condition means going a distance away:  $z \gg kD^2/2 = \pi D^2/\lambda$

$$E(x', y') = \frac{\exp(ikz)}{i\lambda z} \exp\left[ik \frac{x'^2 + y'^2}{2z}\right] \iint \exp\left\{-\frac{ik}{z}(xx' + yy')\right\} t(x, y) E(x, y) dx dy$$

1) 中间一定是 constructive point.

2) 在 fringe 上



Constructive X  
destructive V

P 是 destructive 条件:  $\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \dots)$



for small  $\theta$ :  $\theta = \frac{m\lambda}{a}$

$$\phi = \beta = ka \sin \theta = \frac{2\pi}{\lambda} a \sin \theta$$

$\Rightarrow$  推导出 single slit:

$$I = I_0 \left( \frac{\sin(\frac{1}{2}\phi)}{\frac{1}{2}\phi} \right)^2$$

$$I_0 = (E_m a)^2$$

## Intensity maxima in a single-slit pattern

$$\beta = \frac{2\pi}{\lambda} a \sin \theta$$

$$I = I_0 \left\{ \frac{\sin[\pi a(\sin \theta)/\lambda]}{\pi a(\sin \theta)/\lambda} \right\}^2$$

$$I = I_0 \left[ \frac{\sin(\beta/2)}{\beta/2} \right]^2$$

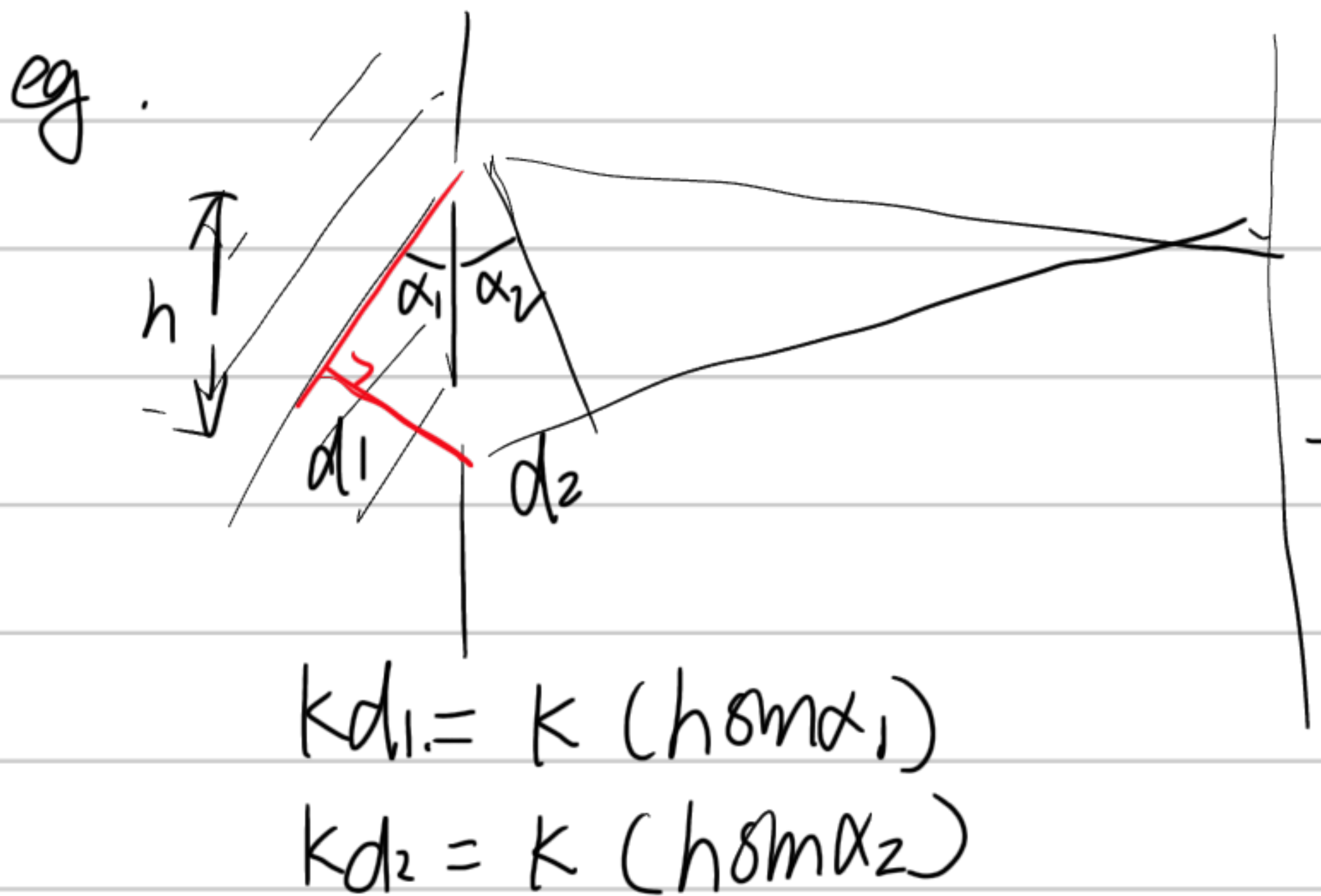
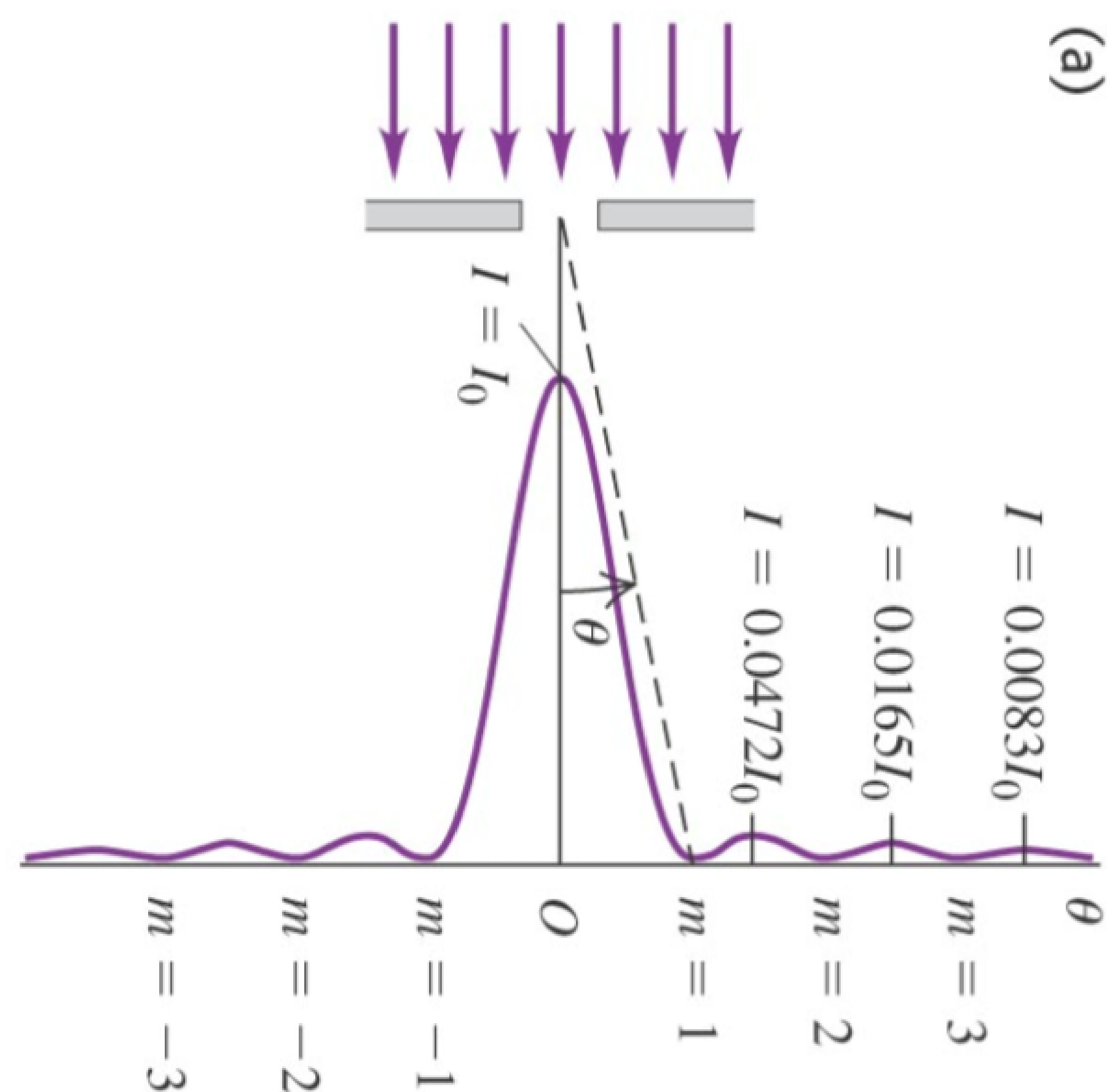
The dark fringes

$$\frac{a \sin \theta}{\lambda} = m \quad (m = \pm 1, \pm 2, \dots)$$

### Intensity Maxima

$$\beta \approx \pm(2m + 1)\pi \quad (m = 0, 1, 2, \dots)$$

from textbook



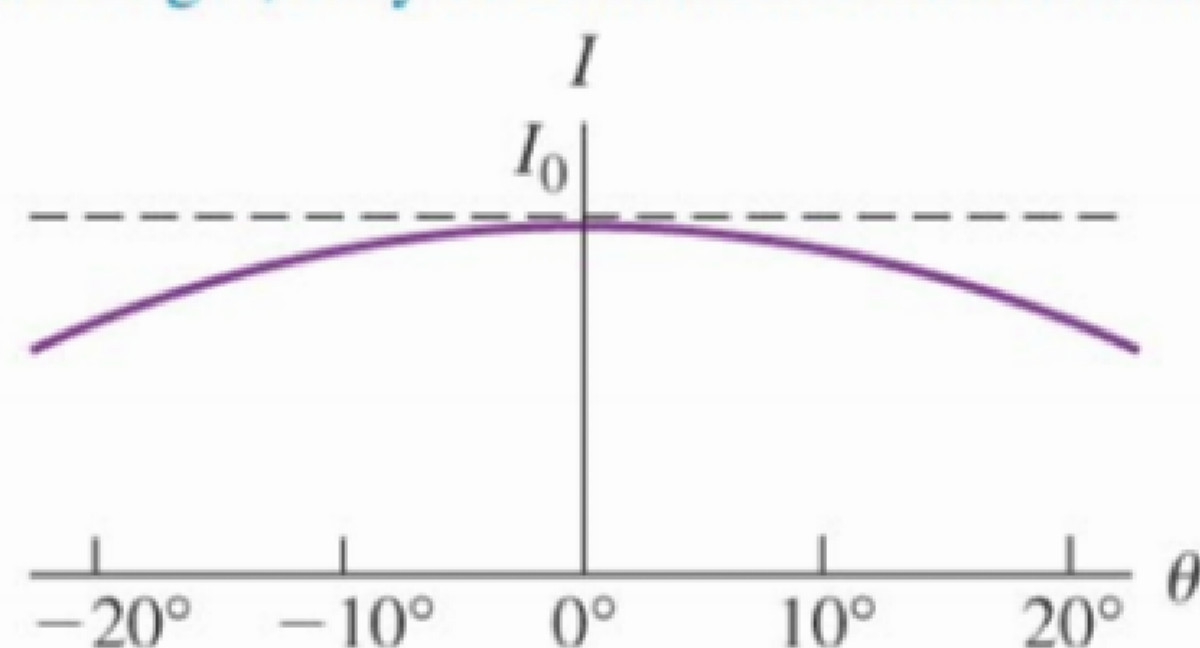


(a)  $a = \lambda$

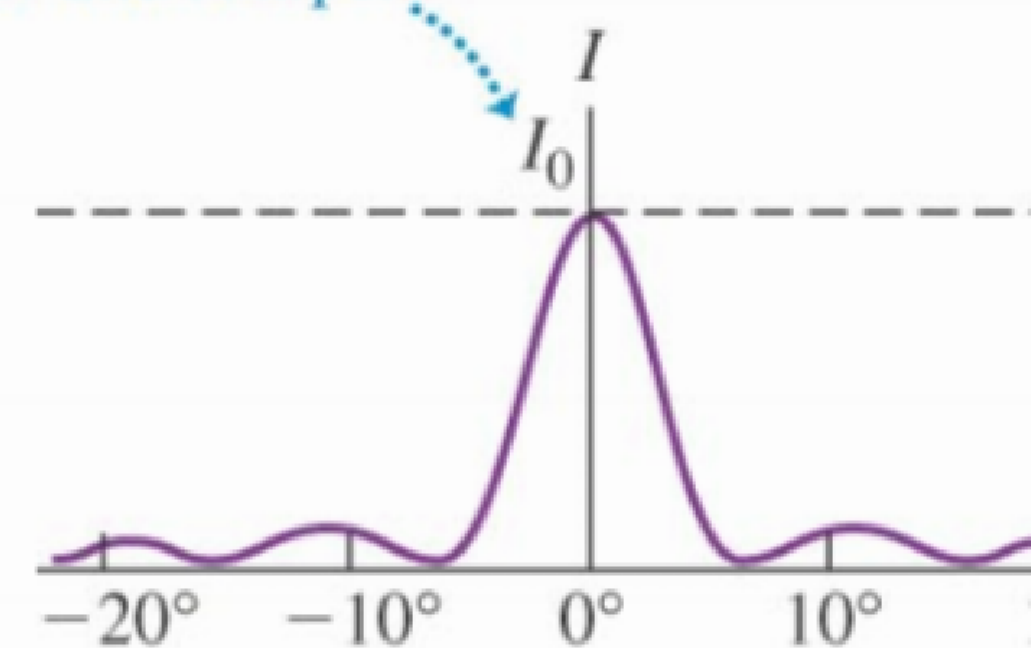
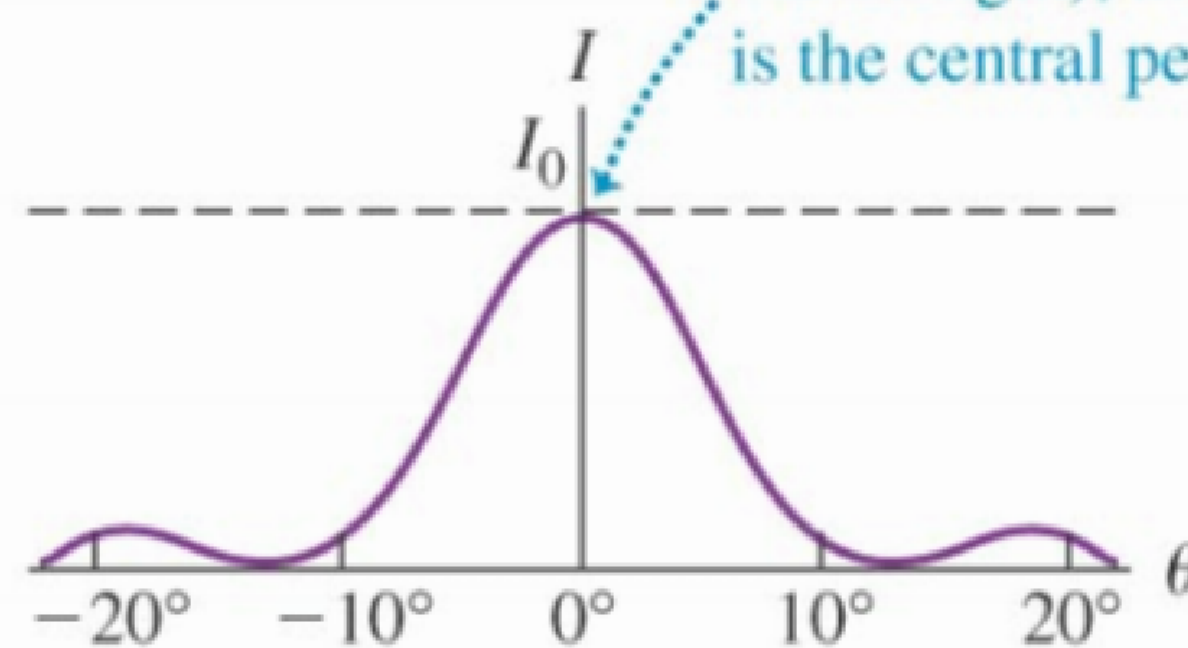
(b)  $a = 5\lambda$

(c)  $a = 8\lambda$

If the slit width is equal to or narrower than the wavelength, only one broad maximum forms.



The wider the slit (or the shorter the wavelength), the narrower and sharper is the central peak.



Young's Double Slit : 要求 slit width small enough.

双缝衍射：用公式时 two slit same width

## Intensity

- $I = I_0 \left( \frac{\sin(\phi/2)}{(\phi/2)} \right)^2 (\cos(\delta/2))^2$
- diffraction and interference
- $\phi = ka \sin(\theta)$        $\delta = kd \sin(\theta)$

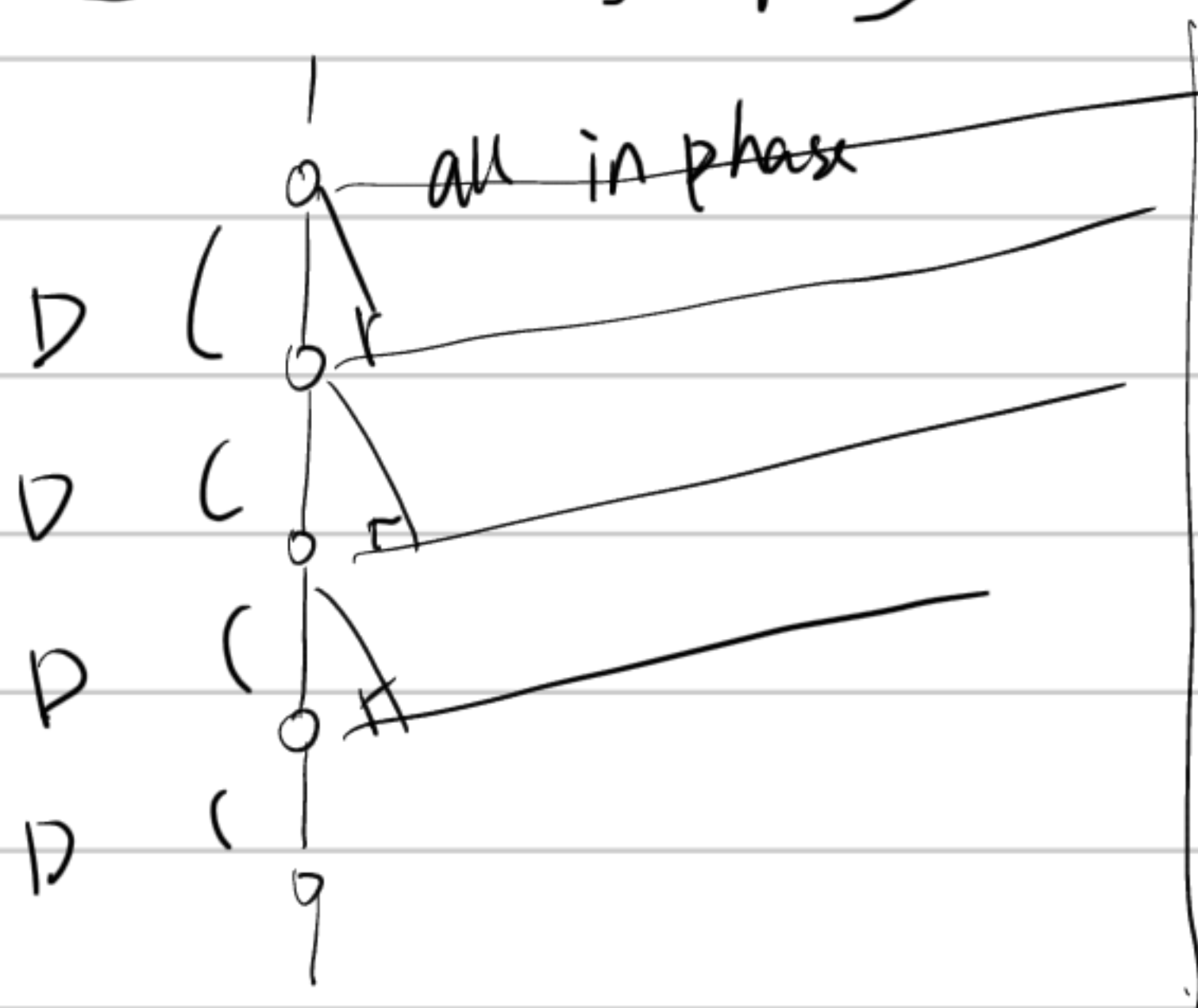
- limit  $a \Rightarrow 0$        $I = I_0 (\cos(\delta/2))^2$       Young's slits

- limit  $d \Rightarrow 0$        $I = I_0 \left( \frac{\sin(\phi/2)}{(\phi/2)} \right)^2$       Single slit

- note:  $\phi/\delta = a/d$

多缝：类似于干涉的讨论

|              |   |
|--------------|---|
| destructive  | x |
| constructive | ✓ |



$$kd \sin \theta = 2\pi m \quad \checkmark$$



# Double Slit Diffraction

双缝

$$E(r, t) = (E_m a) \frac{\sin(\phi / 2)}{(\phi / 2)} \frac{\sin(\delta / 2)}{\sin(\delta / 2)} \sin(kr - \omega t + \phi / 2 + \delta / 2)$$

N 缝

## Diffraction Grating



only one particular color

$$E(r, t) = (E_m a) \frac{\sin(\phi / 2)}{(\phi / 2)} \frac{\sin(N\delta / 2)}{\sin(\delta / 2)} \sin[kr - \omega t + \phi / 2 + (N - 1)\delta / 2]$$

$$I = \frac{I_0}{N^2} \left( \frac{\sin(\phi / 2)}{(\phi / 2)} \right)^2 \left( \frac{\sin(N\delta / 2)}{\sin(\delta / 2)} \right)^2 \quad I_0 = (NE_m a)^2$$

- $N=1 \Rightarrow I=I_0 [\sin(\phi/2)/(\phi/2)]^2$  single slit diffraction
- $N=2 \Rightarrow I=I_0 [\sin(\phi/2)/(\phi/2)]^2 [\cos(\delta/2)]^2$  double slit diffraction
- For N narrow slits  $\phi \Rightarrow 0$ ,  $I = \frac{I_0}{N^2} \left( \frac{\sin(N\delta / 2)}{\sin(\delta / 2)} \right)^2$

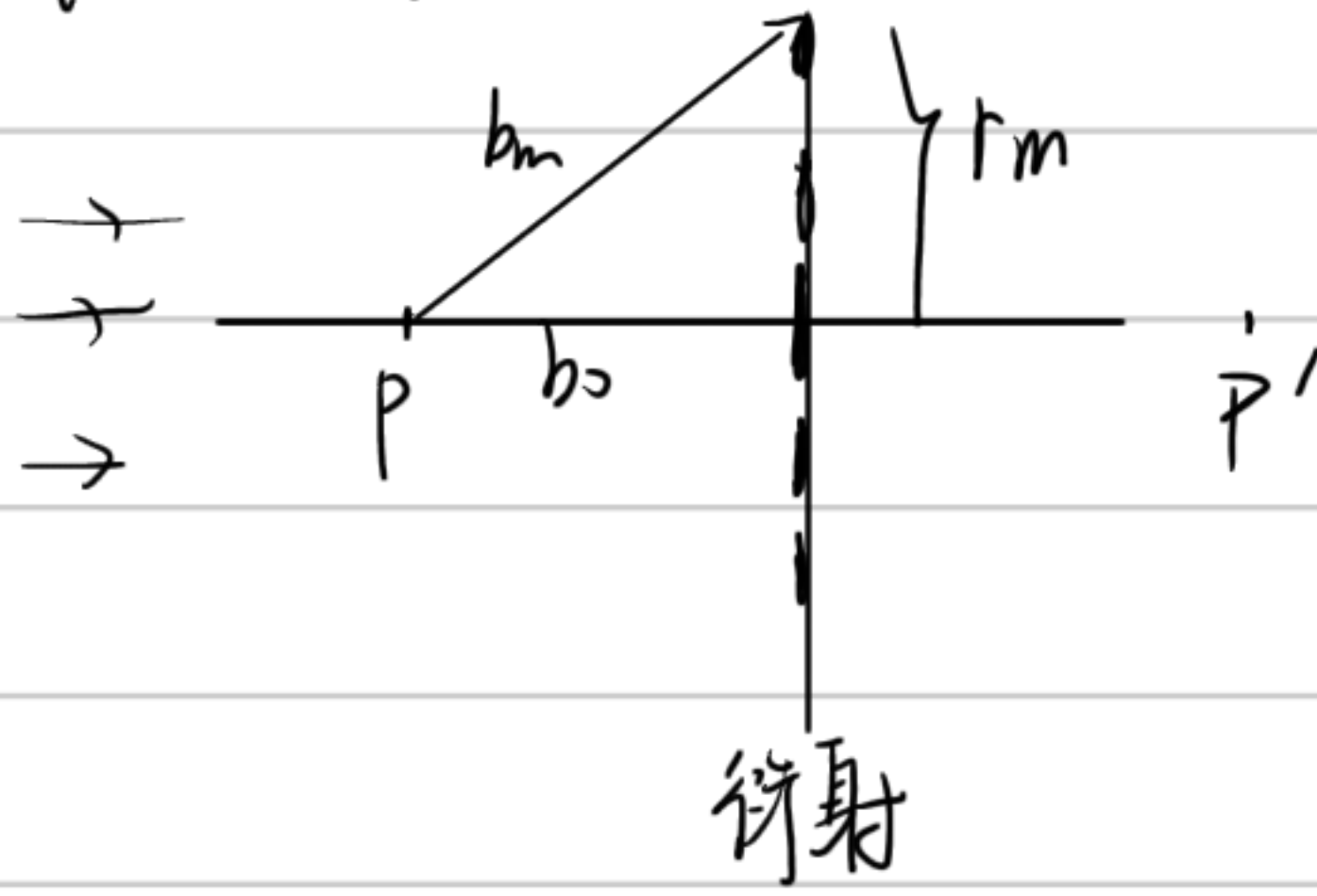
光栅: multiple slit diffraction  
不考虑 coherence time

resolving power ??



# holography and Interference Pattern

全息影像是实像



$$r_m = \sqrt{2m\lambda b_0} \quad (m=1, 2, \dots)$$



Final: electromagnetic induction ★

33. 35. 36 ★ major part.

diffraction / single slit.  
double slit.  
multiple

red light in air faster in air

Q3.



$$\lambda = \frac{\lambda_0}{1.5}$$

$$I = R + T + A$$

↓  
absorption

$$\textcircled{1} 2d = \frac{\lambda}{2} + m\lambda \quad \text{for } \lambda \text{ in the film}$$
$$\lambda = \frac{\lambda_0}{1.5}$$

$$\textcircled{2} 2d = m\lambda \quad \text{for } \lambda \text{ in the film}$$

Qp

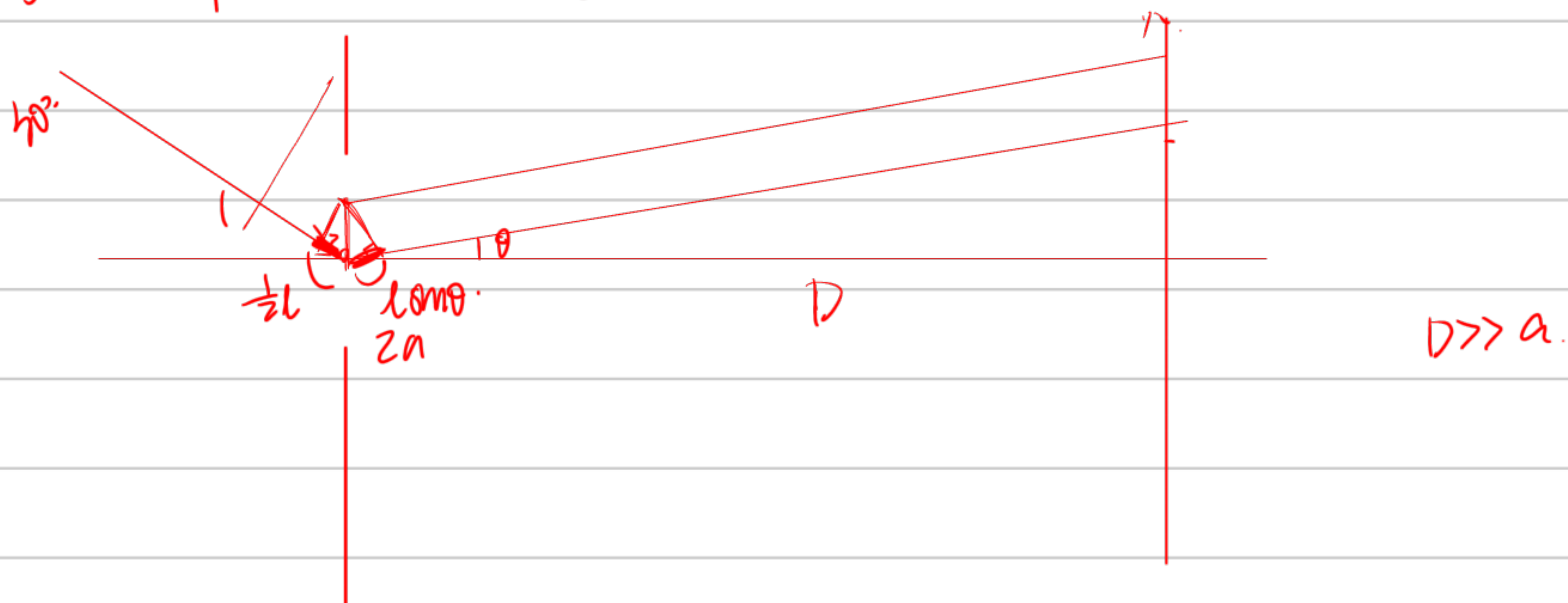
air

water

$$\Delta\theta + k\pi = n\lambda$$

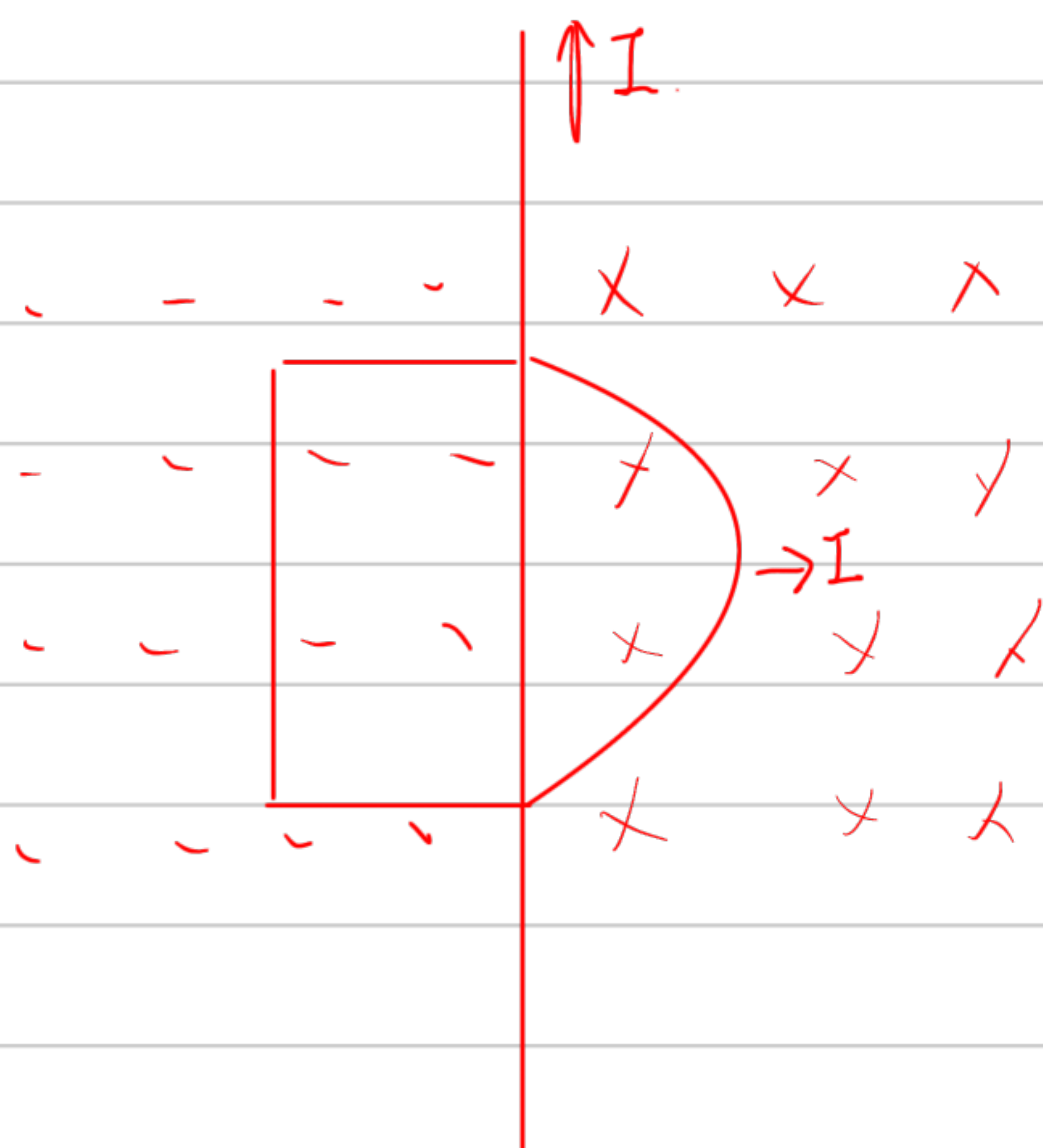


Q5. find intensity minimum



$$k(\frac{1}{2}L + a \sin \theta) = \pi + 2m\pi$$

Q6.



(29) chap

counteracting eddy currents

Meissner effect for

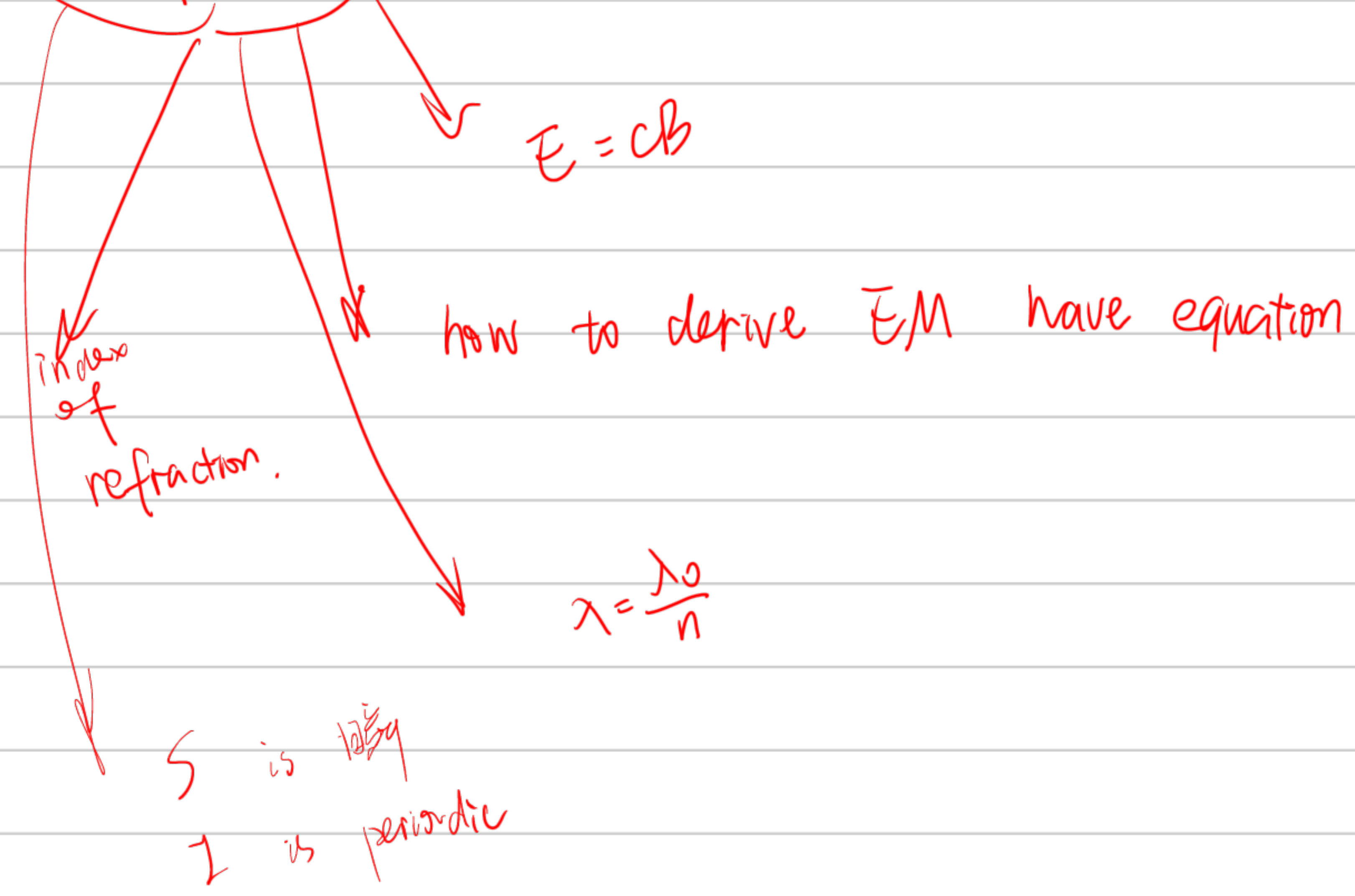
Faraday 17.5k

displacement current

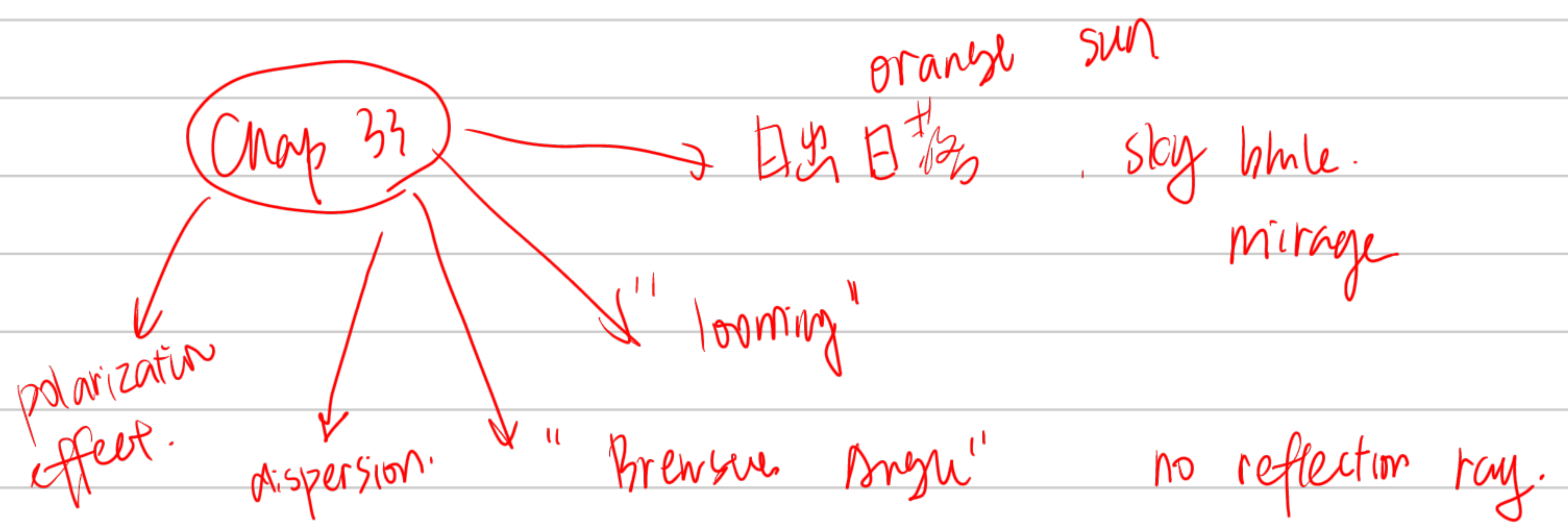
Eddy currents  
inducing E-field  
breaking?



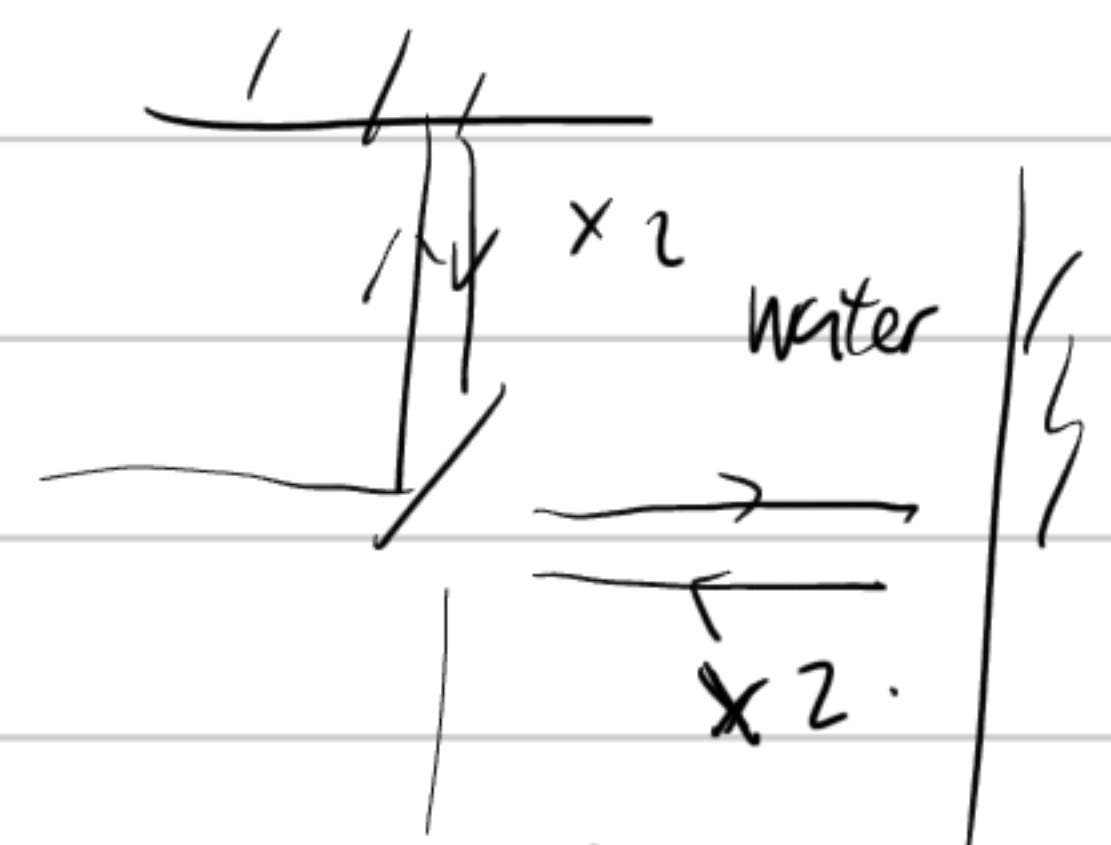
# Chap 32



# Chap 33

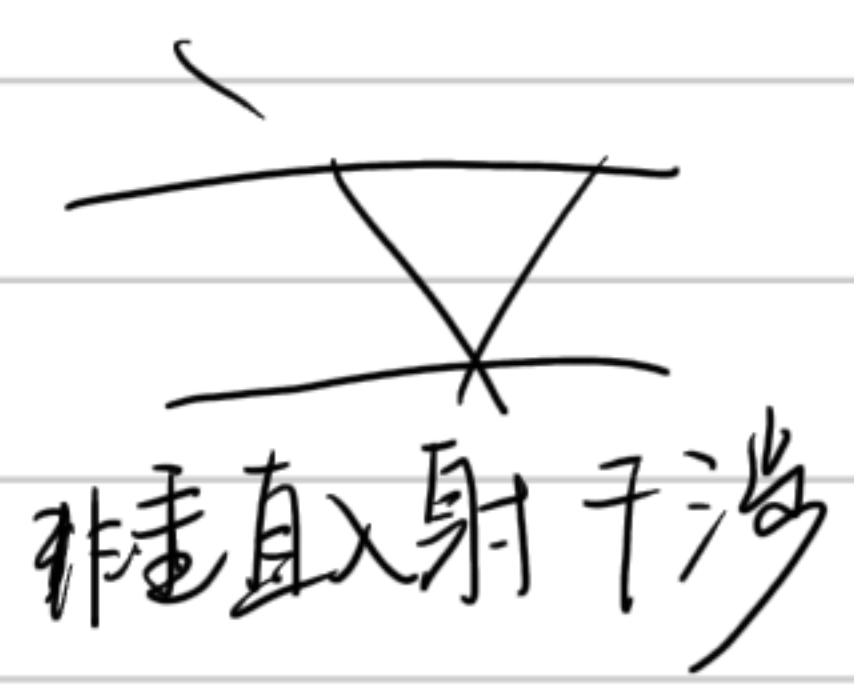


$\cos(kr - \omega t + \theta)$

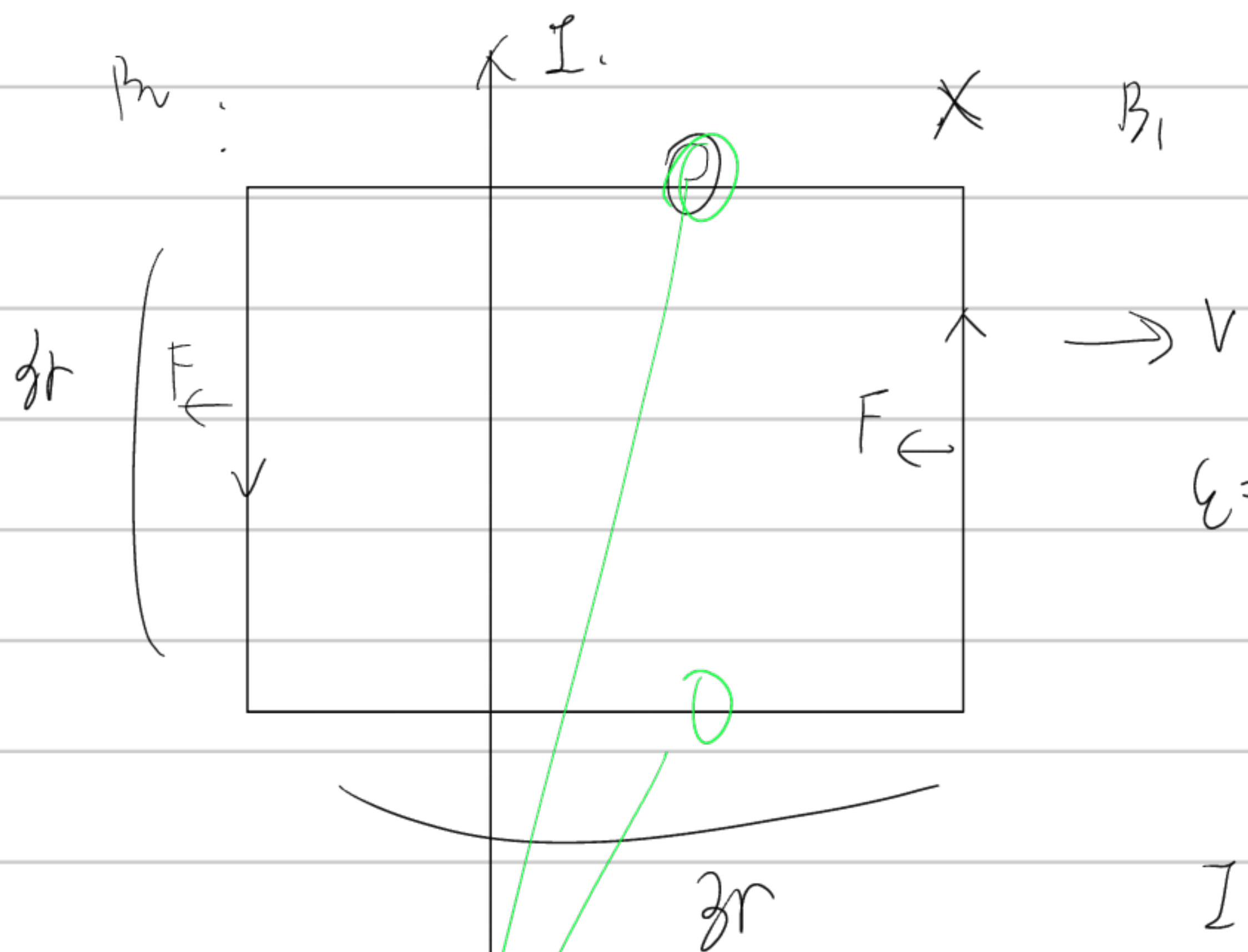


# Chap 34

thin film  
薄膜







$$\begin{aligned} \mathcal{E} &= B_1 \cdot 3r \cdot V + B_2 \cdot 3r \cdot V \\ &= \frac{\mu_0 I}{2\pi \cdot 2} \cdot 3r \cdot V + \frac{\mu_0 I}{2\pi} \cdot 3r \cdot V \\ &= \frac{9}{4} \frac{\mu_0 I V}{\pi} \end{aligned}$$

$$I = \frac{\mathcal{E}}{R} = \frac{9\mu_0 I V}{4\pi R}$$

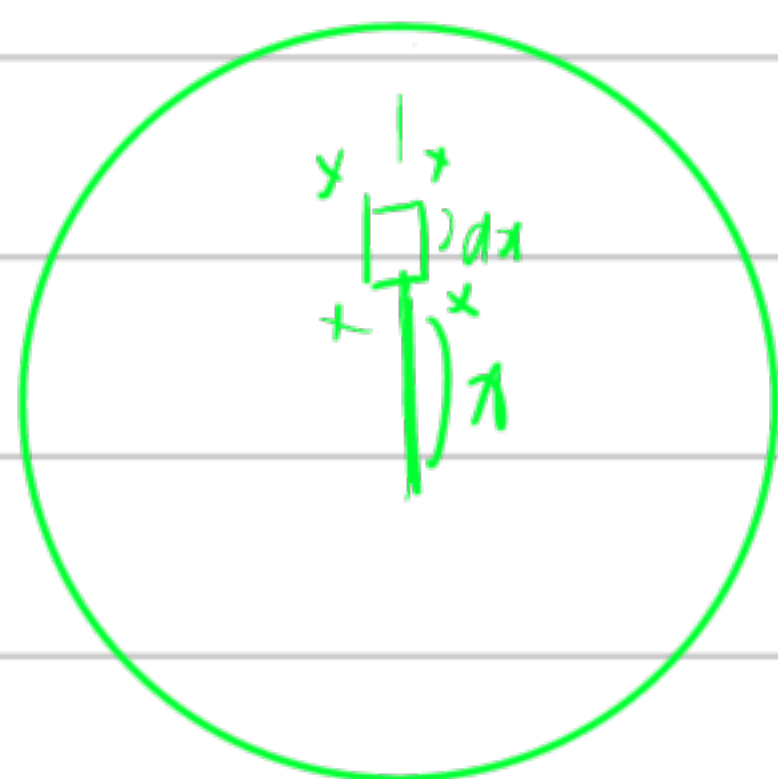
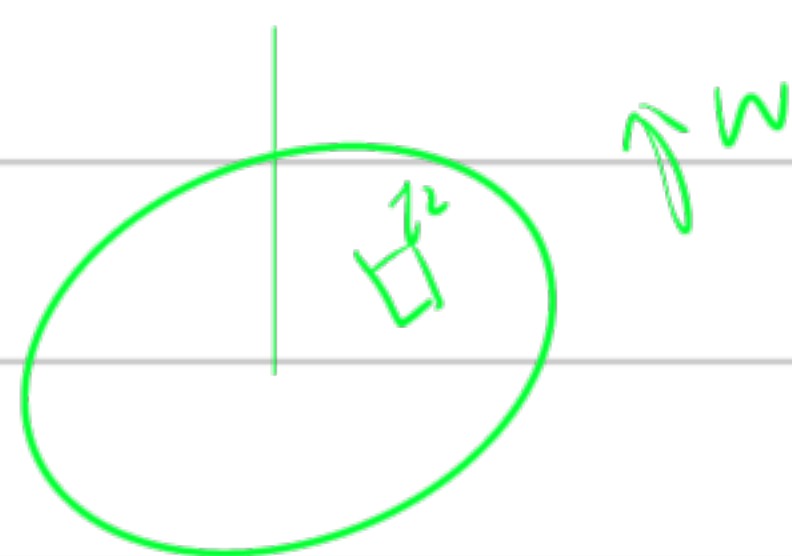
$$F_{\text{net}} = F_1 + F_2 = B_1 \cdot 3r \cdot I + B_2 \cdot 3r \cdot I$$

$$\begin{aligned} &= \frac{\mu_0 I}{2\pi \cdot 2} \cdot 3r \cdot I + \frac{\mu_0 I}{2\pi r} \cdot 3r \cdot I \\ &= \frac{3}{4} \frac{\mu_0 I^2}{\pi R} \cdot \frac{9\mu_0 I V}{4\pi R} \\ &= \frac{81}{16} \frac{\mu_0^2 I^2 V}{\pi^2 R} \end{aligned}$$

$$\mathcal{E} = B_1 V$$

they cancel out

★ F方向



$$dm = dF \cdot x$$

★ 都有 180° phase shift

★ 三角函数方程

★ 光波长表